Lab 1

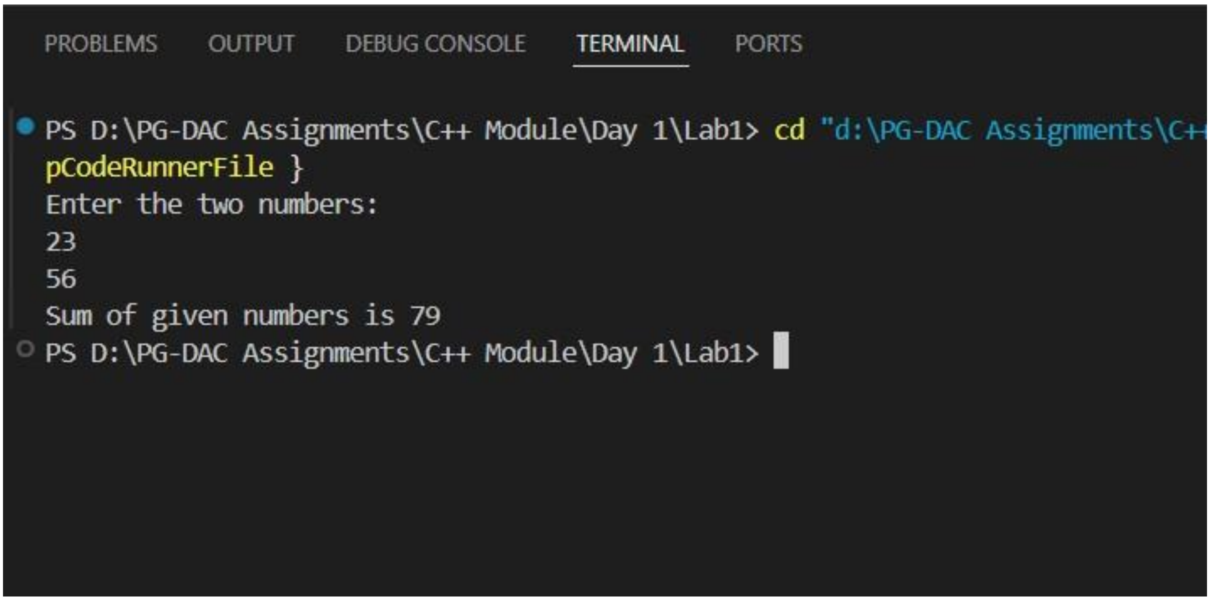
1:write program to test Hello World.

```
#include<bits/stdc++.h> using
namespace std; int main(int argc,
char const *argv[])
{    cout<<"Hello
World!"<<endl;
    return 0;
}
```

```
PS D:\PG-DAC Assignments\C++ Module\Day 1\Lab1> cd "d:\PG-DAC Assignments\C++ Module\Day 1\Lab1\" ; if ($?) { g++ HelloWorld.cpp -o HelloWorld } ;
lloWorld }
Hello World!
PS D:\PG-DAC Assignments\C++ Module\Day 1\Lab1>
```

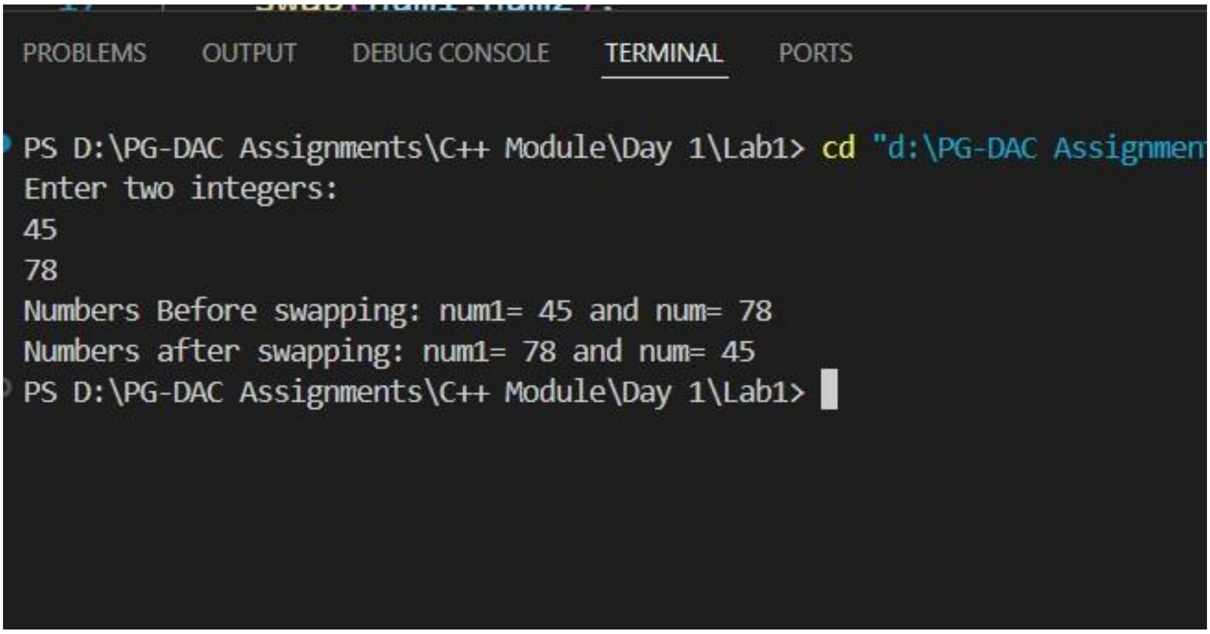
2:Write a program to adddition of two numbers .

```
#include<bits/stdc++.h>
using namespace std;

int main(int argc, char const *argv[])
{    double num1,num2;    cout<<"Enter the two
numbers: "<<endl;    cin>>num1>>num2;
cout<<"Sum of given numbers is
"<<num1+num2<<endl;
    return 0;
}
```



3:Write a program to swap two numbers.

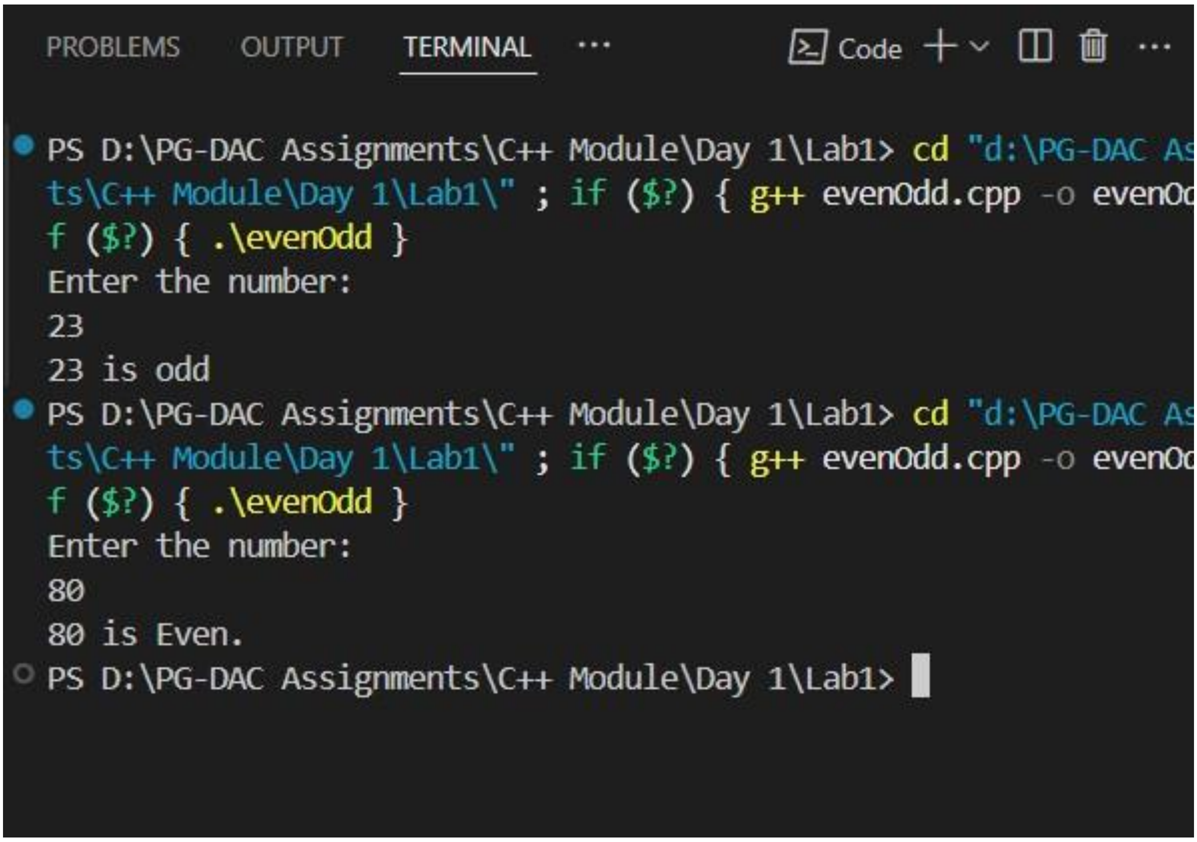
```
#include<bits/stdc++.h>

using namespace std;


void swap(int &num1,int &num2)

{     int temp =

num1;     num1

=num2;     num2

= temp;

}


int main(int argc, char const *argv[])

{     int num1,num2;     cout<<"Enter two integers: "<<endl;

cin>>num1>>num2;     cout<<"Numbers Before swapping: num1= "<<num1<<"

and num= "<<num2<<endl;     swap(num1,num2);     cout<<"Numbers after

swapping: num1= "<<num1<<" and num= "<<num2<<endl;

    return 0;

}
```



4. Write a program to accept an integer and check if it is even or odd.

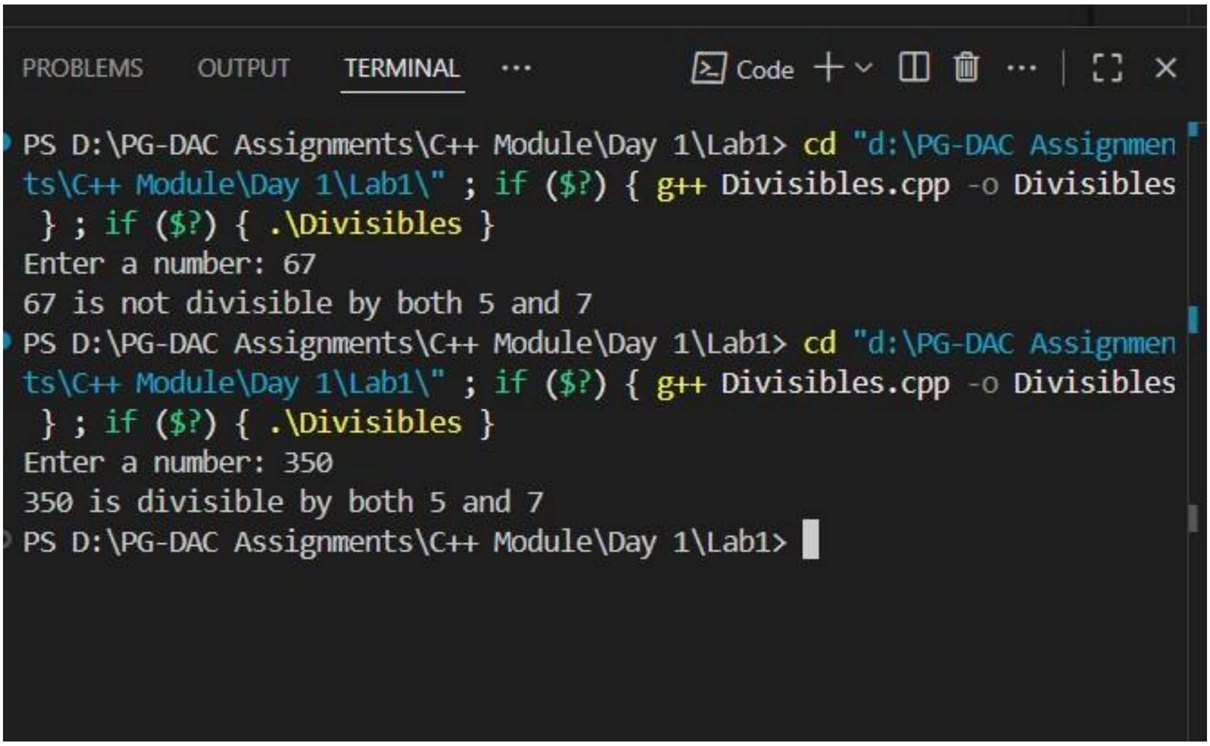
```
//Tell the number is even or
odd #include<bits/stdc++.h>
using namespace std;

int main(int argc, char const *argv[])
{    int num;    cout<<"Enter the
number: "<<endl;    cin>>num;

  if(num % 2== 0)
   {
       cout<<num<<" is Even. "<<endl;
   }
else
   {
       cout<<num<<" is odd "<<endl;
   }
return 0;
}
```

}



5. Write a program to accept a number and check if it is divisible by 5 and 7. #include <iostream> using namespace std;

```
int main() {
   int n;
   cout << "Enter a number: ";
cin >> n;

   if (n % 5 == 0 && n % 7 == 0)       cout <<
n << " is divisible by both 5 and 7";    else
cout << n << " is not divisible by both 5 and 7";

   return 0;
}
```



6. Write a program, which accepts annual basic salary of an employee and calculates and displays the

Income tax as per the following rules.

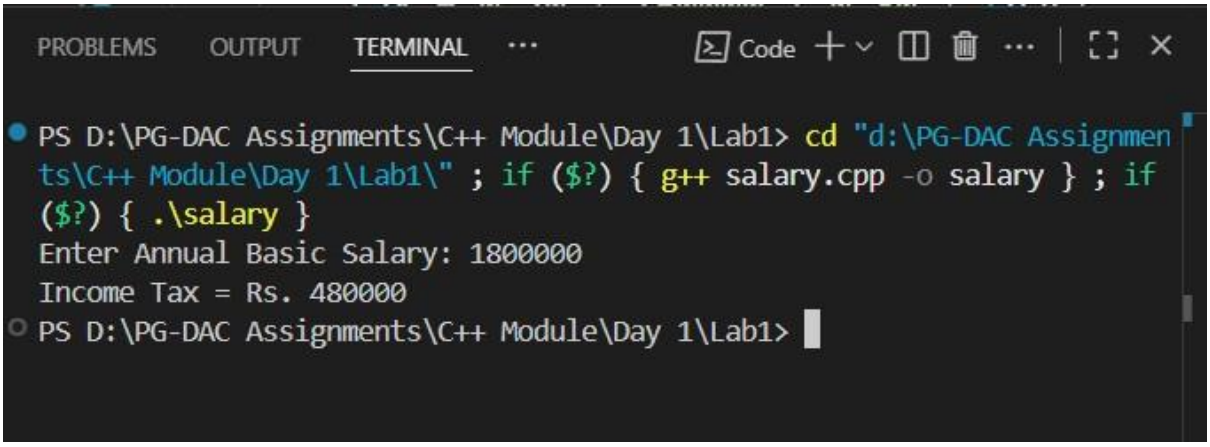
Basic: < 1, 50,000 Tax = 0

1, 50,000 to 3,00,000 Tax = 20%

> 3,00,000 Tax = 30%

```
#include <iostream>
using namespace std;

int main() {     double basicSalary, tax
= 0;     cout << "Enter Annual Basic
Salary: ";     cin >> basicSalary;

    if (basicSalary < 150000) {
        tax = 0;
    }
    else if (basicSalary <= 300000) {
tax = 0.20 * (basicSalary - 150000);
```

```
}      else {         tax = 0.20 * 150000
+ 0.30 * (basicSalary - 300000);
   }

   cout << "Income Tax = Rs. " << tax << endl;

   return 0;
}
```



7. Accept a lowercase character from the user and check whether the character is a vowel or consonant.

(Hint: a, e, i, o, u are

vowels) #include

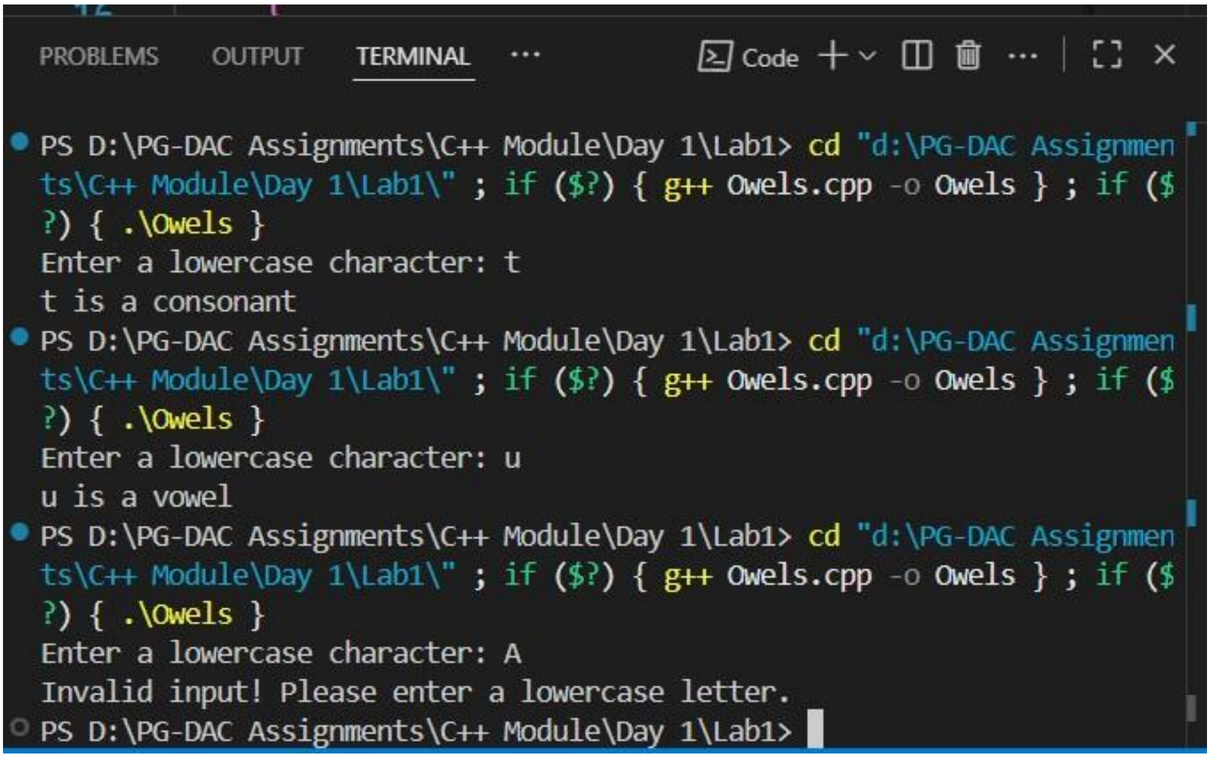
```
<iostream> using
namespace std;

int main() {     char ch;     cout <<
"Enter a lowercase character: ";     cin
>> ch;

   if (ch >= 'a' && ch <= 'z') {         if (ch == 'a' || ch
== 'e' || ch == 'i' || ch == 'o' || ch == 'u')            cout <<
ch << " is a vowel";
```

else cout << ch << " is a consonant"; } else { cout << "Invalid input! Please enter a lowercase letter.";

}

return 0;

}



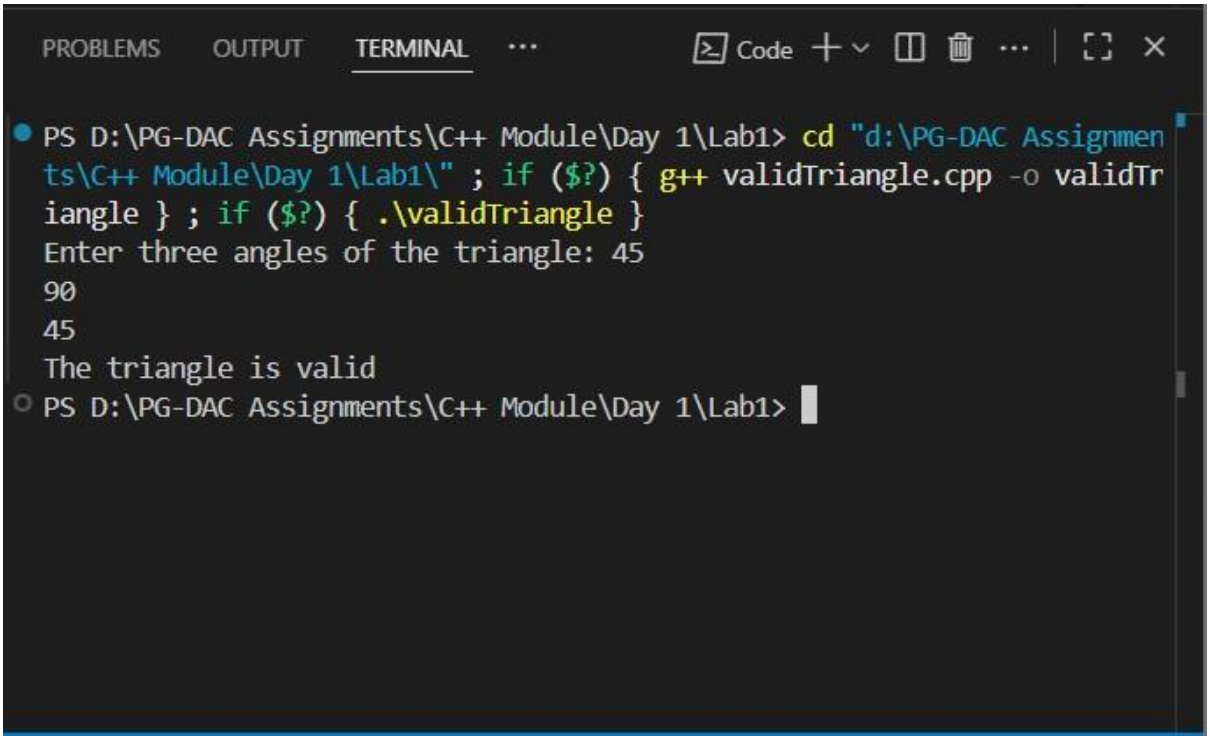
8. Write a program to input angles of a triangle and check whether triangle is valid or not. #include <iostream> using namespace std;

int main() {

int a, b, c; cout << "Enter three angles of the triangle: "; cin >> a >> b >> c;

if (a > 0 && b > 0 && c > 0 && (a + b + c == 180)) cout << "The triangle is valid" << endl;

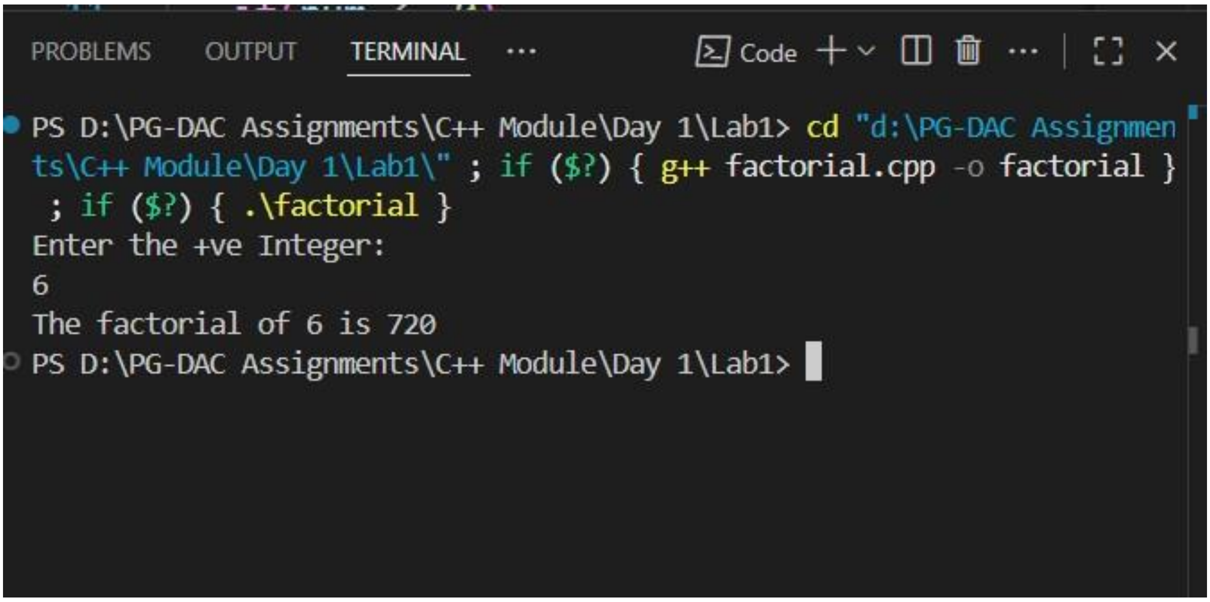
```
else        cout << "The triangle is not valid" <<
endl;


   return 0;
}
```



9:Write a program to find factorial of a given number. ex:no5 fact=5*4*3*2*1=120

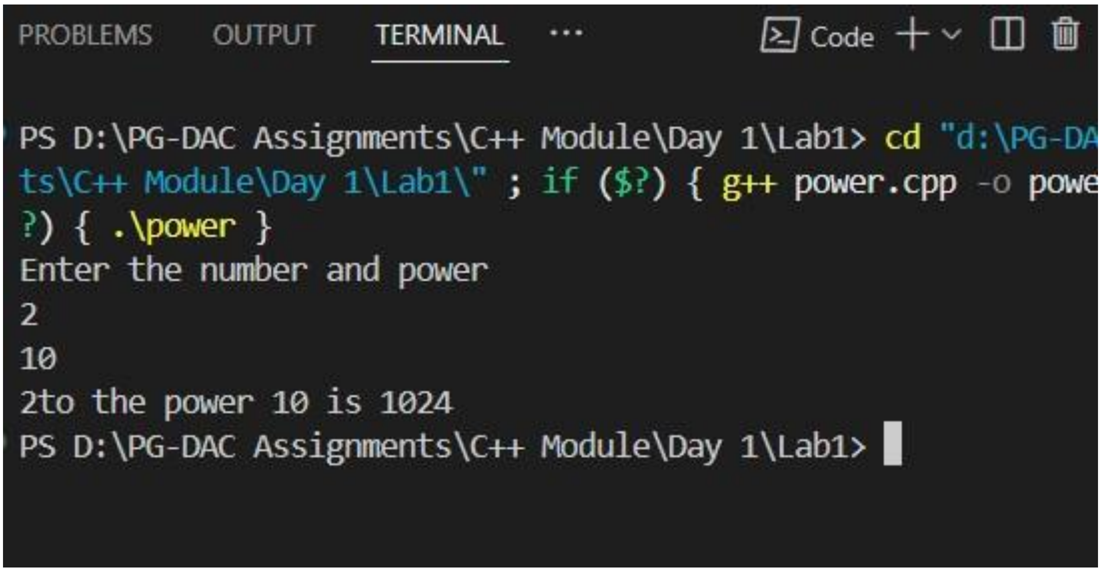
```
//Write a program to find factorial of a given number. ex:no5  fact=5*4*3*2*1=120


#include<bits/stdc++.h>
using namespace std;


int main(int argc, char const *argv[])
{     int num,fact = 1;     cout<<"Enter
the +ve Integer: "<<endl;
cin>>num;    if(num <= 0)    return 1;


  for(int i = 1; i<=num;i++)
fact*=i;
```

```
  cout<<"The factorial of "<<num<<" is "<<fact<<endl;
   return 0;
}
```



10:Write a program to find m to the power n. m=3 and n=4 so 3*3*3*3

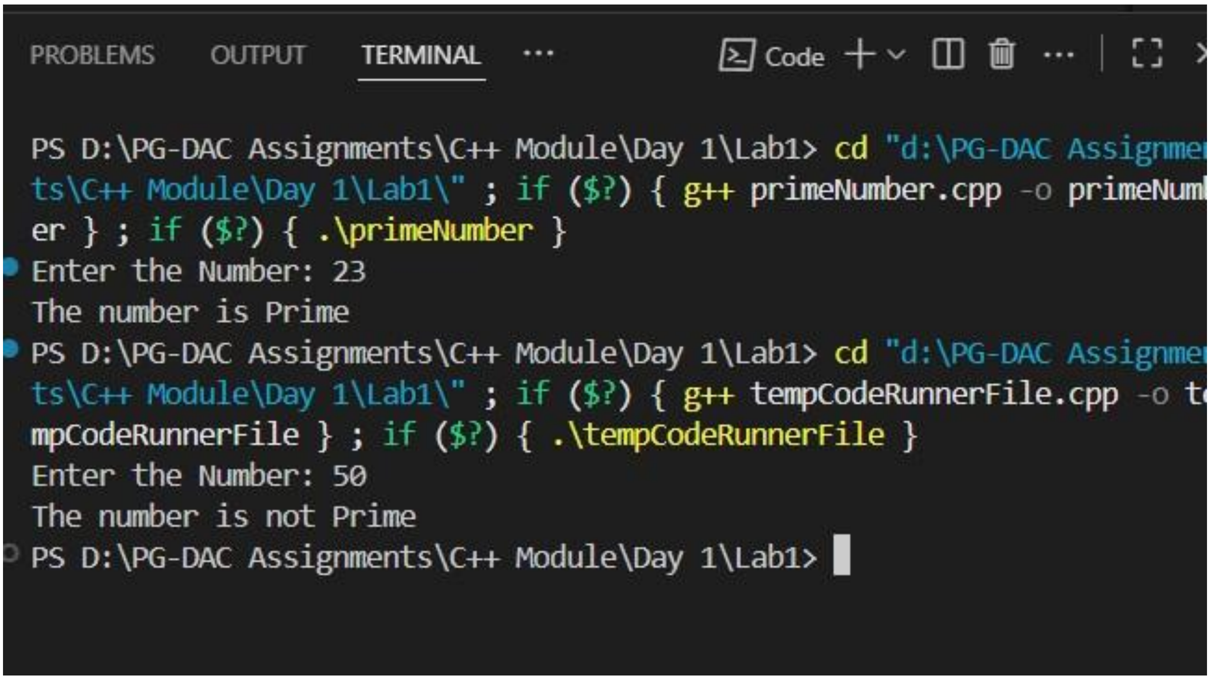
```
#include <bits/stdc++.h>
using namespace std;

int main(int argc, char const *argv[])
{     int num, pw;     cout << "Enter the
number and power" << endl;     cin >> num
>> pw;     if (num <= 0)
    {
        cout << "Enter +ve integer only" << endl;
        return 0;
    }

    cout << num << "to the power " << pw << " is " << pow(num,pw) << endl;
     return 0;
}
```



11:Check if number is a prime number or not.:

```
#include <bits/stdc++.h>
using namespace std;

bool isPrime(int num) {
if (num <= 1) return false;

    for (int i = 2; i <= sqrt(num); i++)
{         if (num % i == 0) return
false;
    }
return true;
}

int main() {     int num;
cout << "Enter the Number:
";     cin >> num;

    if (isPrime(num)) {          cout << "The
number is Prime" << endl;
```

```
    } else {         cout << "The number is not
Prime" << endl;    }


    return 0;
}
```



12:Sum of series :

1+2+3+….+n

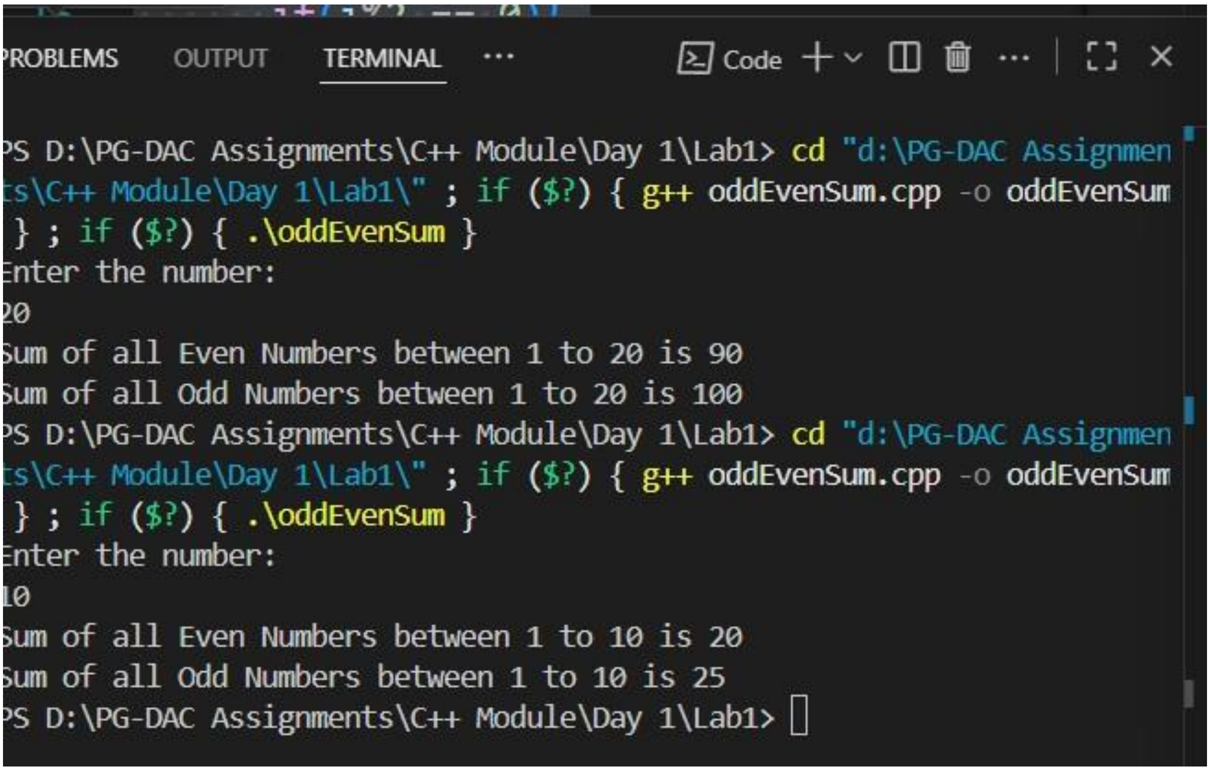
13:Check whether the number is palindrome or not?

```
#include <iostream>
using namespace std;


int main() {     int num,
org, rev = 0, rem;


    cout << "Enter a number: ";
cin >> num;


    org = num;  // store original number
```

```
  while (num > 0) {
rem = num % 10;
rev = rev * 10 + rem;
num = num / 10;
   }



   if (org == rev) {        cout << org << " is a
Palindrome number." << endl;
   } else {        cout << org << " is NOT a Palindrome
number." << endl;
   }

   return 0;
}
```

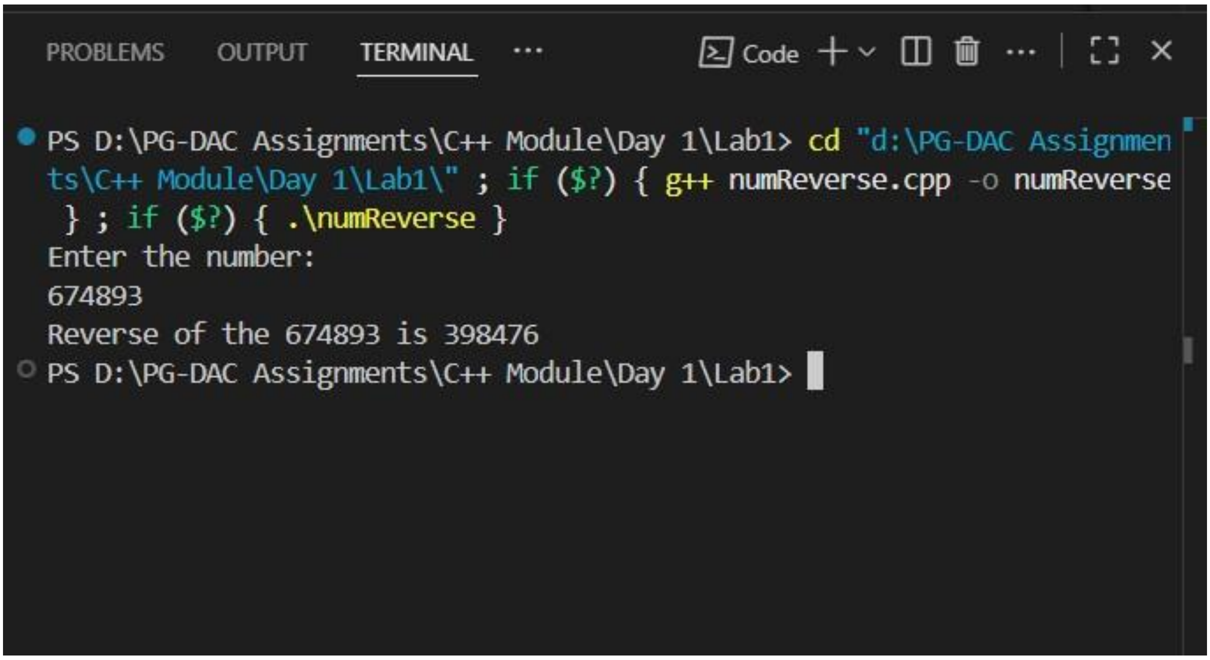
14:Write a program to find sum of all even and odd numbers between 1 to n.

```
#include<bits/stdc++.h>
using namespace std;

int main(int argc, char const *argv[])
{    int num,evenSum = 0,oddSum
= 0;    cout<<"Enter the number:
"<<endl;    cin>>num;    if(num
<0)
   {
    cout<<"Enter the +ve number only"<<endl;
    return 0;
   }    for(int i = 1;
i<num;i++)
   {       if(i%2
== 0){
evenSum += i;
      }else
      {
        oddSum += i;
      }
   }
                    cout<<"Sum of all Even Numbers between 1 to "<<num<<" is
 "<<evenSum<<endl;      cout<<"Sum of all Odd Numbers between 1 to "<<num<<"
                                                       is "<<oddSum<<endl;
    return 0;
}
```



15: Write a program to enter a number and print its reverse.

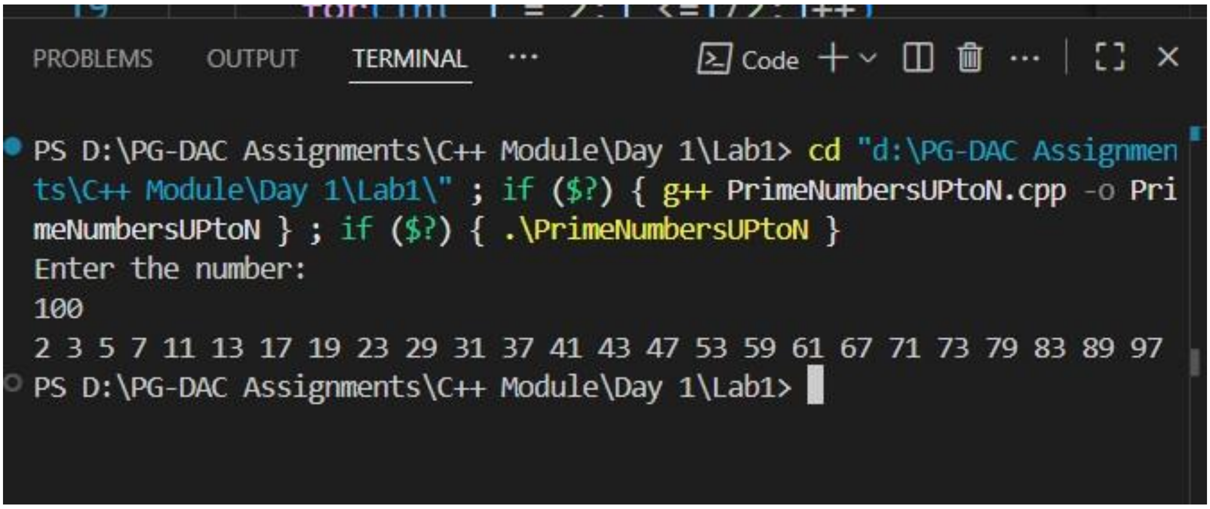
```
#include<bits/stdc++.h>

using namespace std;


int main(int argc, char const *argv[])

{    int num,temp,org,rev=0;

cout<<"Enter the number:

"<<endl;    cin>>num;     org

=num;    while (num > 0) {

temp = num % 10;                  rev

= rev * 10 + temp;           num =

num / 10;

    }

    cout<<"Reverse of the "<<org<<" is "<<rev<<endl;

    return 0;

}
```



16:Write a program to print all Prime numbers between 1 to n.

```
#include<bits/stdc++.h>
using namespace std;

int main(int argc, char const *argv[])
{    int num;    cout<<"Enter the
number: "<<endl;    cin>>num;
if(num <= 1)
  {
    cout<<"Not a Prime Number!"<<endl;
    return 0;
  }

  for(int i = 2;i<=num;i++)
  {      int flag=0;
for(int j = 2;j <=i/2;j++)
{          if(i%j==0){
flag=1;          break;
```

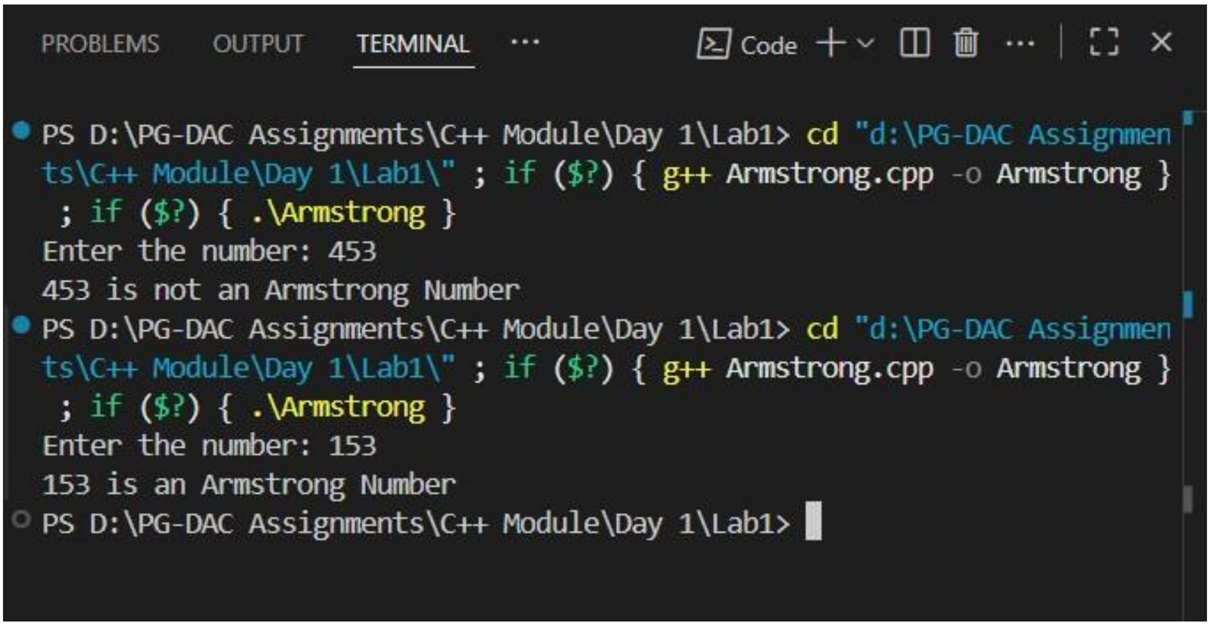
```
        }
      }
if(flag==0)
      {
        cout<<i<<endl;
      }
    }

    return 0;
}
```



17:Write a program to check entered number is Armstrong number or not.

```
#include <bits/stdc++.h>
using namespace std;

int main() {     int num;
cout << "Enter the number:
";     cin >> num;

    int org = num;
int cnt = 0;     int
temp = num;
```

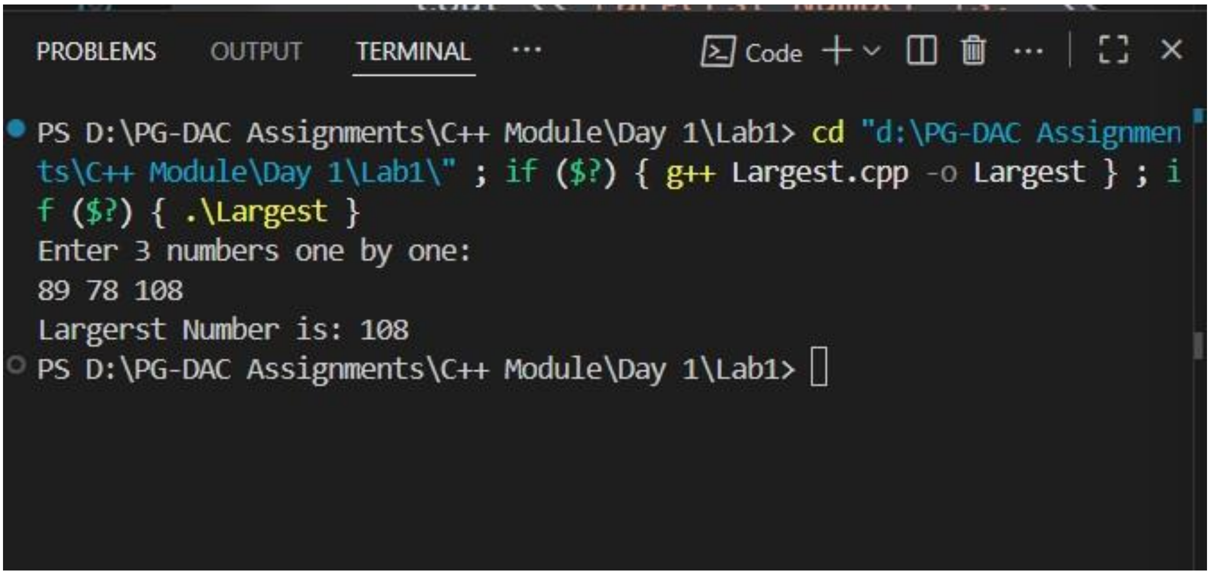
```
    while (temp != 0)
{         cnt++;
temp /= 10;
    }

    long long result = 0;
temp = num;

    while (temp != 0) {         int digit =
temp % 10;         result +=
(int)round(pow(digit, cnt));         temp
/= 10;
    }

    if (result == org) {         cout << org << " is an
Armstrong Number" << endl;
    } else {         cout << org << " is not an Armstrong
Number" << endl;
    }

    return 0;
}
```



18:Write a program to find greatest of three numbers using nested if-else.

```
#include <iostream>
using namespace std;

int main() {
   int a, b, c;
   cout<<"Enter 3 numbers one by one:
"<<endl;     cin >> a >> b >> c;

   if (a >= b) {         if (a >= c)
cout <<"Largerst Number is: "<< a;
        else
           cout <<"Largerst Number is: "<<c;
   } else {         if (b >= c)           cout
<<"Largerst Number is: "<< b;
        else
           cout <<"Largerst Number is: "<< c;
   }

   return 0;
```

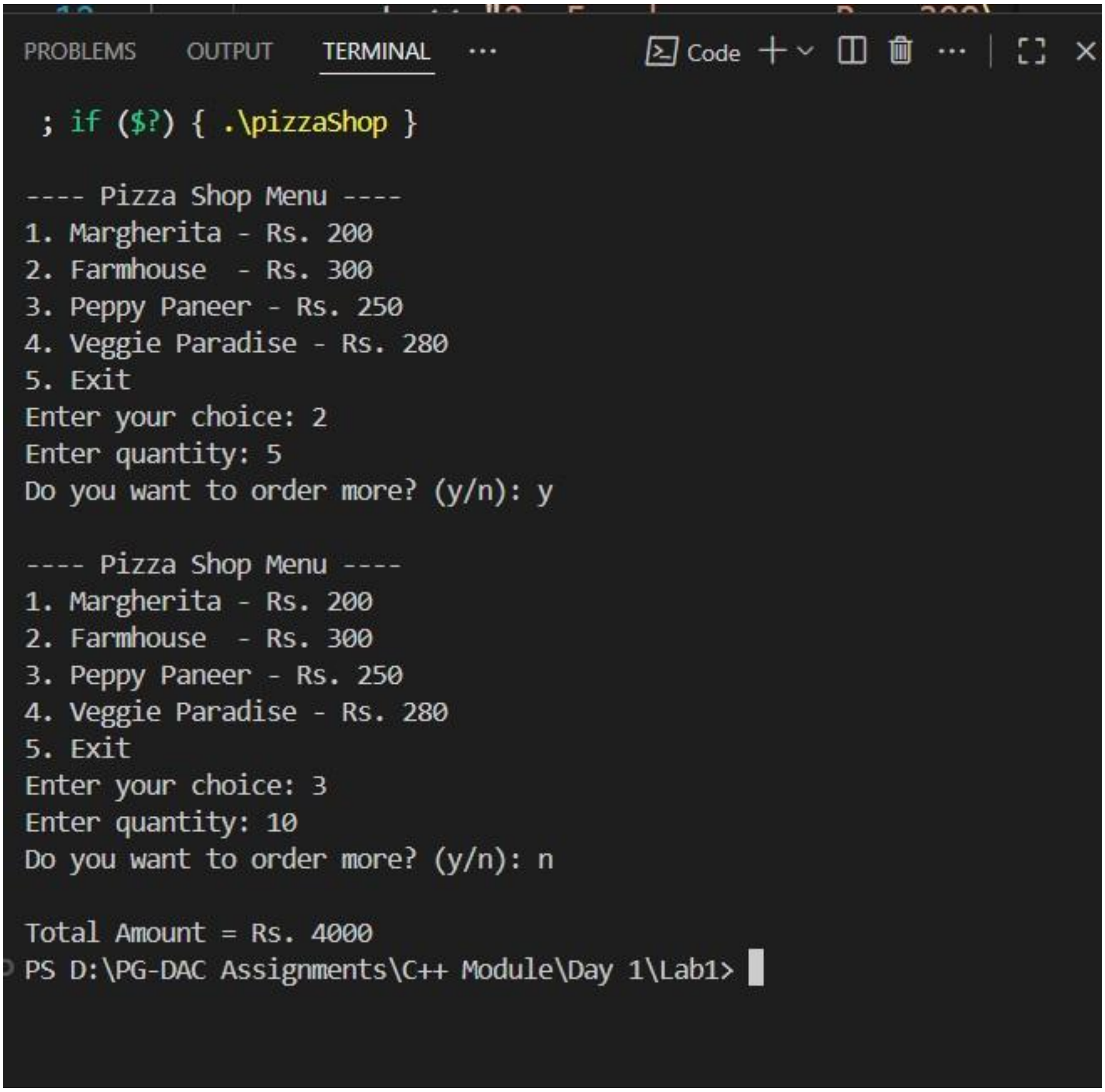
}



19:Create menu driven program for Pizza Shop.And display total amount,

```
#include <iostream>

using namespace std;


int main() {

int choice, qty;

int total = 0;

char ch;


    do {         cout << "\n---- Pizza Shop

Menu ----\n";         cout << "1. Margherita

- Rs. 200\n";         cout << "2. Farmhouse

- Rs. 300\n";         cout << "3. Peppy

Paneer - Rs. 250\n";         cout << "4.

Veggie Paradise - Rs. 280\n";         cout

<< "5. Exit\n";         cout << "Enter your

choice: ";         cin >> choice;
```

```
        if (choice >= 1 && choice <=
4) {            cout << "Enter
quantity: ";            cin >> qty;
        }

        switch (choice) {            case 1: total +=
200 * qty; break;            case 2: total += 300
* qty; break;            case 3: total += 250 *
qty; break;            case 4: total += 280 * qty;
break;            case 5: cout << "Exiting...\n";
break;            default: cout << "Invalid
choice!\n"; break;
        }

        if (choice != 5) {            cout << "Do you
want to order more? (y/n): ";            cin >> ch;
} else {            ch = 'n';
        }

    } while (ch == 'y' || ch == 'Y');

    cout << "\nTotal Amount = Rs. " << total << endl;

    return 0;
}
```

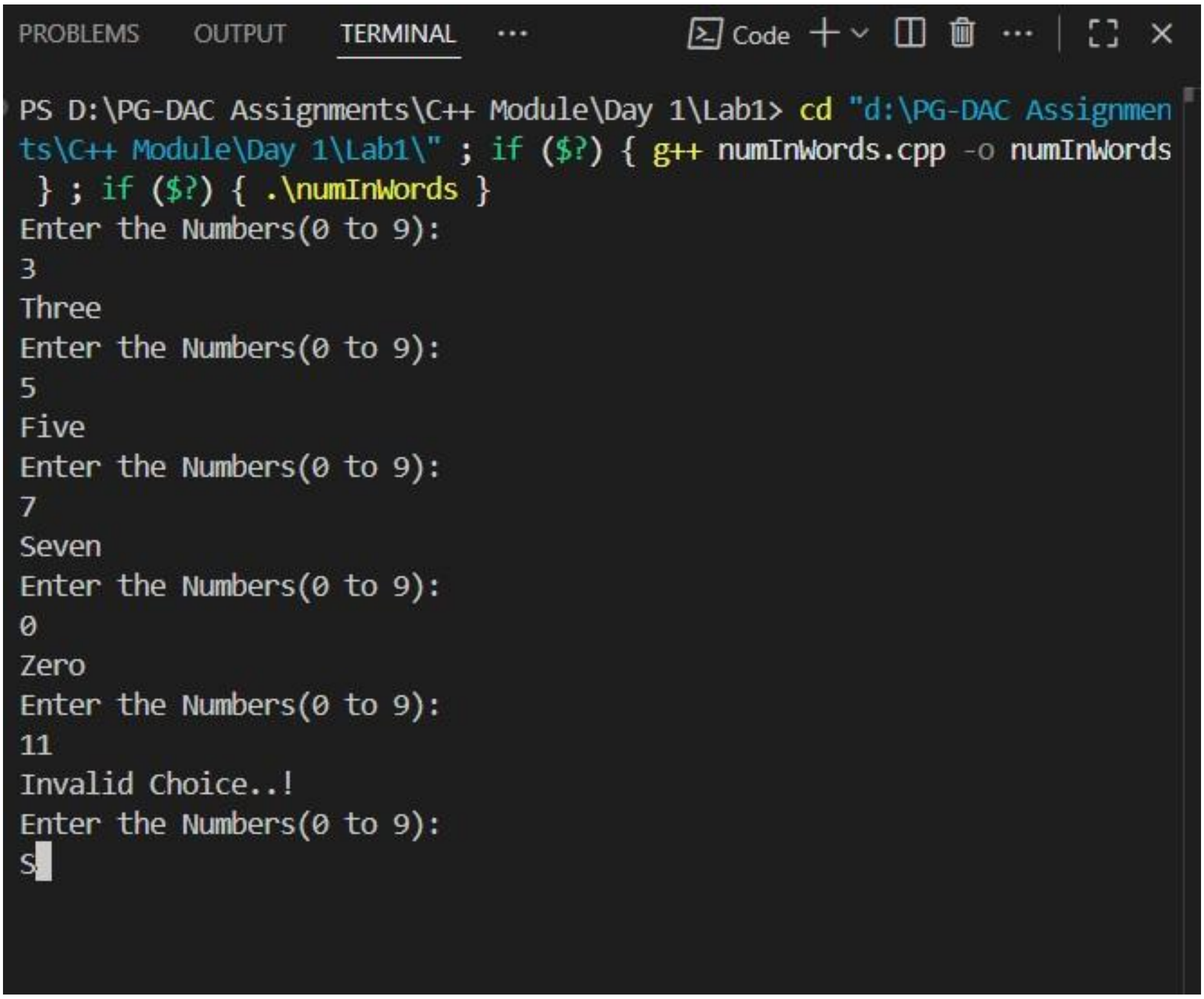


20:Accept a single digit from the user and display it in words. For example, if digit entered is 9, display Nine.

```
#include <bits/stdc++.h>

using namespace std;


int main(int argc, char const *argv[])

{    int

num;

do

   {
```

```
        cout << "Enter the Numbers(0 to 9): " <<
endl;         cin >> num;


        switch (num)
        {
case 0:
            cout << "Zero" << endl;
break;
         case
1:
            cout << "One" << endl;
break;
         case
2:
            cout << "Two" << endl;
            break;
         case
3:
            cout << "Three" << endl;
break;
         case
4:
            cout << "Four" <<
endl;             break;
case 5:             cout <<
"Five" << endl;
break;         case 6:
cout << "Six" << endl;
break;         case 7:
```

```
            cout << "Seven" <<
endl;              break;          case
8:
            cout << "Eight" <<
endl;              break;          case
9:
            cout << "Nine" << endl;
break;

default:
            cout << "Invalid Choice..!" << endl;
        }

    } while (num != 10);
return 0;
}
```



```
PROBLEMS   OUTPUT   TERMINAL   ...                    Code  +

PS D:\PG-DAC Assignments\C++ Module\Day 1\Lab1> cd "d:\PG-DAC Assignmen
ts\C++ Module\Day 1\Lab1\" ; if ($?) { g++ numInWords.cpp -o numInWords
 } ; if ($?) { .\numInWords }
Enter the Numbers(0 to 9):
3
Three
Enter the Numbers(0 to 9):
5
Five
Enter the Numbers(0 to 9):
7
Seven
Enter the Numbers(0 to 9):
0
Zero
Enter the Numbers(0 to 9):
11
Invalid Choice..!
Enter the Numbers(0 to 9):
S
```

21. Write a program, which accepts two integers and an operator as a character (+ - * /), performs the

corresponding operation and displays the result.

```
#include<bits/stdc++.h>
using namespace std;


int main(int argc, char const *argv[])
{    char ch;    long
long num1,num2;
do{


     cout<<"Enter First Numbers: "<<endl;
cin>>num1;       cout<<"Enter the Second
Number: "<<endl;       cin>>num2;       cout<<"--
```

```
-------Menu----------"<<endl;
cout<<"Additon"<<endl;
cout<<"Substraction"<<endl;
cout<<"Multiplication"<<endl;
cout<<"Division"<<endl;        cout<<"Enter the
Operation: "<<endl;        cout<<" \n-----------------
---------------- \n"<<endl;        cout<<"Enter the
Operation: "<<endl;        cin>>ch;
        cout<<"\n"<<num1<<" "<<ch<<" "<<num2<<" \n"<<endl;
        switch (ch)
        {
case '+':
                    cout<<"sum is "<<num1+num2<<endl;
                     break;
            case
'-':
                     cout<<"Substraction is "<<abs(num1-num2)<<endl;                    break;
            case
'*':
                     cout<<"Multiplication is "<<num1*num2<<endl;
                     break;
                        case '/':
if(num2<=0)
                      {
cout<<"Error......."<<endl;
                      }else{
                      cout<<"Division is "<<num1/num2<<endl;
                      }
break;
```

```
default:
            cout<<"Invalid Operation!"<<endl;
    }

  }while(ch != '\0');
return 0;
}
```

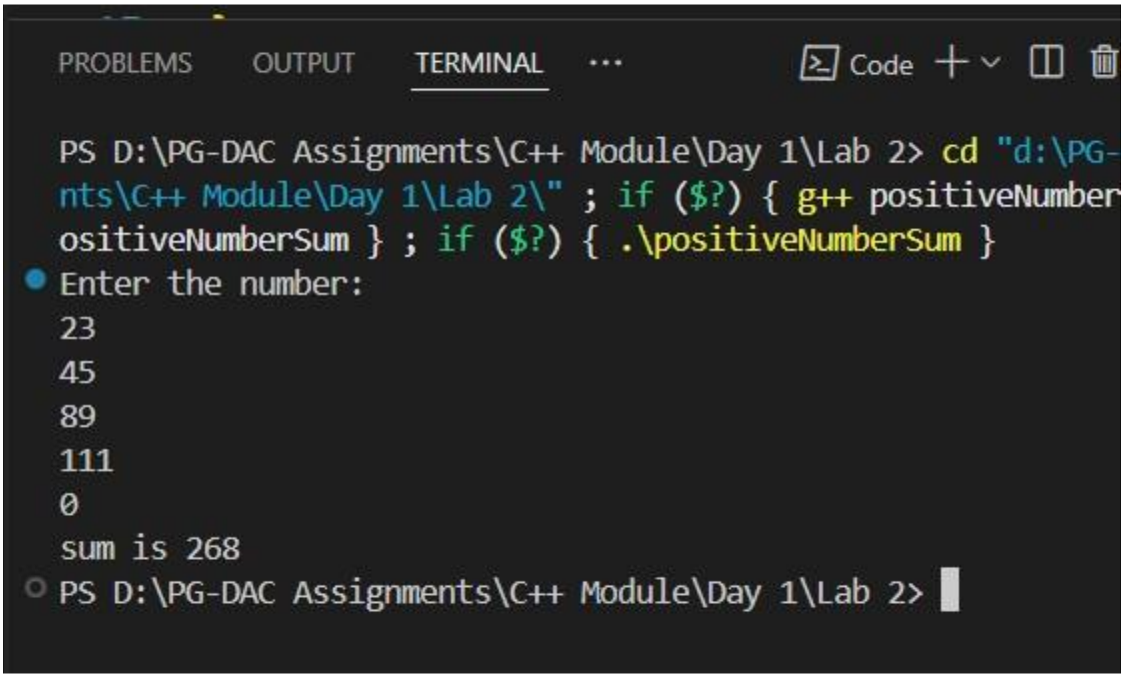
PROBLEMS OUTPUT TERMINAL ··· Code

```
PS D:\PG-DAC Assignments\C++ Module\Day 1\Lab1> cd "d:\PG-DAC Assignments\C++ Module\Day 1\Lab1\" ; if ($?) { g++ calculator.cpp -o calculator } ; if ($?) { .\calculator }
Enter First Numbers:
124
Enter the Second Number:
896
---------Menu----------
Additon
Substraction
Multiplication
Division
Enter the Operation:

-----------------------------------


Enter the Operation(+,-,*,/):
*


124 * 896

Multiplication is 111104
Enter First Numbers:
7
Enter the Second Number:
98
---------Menu----------
Additon
Substraction
Multiplication
Division
Enter the Operation:

-----------------------------------


Enter the Operation(+,-,*,/):
+

7 + 98

sum is 105
```

Lab 2

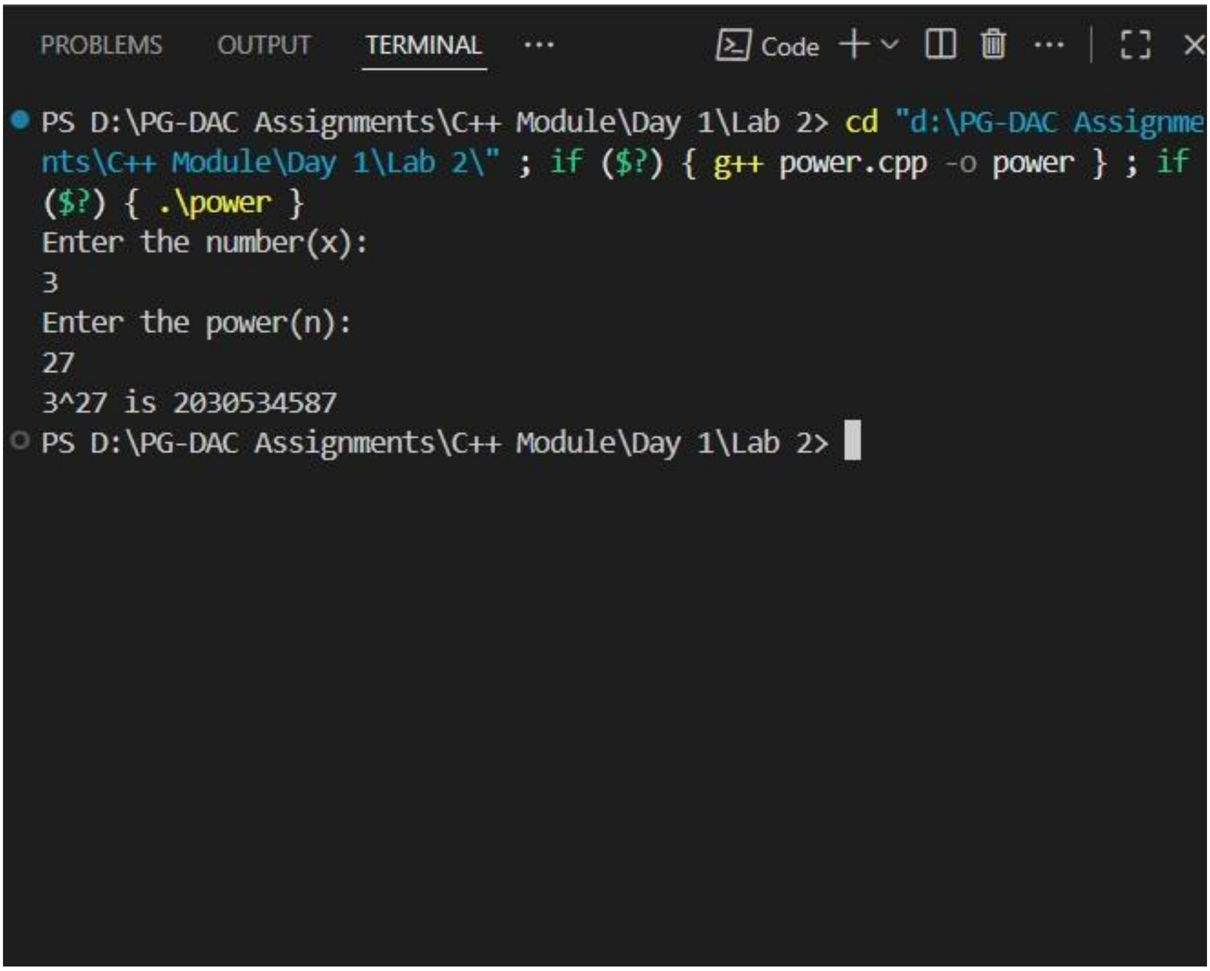
1:Write a program that accepts numbers continuously as long as the number is positive and prints the sum of the given numbers. #include<bits/stdc++.h> using namespace std;

```
int main(int argc, char const *argv[])
{    int num,sum = 0;
cout<<"Enter the number:
"<<endl;
   do{        cin>>num;
if(num%2 >=0)         sum +=
num;    }while(num != 0);
cout<<"sum is
"<<sum<<endl;
   return 0;
}
```



2. Write a program to accept two integers x and n and compute x raised to n. #include<bits/stdc++.h> using namespace std;

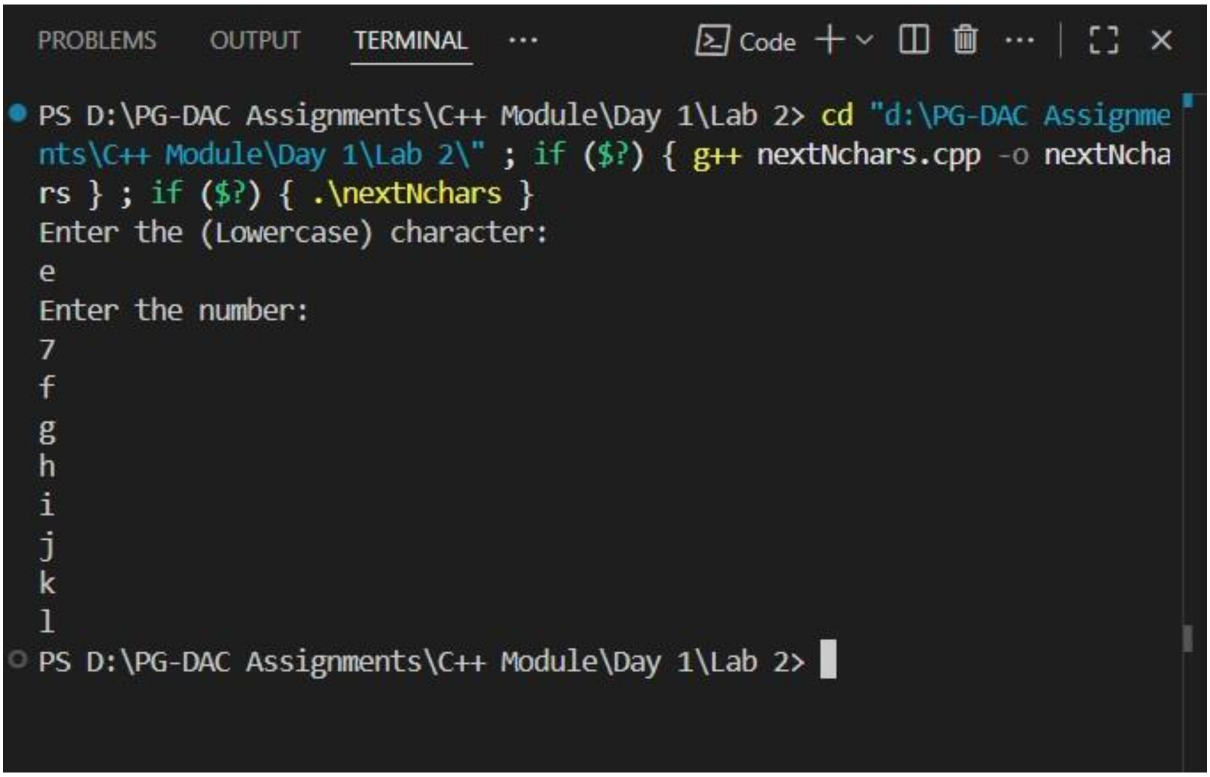
```
int main(int argc, char const *argv[])
{    int x,n,result = 1;    cout<<"Enter
the number(x): "<<endl;    cin>>x;
cout<<"Enter the power(n):
"<<endl;    cin>>n;    if(n<0)
   {
    cout<<"Enter +ve integer only"<<endl;
    return   0;            }
for(int i =0; i<n;i++)
    {
result *=x;
    }
    cout<<x<<"^"<<n<<" is "<<result<<endl;
    return 0;
}
```



3. Write a program to accept a character, an integer n and display the next n characters. #include<bits/stdc++.h> using namespace std;

```
int main(int argc, char const *argv[])
{    char ch;    int num;    cout<<"Enter the
(Lowercase) character: "<<endl;    cin>>ch;
cout<<"Enter the number: "<<endl;
cin>>num;    if(num<0)
   {
       cout<<"Enter only +ve integer."<<endl;
       return 0;
   }  for(int i =1; i<=
  num;i++)
  {
```

```
        cout<<(char)(ch+i)<<" "<<endl;
    }
return 0;
}
```

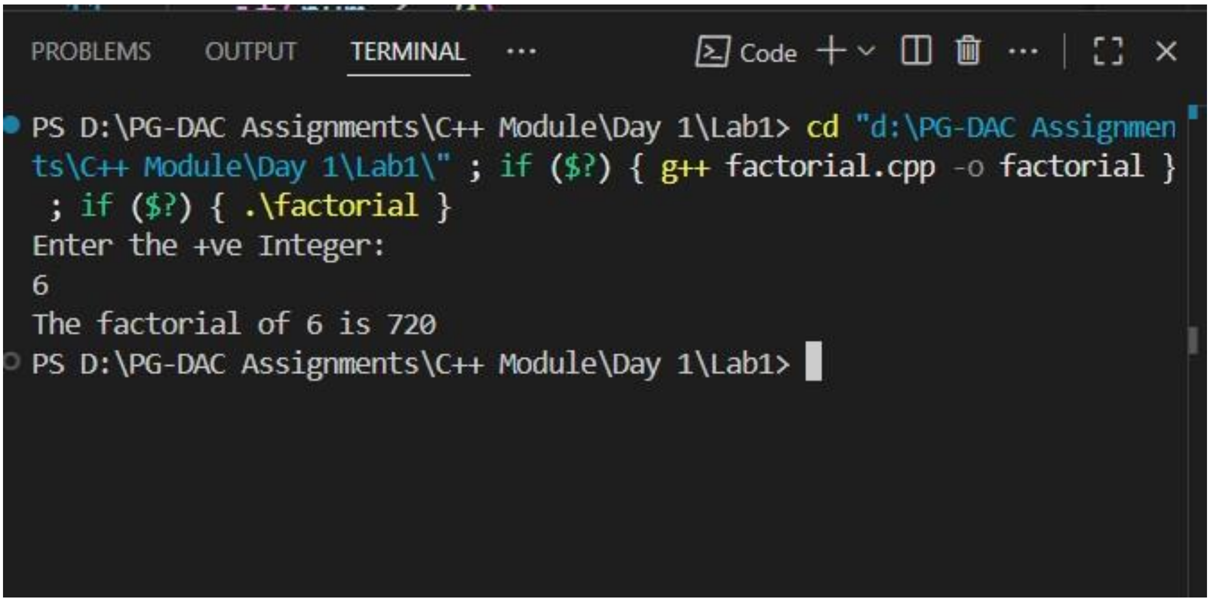


4. Write a program to calculate factorial of a number.

For e.g. factorial of 5 = 5! = 5 *4*3*2*1 = 120

```
#include<bits/stdc++.h>
using namespace std;

int main(int argc, char const *argv[])
{    int num,fact = 1;    cout<<"Enter
the +ve Integer: "<<endl; cin>>num;
if(num <= 0) return 1;

  for(int i = 1; i<=num;i++)     fact*=i;
cout<<"The factorial of "<<num<<" is
"<<fact<<endl;
```

```
    return 0;
}
```



5. Write a program to calculate factors of a given number. #include<bits/stdc++.h> using namespace std;

```
int main(int argc, char const *argv[])
{    int num,flag;    cout<<"Enter
the number: "<<endl;
cin>>num;    if(num< 0)
   {
    cout<<"Enter only +ve integer only."<<endl;
   return 0;
   }   cout<<"Factors of "<<num<<" are:
"<<endl;    for(int i =1;i<=num;i++)
   {          flag = 0;
if(num%i == 0)
            flag = 1;
if(flag == 1)
```

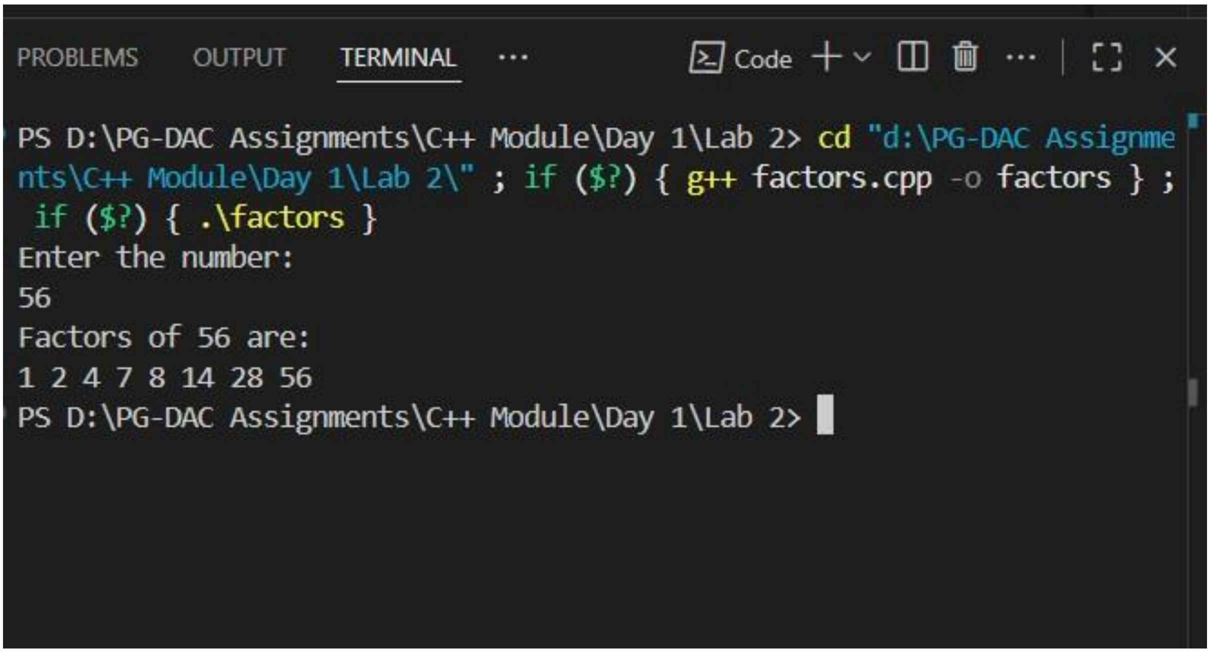
{

cout<<i<” ”;

} }

return 0;

}



6. Accept two numbers and calculate GCD of them. #include <iostream> using namespace std;

int main() {

int a, b;

cout << "Enter two numbers: ";

cin >> a >> b;

while (b != 0)

{ int temp

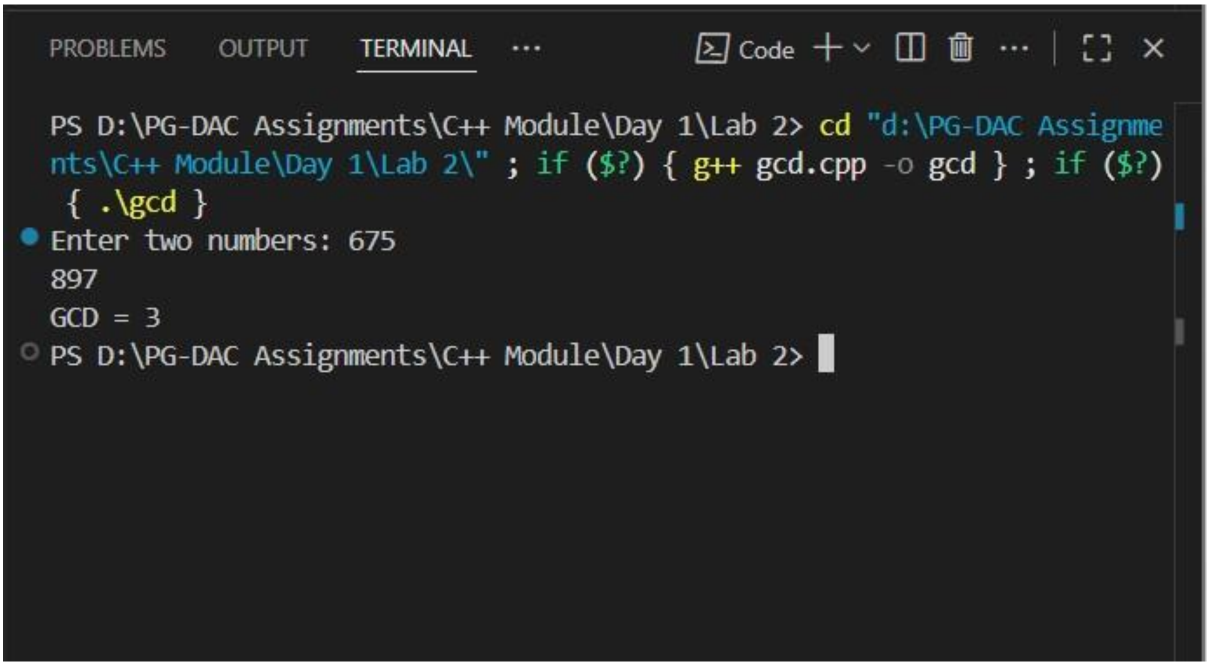
= b; b = a

% b; a =

temp;

}

```
    cout << "GCD = " << a << endl;
    return 0;
}
```



7. Write a menu driven program to do following operations :

a) Compute area of circle

b) Compute area of rectangle

c) Compute area of triangle

d) Exit

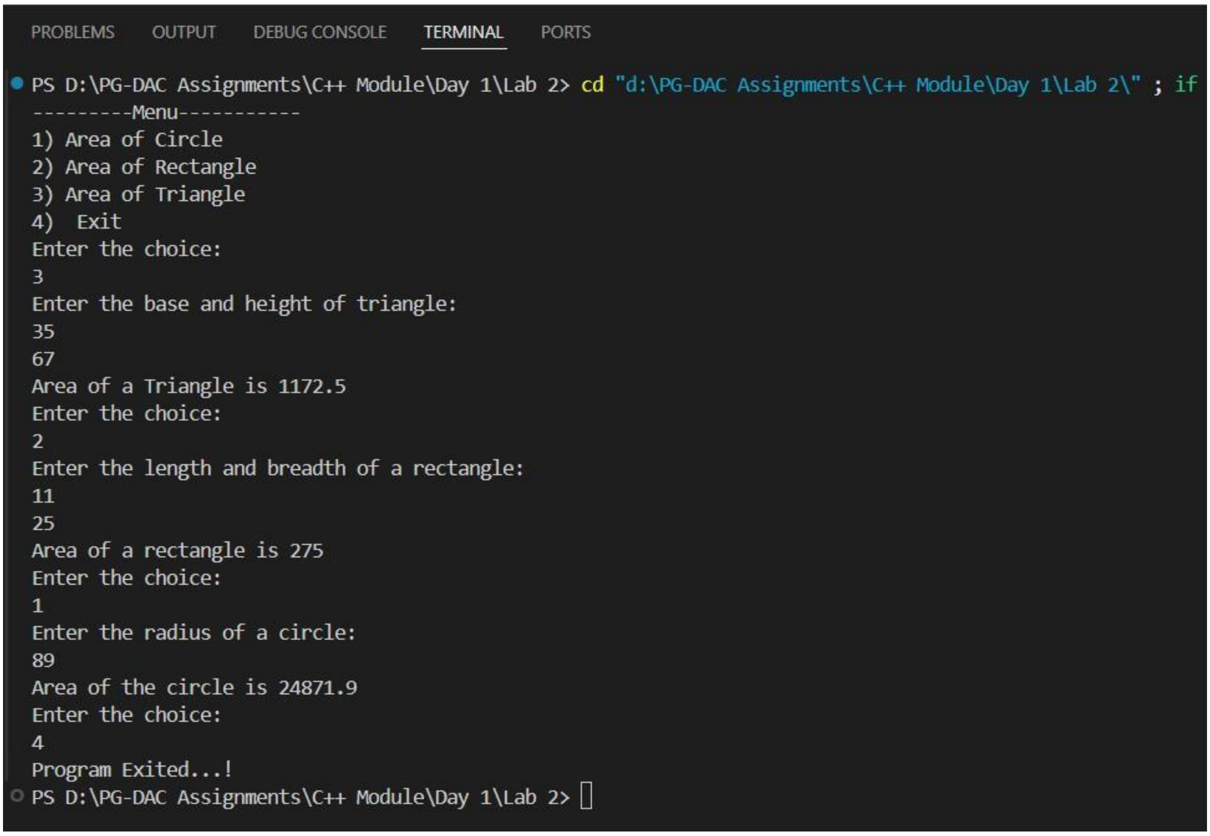
Display menu, ask choice to the user, depending on choice accept the parameters and perform the operation. Continue this process until user selects exit option.

```
#include
<bits/stdc++.h>
#define PI 3.14 using
namespace std;

int main(int argc, char const *argv[])
```

```
{     int choice;     float radius, length,
breadth, base, height;     cout << "--------
-Menu-----------" << endl;     cout << "1)
Area of Circle " << endl;     cout << "2)
Area of Rectangle " << endl;     cout <<
"3) Area of Triangle " << endl;     cout
<< "4)  Exit " << endl;     do
   {
        cout << "Enter the choice: " <<
endl;        cin >> choice;        switch
(choice)
        {
case 1:
            cout << "Enter the radius of a circle: " <<
endl;            cin >> radius;            if (radius <=
0)
            {
                cout << "Enter valid Radius..!" << endl;
                return 0;
            }
else
            {
                cout << "Area of the circle is " << PI * radius * radius << endl;
            }
            break;

       case 2:
            cout << "Enter the length and breadth of a rectangle: "
<< endl;            cin >> length >> breadth;            if (length <= 0
|| breadth <= 0)
```

```
            {
                cout << "Enter Valid Length and Breadth" << endl;
            }
else
            {
                cout << "Area of a rectangle is " << length * breadth << endl;
            }
break;
         case
3:
            cout << "Enter the base and height of triangle: " <<
endl;            cin >> height >> base;            if (height <= 0
|| base <= 0)
            {
                cout << "Enter the valid height and base." << endl;
                return 0;
            }
else
            {
                cout << "Area of a Triangle is " << 0.5* base * height << endl;
            }
break;
case 4:
            cout << "Program Exited...!" << endl;
            break;

        default:
           cout << "Invalid Choice " << endl;
       }
   } while (choice != 4);
```

```
  return 0;
}
```

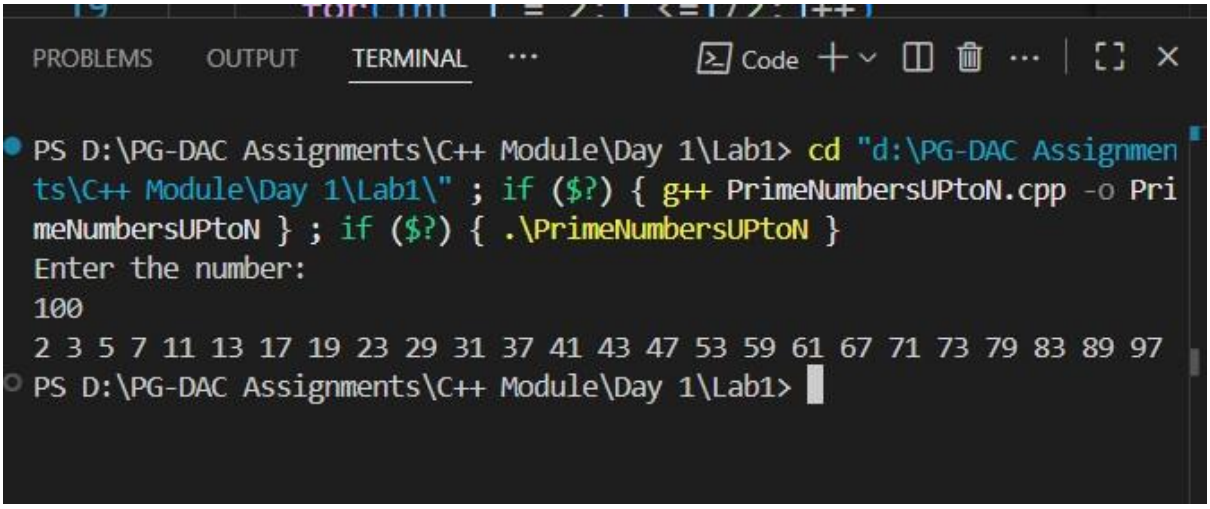


```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\PG-DAC Assignments\C++ Module\Day 1\Lab 2> cd "d:\PG-DAC Assignments\C++ Module\Day 1\Lab 2\" ; if
---------Menu-----------
1) Area of Circle
2) Area of Rectangle
3) Area of Triangle
4)  Exit
Enter the choice:
3
Enter the base and height of triangle:
35
67
Area of a Triangle is 1172.5
Enter the choice:
2
Enter the length and breadth of a rectangle:
11
25
Area of a rectangle is 275
Enter the choice:
1
Enter the radius of a circle:
89
Area of the circle is 24871.9
Enter the choice:
4
Program Exited...!
PS D:\PG-DAC Assignments\C++ Module\Day 1\Lab 2>
```

8. Write a program to print all prime numbers between 1 to n

```
#include<bits/stdc++.h>
using namespace std;

int main(int argc, char const *argv[])
{    int num;    cout<<"Enter the
number: "<<endl;    cin>>num;

  if(num <= 1)
  {
   cout<<"Not a Prime Number!"<<endl;
   return 0;
  }
```

```
for(int i = 2;i<=num;i++)
{       int flag=0;
for(int j = 2;j <=i/2;j++)
    {
if(i%j==0){
flag=1;
break;          }
    }
if(flag==0)
    {
      cout<<i<<endl;
    }
}


 return 0;
}
```



}

Lab 3

1:Write a program to create an array of integers and perform following operations on that array like

finding the sum, average, maximum and minimum number in that array. Accept the numbers of the array from user. #include <bits/stdc++.h> using namespace std; #define MAX 500

```
int main(int argc, char const *argv[])
{     int size, choice, index = 0, total, mini,
maxi;     int arr[MAX];     cout << "Enter the
size of the array: " << endl;     cin >> size;
if (size <= 0)
    {
        cout << "Please Enter +ve size only..!" << endl;
return 0;
    }
```

```
    cout << "--------------Menu------------" << endl;     cout <<
"1) Input Array Elements " << endl;     cout << "2) Display
Array " << endl;     cout << "3) Sum of Array Elements " <<
endl;     cout << "4) Average of Array Elements " << endl;
cout << "5) Maximum and Minimum element in array " <<
endl;    cout << "6) Exit" << endl;

   do
{ cout
<<
"Ente
r the
choic
e: "
<<
endl;
cin
>>
choic
e;
switc
h
(choic
e)
        { case 1:
if (index < size)
            {
                cout << "Enter the Elements in array: " <<
endl;                 cin >> arr[index++];
```

```
            }
else
            {
                cout << "Sorry! Array is Full." << endl;
            }
break;
         case 2:
if (index <= 0)
            {
                cout << "Your array is Empty.Please Enter the Elements.\n"
                     << endl;
            }
else
            {
                cout << "Your array is " <<
endl;                  for (int i = 0; i < index;
i++)
                {                      cout <<
 arr[i] << endl;
               }
}
break; case
3:

            total = 0;
        if (index <= 0)
            {
                cout << "Your array is Empty. So sum is zero"
                     << endl;
            }            else            {
for (int i = 0; i < index; i++)
```

```
{
total += arr[i];
}
cout << "Total sum of the Array Elements is " << total << endl;
}
break;
case
4:

if (index <= 0)
{
cout << "Your array is Empty. So Average sum of Array elements is zero" <<
endl;
}
else
{
cout << "Total Average sum of the Array Elements is " << (float)total / index
<< endl;
}
break;

case 5:

if (index <= 0)
{
cout << "Array is Empty..!" << endl;
}
else
{                maxi = arr[0];
mini = arr[0];               for (int i =
1; i < index; i++)
```

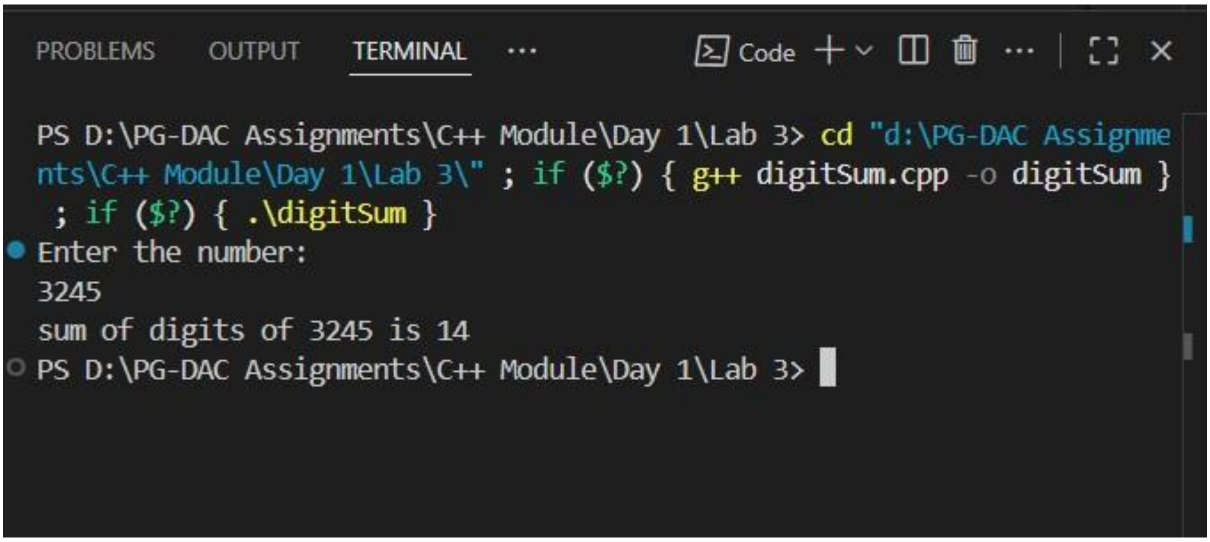
```
            {                   if
(arr[i] > maxi)
maxi = arr[i];
if (arr[i] < mini)
mini = arr[i];
            }
            cout << "The Largest Element in array is " << maxi
                << " and Smallest Element in array is " << mini << endl;
         }
break;
      case
6:
         cout << "Program Exited..!" << endl;
         break;

default:
         cout << "Invalid Choice..!\n please Enter the Valid Choice..." << endl;
      }
   } while (choice < 6);
return 0;
}
```

```
PROBLEMS    OUTPUT    TERMINAL    ...                Code
Enter the size of the array:
5
-------------Menu-----------
1) Input Array Elements
2) Display Array
3) Sum of Array Elements
4) Average of Array Elements
5) Maximum and Minimum element in array
6) Exit
Enter the choice:
1
Enter the Elements in array:
12
Enter the choice:
1
Enter the Elements in array:
24
Enter the choice:
1
Enter the Elements in array:
48
Enter the choice:
1
Enter the Elements in array:
96
Enter the choice:
2
Your array is
12
24
48
96
Enter the choice:
3
Total sum of the Array Elements is 180
Enter the choice:
4
Total Average sum of the Array Elements is 45
Enter the choice:
5
The Largest Element in array is 96 and Smallest Element in array is 12
```

2: Write a program to Accept a number and display its sum of digits.:ex 568 5+6+8

```
#include<bits/stdc++.h>

using namespace std;
```

```
int main(int argc, char const *argv[])
{    int num,org, temp = 0;
cout<<"Enter the number:
"<<endl;    cin>>num;    org =
num;    while(num != 0)
  {
    temp += num%10;
num = num/10;
  }
  cout<<"sum of digits of "<<org<<" is "<<temp<<endl;
   return 0;
}
```

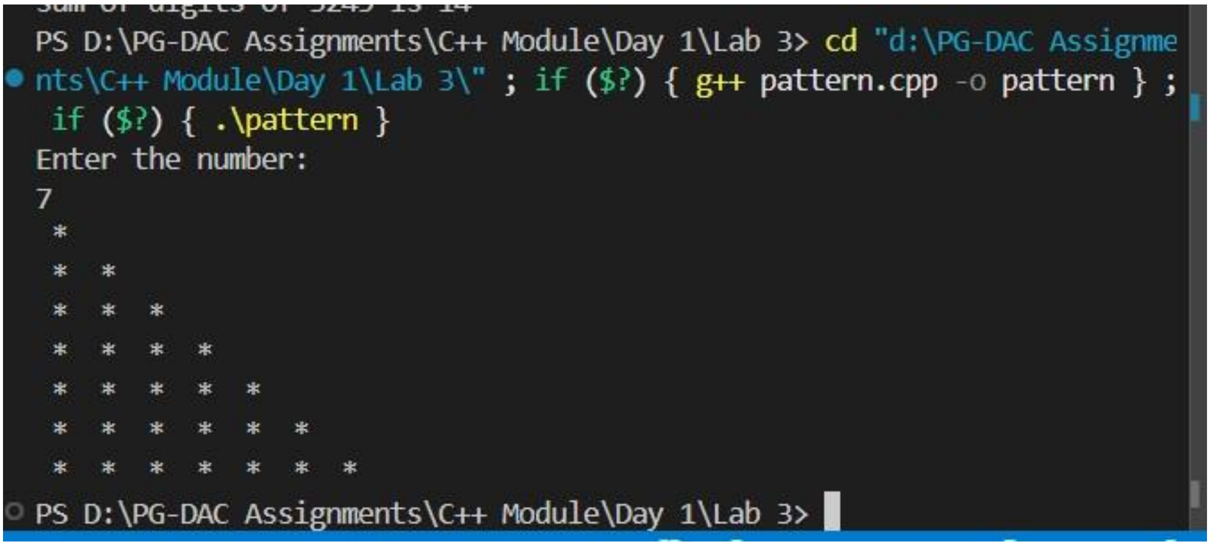


6:Write a program to print following pattern.

```
*
* *
* * *
* * * *
* * * * *
```

```
#include<bits/stdc++.h>
```

```
using namespace std;

int main(int argc, char const *argv[])
{    int num;    cout<<"Enter the
number: "<<endl;    cin>>num;
if(num<=0)
   {
     cout<<"Enter the valid number..!"<<endl;
     return 0;
   }    for(int i = 0; i<
num;i++)
   {       for(int j
=0;j<=i;j++)
      {
cout<<" * ";
      }
      cout<<endl;
   }
return 0;
}
```



```
PS D:\PG-DAC Assignments\C++ Module\Day 1\Lab 3> cd "d:\PG-DAC Assignme
nts\C++ Module\Day 1\Lab 3\" ; if ($?) { g++ pattern.cpp -o pattern } ;
 if ($?) { .\pattern }
Enter the number:
7
 *
 *  *
 *  *  *
 *  *  *  *
 *  *  *  *  *
 *  *  *  *  *  *
 *  *  *  *  *  *  *
PS D:\PG-DAC Assignments\C++ Module\Day 1\Lab 3>
```

7:Write a program to create student class with data members rollno, marks1,mark2,mark3.

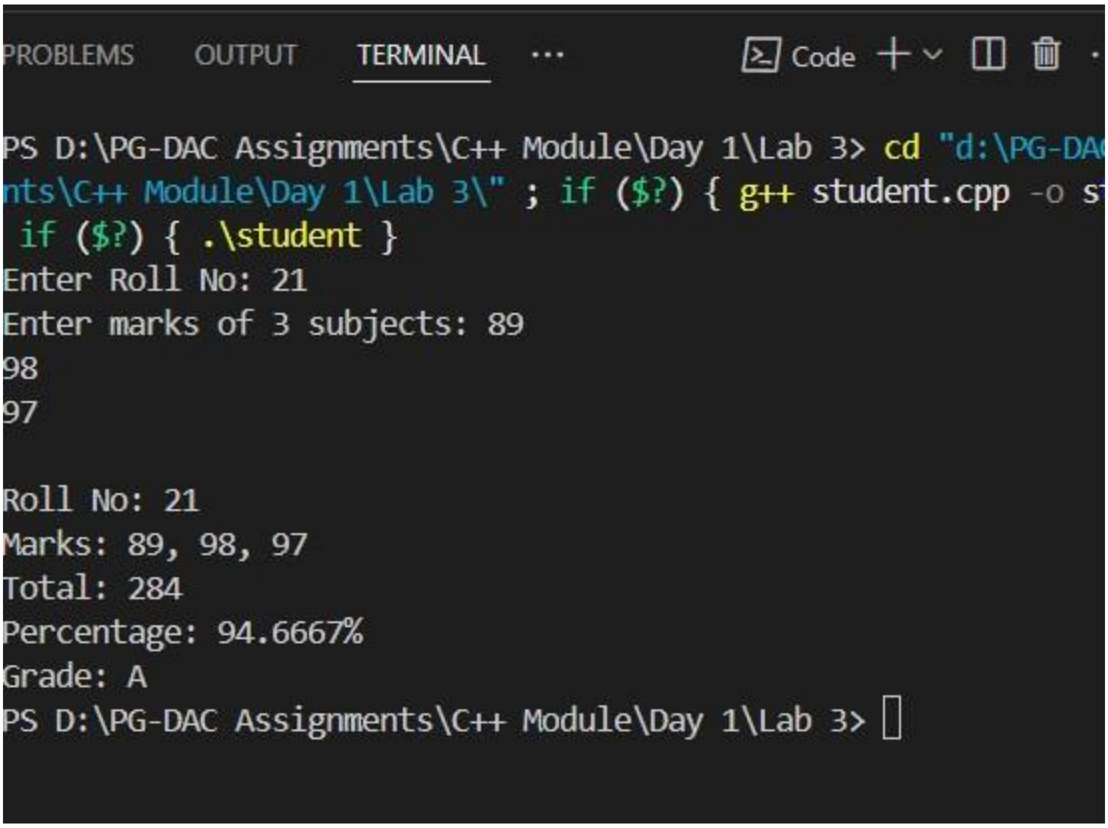
Accept data (acceptInfo()) and display using display member function.

Also display total,percentage and grade.

```
#include <iostream>
using namespace std;

class Student {
private:
   int rollno;     float marks1,
marks2, marks3;

public:
   void setRollNo(int r) { rollno = r; }
void setMarks1(float m) { marks1 = m;
}     void setMarks2(float m) { marks2
= m; }     void setMarks3(float m) {
marks3 = m; }

   int getRollNo() { return rollno; }
float getMarks1() { return marks1; }
float getMarks2() { return marks2; }
float getMarks3() { return marks3; }

   void acceptInfo() {
      int r;
      float m1, m2, m3;        cout <<
"Enter Roll No: ";        cin >> r;
setRollNo(r);        cout << "Enter
```

```
marks of 3 subjects: ";         cin >> m1
>> m2 >> m3;        setMarks1(m1);

        setMarks2(m2);
setMarks3(m3);
    }

    void display() {        float total =
marks1 + marks2 + marks3;        float
percentage = total / 3;        char grade;

        if (percentage >= 75)
grade = 'A';        else if
(percentage >= 60)
grade = 'B';        else if
(percentage >= 50)
grade = 'C';        else if
(percentage >= 35)
grade = 'D';        else
grade = 'F';

        cout << "\nRoll No: " << getRollNo() << endl;        cout << "Marks: " <<
getMarks1() << ", " << getMarks2() << ", " << getMarks3() << endl;        cout <<
"Total: " << total << endl;        cout << "Percentage: " << percentage << "%" << endl;
cout << "Grade: " << grade << endl;
    }
};

int main() {
Student s;
    s.acceptInfo();
```

```
    s.display();
    return 0;
}
```



```
PROBLEMS    OUTPUT    TERMINAL    ...    Code

PS D:\PG-DAC Assignments\C++ Module\Day 1\Lab 3> cd "d:\PG-DA
nts\C++ Module\Day 1\Lab 3\" ; if ($?) { g++ student.cpp -o s
 if ($?) { .\student }
Enter Roll No: 21
Enter marks of 3 subjects: 89
98
97

Roll No: 21
Marks: 89, 98, 97
Total: 284
Percentage: 94.6667%
Grade: A
PS D:\PG-DAC Assignments\C++ Module\Day 1\Lab 3>
```

Day 2 Lab

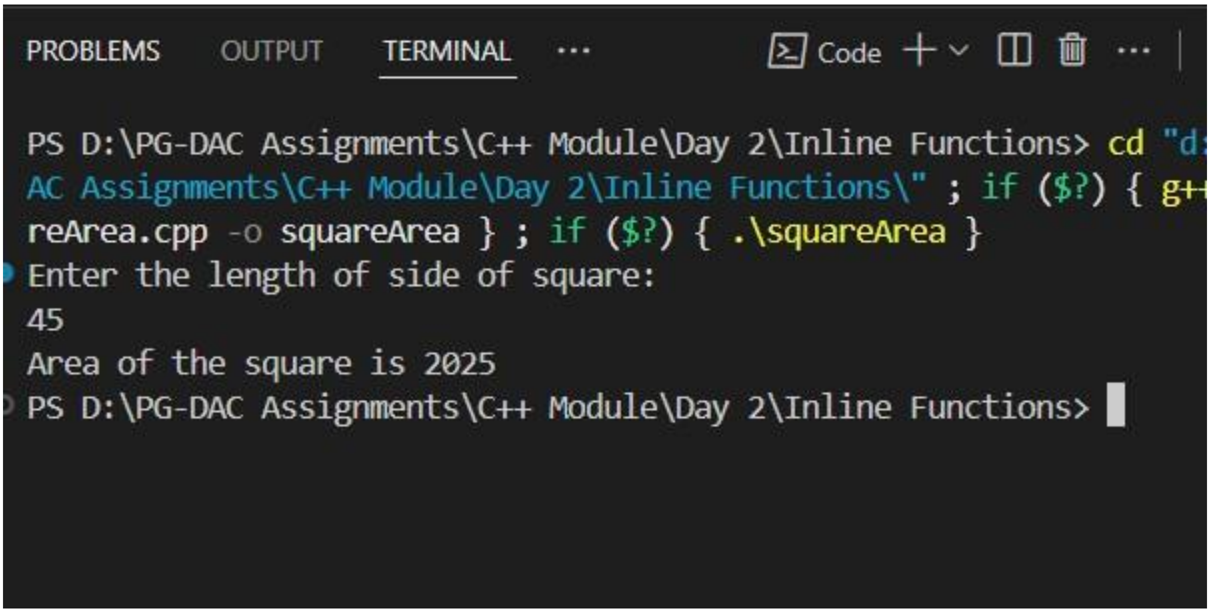
Assignment: Inline Functions in C++

Task

Write a program using inline functions to calculate:

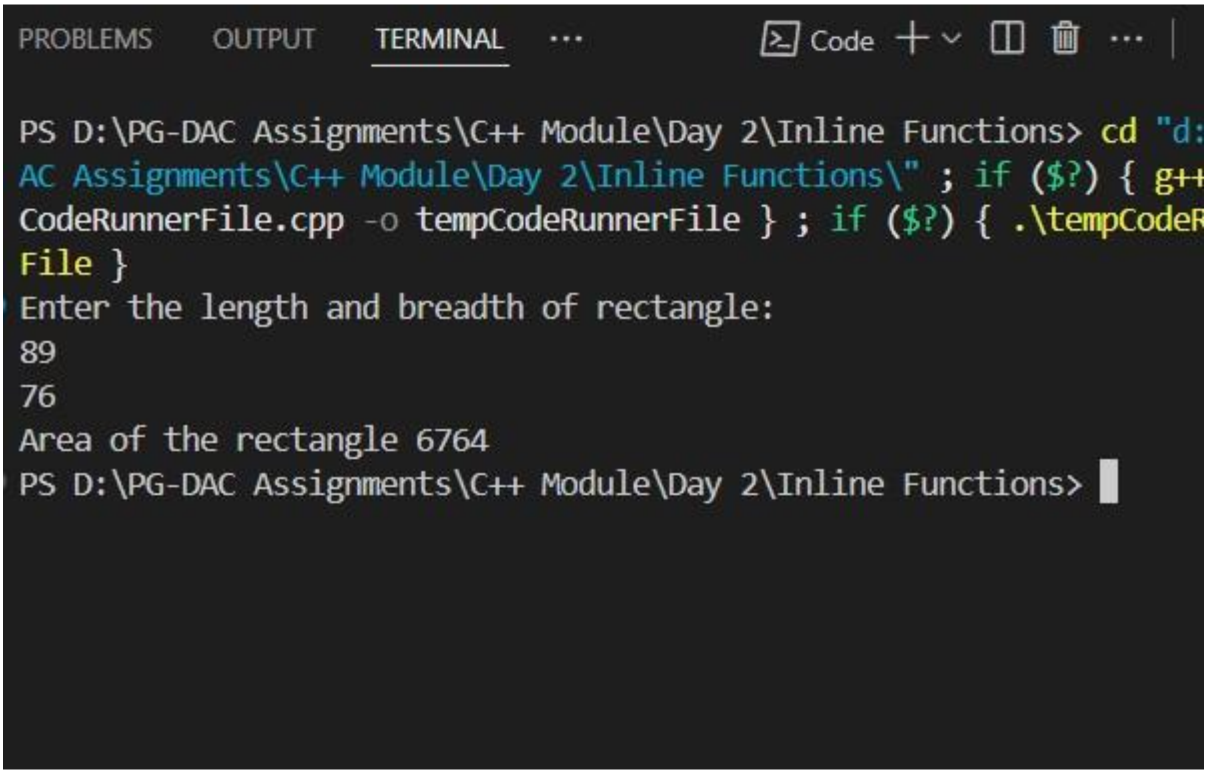
Area of a square (side × side)

```
#include<bits/stdc++.h> using
namespace std; inline int area(int
&side)
{    return
side*side;
} int main(int argc, char const
*argv[])
{    int side;    cout<<"Enter the length of side of
square: "<<endl;    cin>>side;    cout<<"Area of the
square is "<<area(side)<<endl;
    return 0;
}
```



Area of a rectangle (length × breadth)

```
#include<bits/stdc++.h> using

namespace std;


inline int area(int length,int breadth)

{     return

length*breadth;

} int main(int argc, char const

*argv[])

{    int length,breadth;    cout<<"Enter the length and breadth of

rectangle: "<<endl;    cin>>length>>breadth;    cout<<"Area of

the rectangle "<<area(length,breadth)<<endl;

    return 0;

}
```



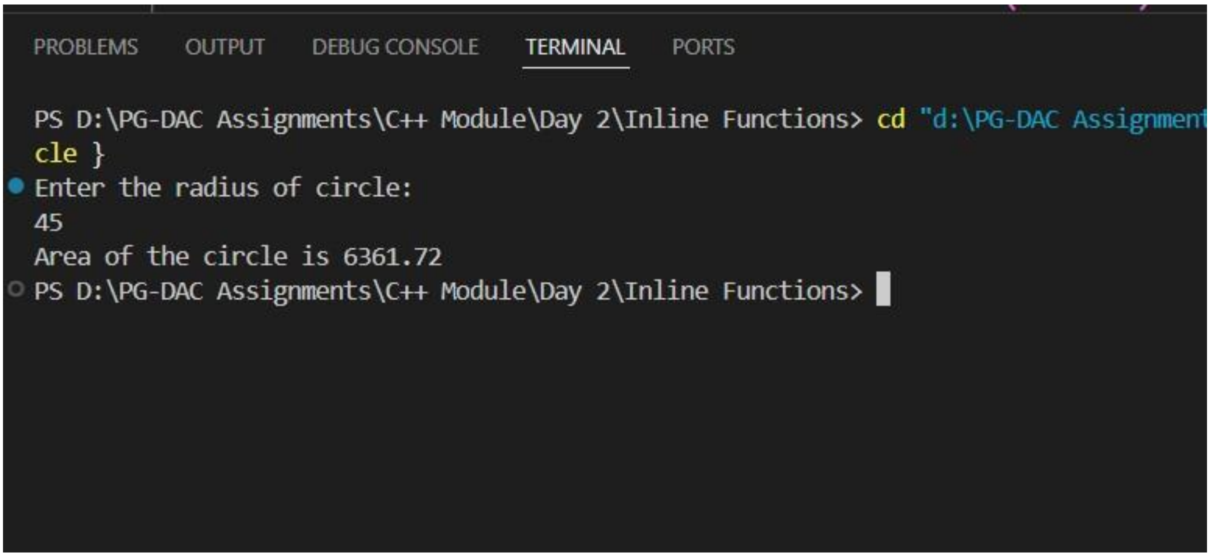
Area of a circle (π × r × r, use π = 3.14159)

```
#include <bits/stdc++.h> using
namespace std;
#define PI 3.14159

inline float area(float radius)
{     return PI* radius *
radius;
}

int main(int argc, char const *argv[])
{     float radius;     cout << "Enter the radius of
circle: " << endl;     cin >> radius;     cout <<
```

"Area of the circle is " << area(radius) << endl;

return 0;

}



Array_Lab

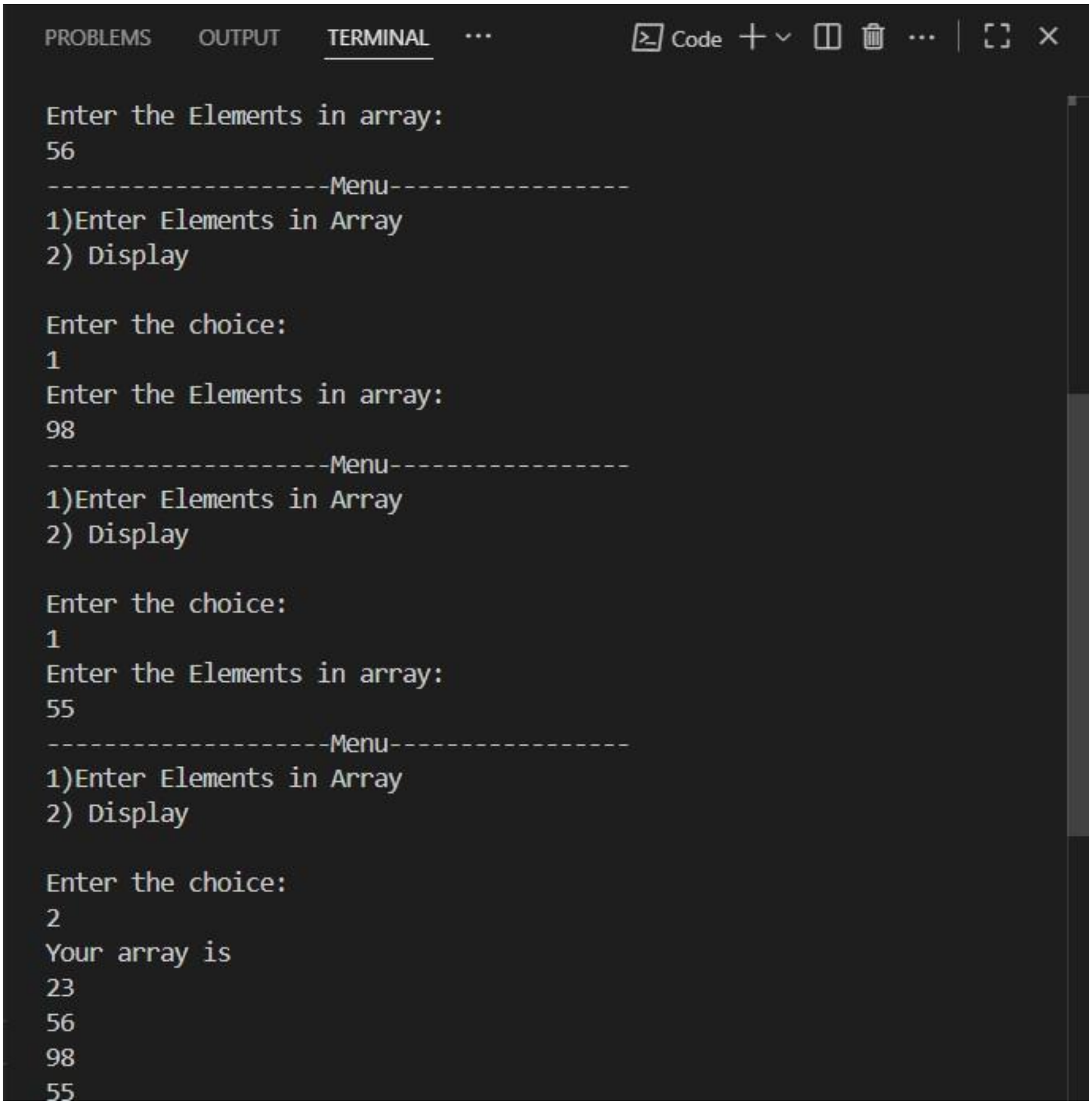
Basic Array Lab Assignments

1. Array Input & Output**

*** Take `n` elements from the user and display them.**

```
#include<bits/stdc++.h>
using namespace std; #define
MAX 500
int main(int argc, char const *argv[])
{     int size,choice,index = 0;     int arr[MAX];
cout<<"Enter the size of the array: "<<endl;
cin>>size;     if(size <= 0)
   {
       cout<<"Please Enter +ve size only..!"<<endl;
       return 0;
   }

   do
   {

          cout<<"--------------------Menu-----------------"<<endl;
cout<<"1)Enter Elements in Array "<<endl;             cout<<"2)
Display "<<endl;             cout<<endl;        cout<<"Enter the
choice: "<<endl;

       cin>>choice;
switch(choice)
       {
case 1:
              if(index< size){                  cout<<"Enter the
Elements in array: "<<endl;
cin>>arr[index++];
```

```
                }else
                {
                  cout<<"Sorry! Array is Full."<<endl;
                }
break;
                  case      2:
if(index <= 0)
                 {
                  cout<<"Your array is Empty.Please Enter the Elements.\n"<<endl;
                 }
else
                 {
                  cout<<"Your array is "<<endl;
for(int i = 0; i< index;i++)
                    {
cout<<arr[i]<<endl;
                    }
}
break;
default:
                cout<<"Invalid Choice..!\n please Enter the Valid Choice..."<<endl;
       }
   } while(choice <3);
return 0;
}
```



```
PROBLEMS    OUTPUT    TERMINAL    ...                Code  +  ...

Enter the Elements in array:
56
--------------------Menu----------------
1)Enter Elements in Array
2) Display

Enter the choice:
1
Enter the Elements in array:
98
--------------------Menu----------------
1)Enter Elements in Array
2) Display

Enter the choice:
1
Enter the Elements in array:
55
--------------------Menu----------------
1)Enter Elements in Array
2) Display

Enter the choice:
2
Your array is
23
56
98
55
```

2. Sum & Average of Array**

*** Input marks of `n` students, find total & average.**

3. Find Maximum and
Minimum**

*** Input array and print the largest & smallest element.**

#include <bits/stdc++.h> using

namespace std; #define MAX

500

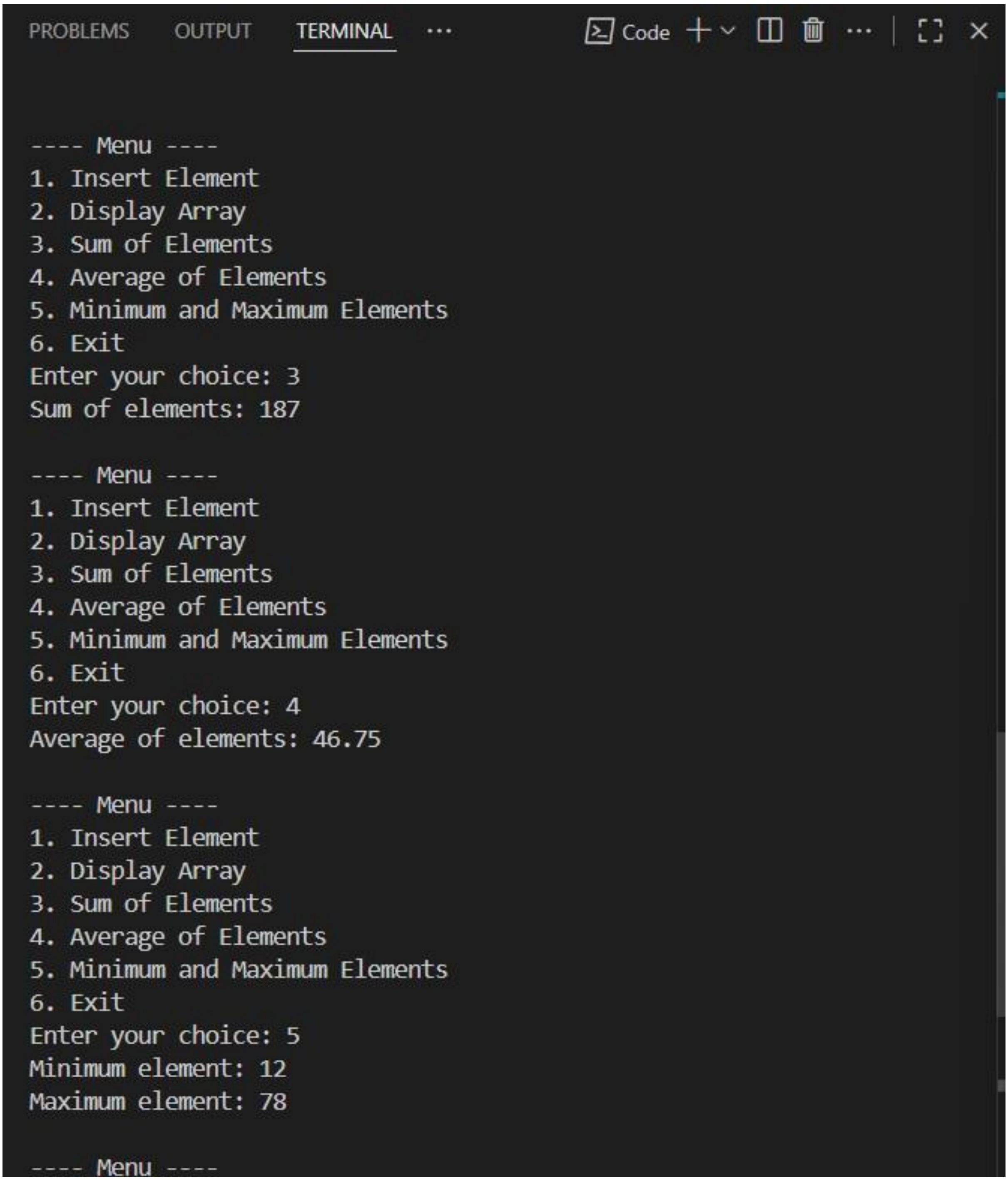
```
int main() {     int size,
choice, index = 0;     int
arr[MAX];

    cout << "Enter the size of the array: ";
cin >> size;

    if (size <= 0) {         cout << "Please enter a positive
size only!" << endl;         return 0;
    }

    do {         cout << "\n---- Menu ----\n";         cout <<
"1. Insert Element\n";         cout << "2. Display
Array\n";         cout << "3. Sum of Elements\n";
cout << "4. Average of Elements\n";         cout << "5.
Minimum and Maximum Elements\n";
        cout << "6. Exit\n";         cout
<< "Enter your choice: ";         cin
>> choice;

        switch (choice) {
case 1:
                if (index < size) {
cout << "Enter element: ";
cin >> arr[index++];

                } else {                     cout <<
"Array is full!" << endl;
                }
break;
             case 2:
```

```
            if (index <= 0) {                      cout << "Array is empty. Please
insert elements first." << endl;
            } else {                     cout <<
"Array elements: ";                      for (int i
= 0; i < index; i++)                        cout
<< arr[i] << " ";                    cout <<
endl;
            }
break;
        case 3:
            if (index <= 0) {                      cout << "Array is
empty. Sum is 0." << endl;
            } else {                     int total = 0;                      for
(int i = 0; i < index; i++)                          total += arr[i];
cout << "Sum of elements: " << total << endl;
            }
break;
        case 4:
            if (index <= 0) {                      cout << "Array is
empty. Average is 0." << endl;                 } else {
int total = 0;                      for (int i = 0; i < index; i++)
total += arr[i];                      cout << "Average of elements: "
<< (float)total / index << endl;
            }
break;
        case 5:
            if (index <= 0) {                      cout <<
"Array is empty." << endl;
            } else {                     int mini =
arr[0], maxi = arr[0];                      for (int i =
1; i < index; i++) {                        if (arr[i]
```

```
< mini)                    mini = arr[i];
if (arr[i] > maxi)                    maxi =
arr[i];
            }
            cout << "Minimum element: " << mini << endl;
cout << "Maximum element: " << maxi << endl;
          }
break;
        case 6:
          cout << "Exiting program..." << endl;
          break;
        default:
          cout << "Invalid choice! Please enter a valid choice." << endl;
      }

  } while (choice != 6);

  return 0;
}
```

PROBLEMS OUTPUT TERMINAL ··· Code

```
PS D:\PG-DAC Assignments\C++ Module\Day 2\Array Lab> cd "d:\PG-DAC Assi
gnments\C++ Module\Day 2\Array Lab\" ; if ($?) { g++ totalAvg.cpp -o to
talAvg } ; if ($?) { .\totalAvg }
Enter the size of the array: 4

---- Menu ----
1. Insert Element
2. Display Array
3. Sum of Elements
4. Average of Elements
5. Minimum and Maximum Elements
6. Exit
Enter your choice: 1
Enter element: 12

---- Menu ----
1. Insert Element
2. Display Array
3. Sum of Elements
4. Average of Elements
5. Minimum and Maximum Elements
6. Exit
Enter your choice: 1
Enter element: 24

---- Menu ----
1. Insert Element
2. Display Array
3. Sum of Elements
4. Average of Elements
5. Minimum and Maximum Elements
6. Exit
Enter your choice: 45
```



4. Search an Element (Linear Search)**

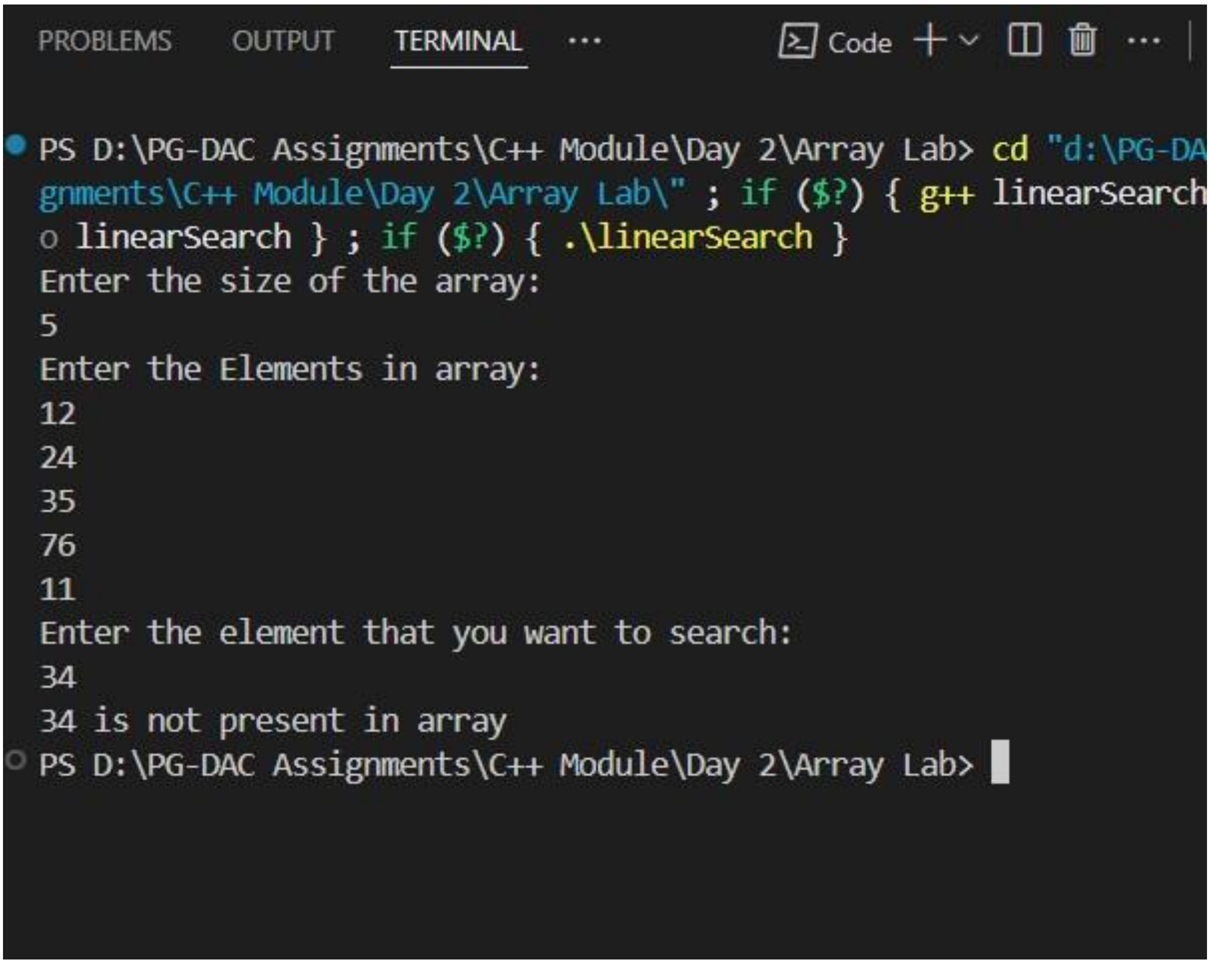
*** Input an array and search if a number exists.**

#include<bits/stdc++.h>

#define MAX 500 using

namespace std;

```
int main(int argc, char const *argv[])
{    int size,element,arr[MAX],flag;
cout<<"Enter the size of the array: "<<endl;
cin>>size;    if(size <= 0)
   {
    cout<<"Ente the +ve size only."<<endl;
return 0;
   }
   cout<<"Enter the Elements in array: "<<endl;
for(int i =0;i<size;i++)
   {
cin>>arr[i];
   }
   cout<<"Enter the element that you want to search: "<<endl;
cin>>element;    for(int i =0;i<size;i++)
   {         flag = 0;
if(arr[i]==element)
       {
flag = 1;
break;
       }
   }

   if(flag == 1)    cout<<element<<" is present in
array."<<endl;
   else
   cout<<element<<" is not present in array"<<endl;
    return 0;
}
```

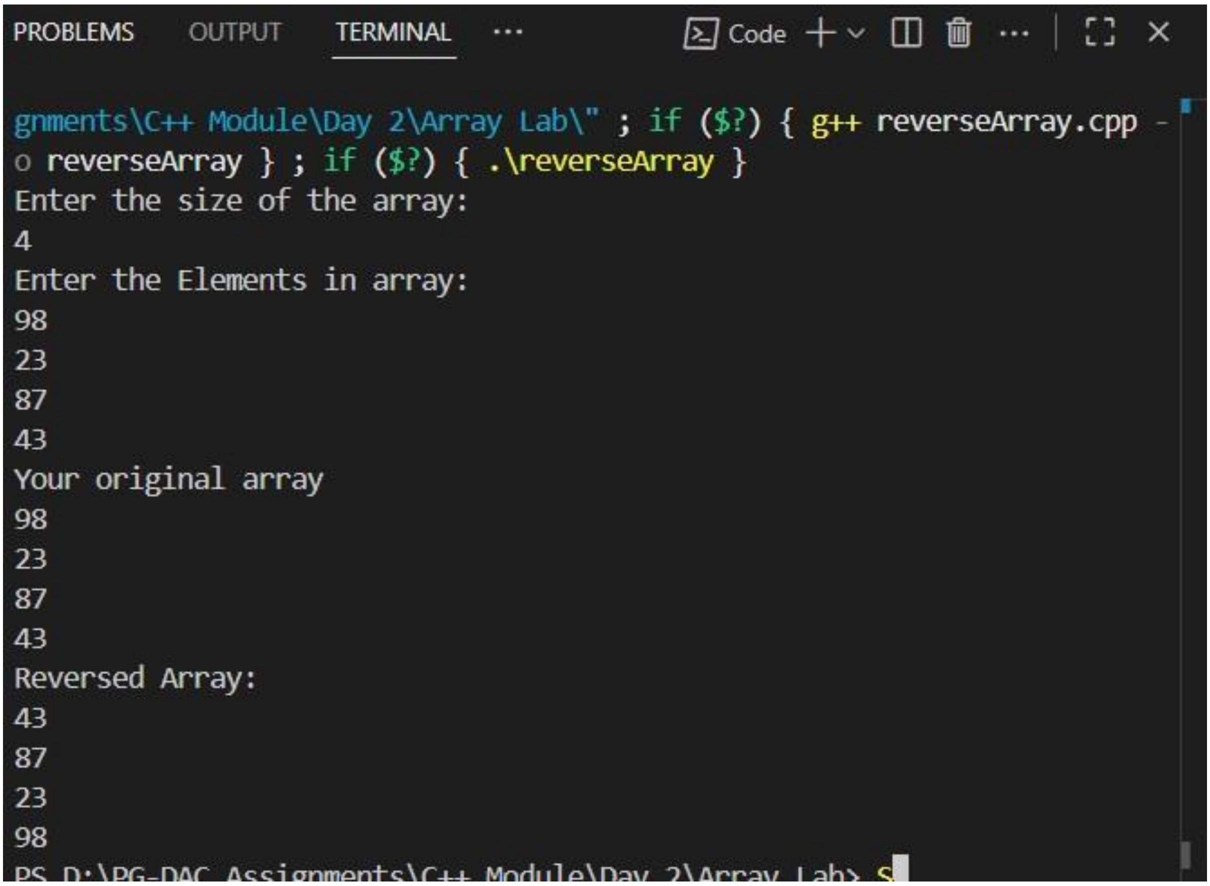


5. Reverse Array**

*** Reverse an array and display the result.**

```
#include<bits/stdc++.h>
#define MAX 500 using
namespace std;

int main(int argc, char const *argv[])
{    int
size,element,arr[MAX],flag;
cout<<"Enter the size of the
array: "<<endl; cin>>size; if(size
<= 0)
```

```
  {    cout<<"Ente the +ve size
only."<<endl;    return 0;
  }
  cout<<"Enter the Elements in array: "<<endl;
for(int i =0;i<size;i++)
  {
cin>>arr[i];
  }
  cout<<"Your original array "<<endl;
for(int i =0;i<size;i++)
  {
    cout<<arr[i]<<endl;
  }
  cout<<"Reversed Array: "<<endl;
for(int i = size-1;i>=0;i--)
  {
   cout<<arr[i]<<endl;
  }
return 0;
}
```



Intermediate Array Lab Assignments

6. Count Even and Odd
Numbers**

*** Count how many even and odd numbers are in the array.**

#include<bits/stdc++.h>

#define MAX 500 using

namespace std;

int main(int argc, char const *argv[])

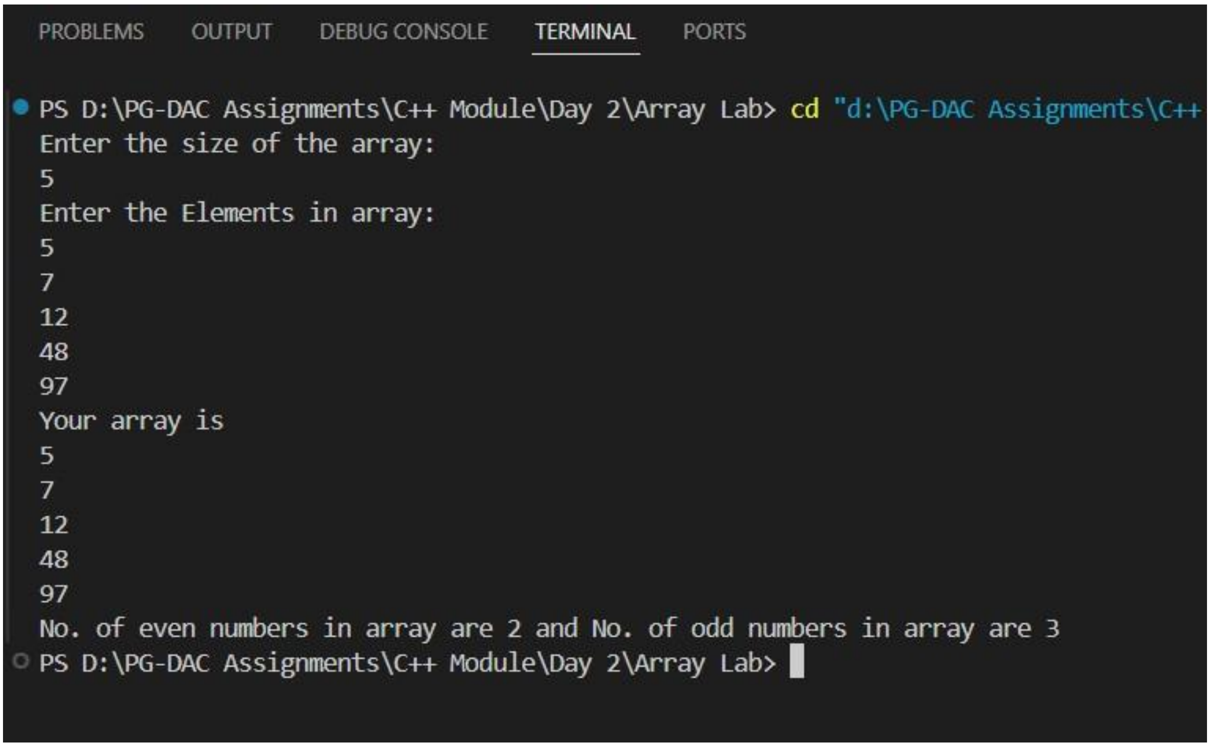
{ int size,element,arr[MAX],even = 0,odd=

0; cout<<"Enter the size of the array:

"<<endl; cin>>size; if(size <= 0)

{

```
  cout<<"Ente the +ve size only."<<endl;
return 0; } cout<<"Enter the Elements in array:
"<<endl;    for(int i =0;i<size;i++)
  {
cin>>arr[i];
  }
  cout<<"Your array is "<<endl;
for(int i =0 ;i < size;i++)
  {
     cout<<arr[i]<<endl;
if(arr[i]%2==0)
     {
even++;
     }
else{
odd++;
     }
  }

  cout<<"No. of even numbers in array are "<<even<<" and No. of odd numbers in array are
"<<odd<<endl;

   return 0;
}
```



```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\PG-DAC Assignments\C++ Module\Day 2\Array Lab> cd "d:\PG-DAC Assignments\C++
Enter the size of the array:
5
Enter the Elements in array:
5
7
12
48
97
Your array is
5
7
12
48
97
No. of even numbers in array are 2 and No. of odd numbers in array are 3
PS D:\PG-DAC Assignments\C++ Module\Day 2\Array Lab>
```

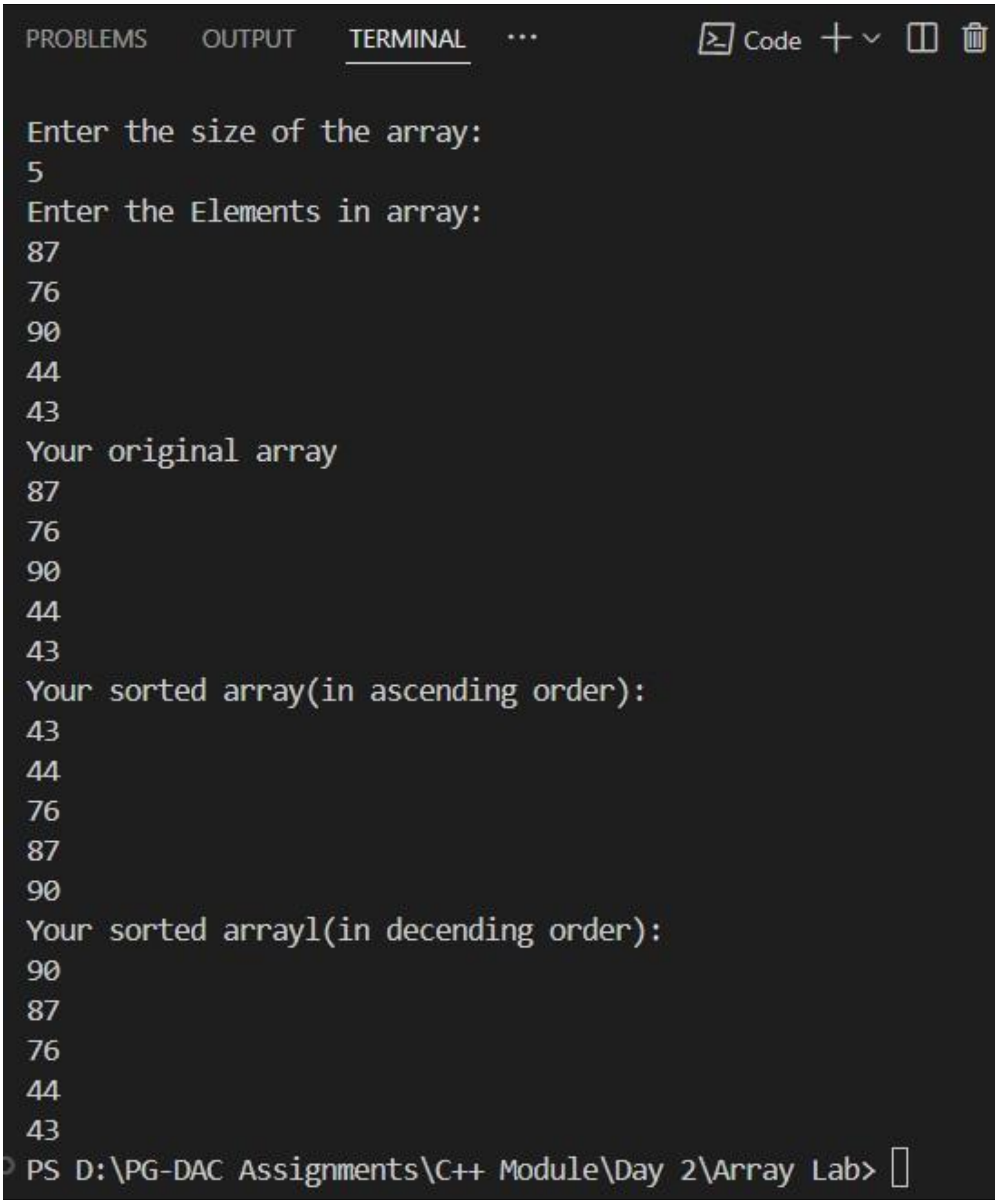
7. Sort Array (Ascending & Descending)**

*** Implement a simple bubble sort.**

```
#include<bits/stdc++.h>

#define MAX 500 using

namespace std;


int main(int argc, char const *argv[])

{    int size,element,arr[MAX],temp;

cout<<"Enter the size of the array: "<<endl;

cin>>size;    if(size <= 0)

   {

    cout<<"Ente the +ve size only."<<endl;

return 0;

   }

   cout<<"Enter the Elements in array: "<<endl;

   for(int i =0;i<size;i++)
```

```
  {
  cin>>arr[i];
  }    cout<<"Your original array
"<<endl;    for(int i =0;i<size;i++)
  {
    cout<<arr[i]<<endl;
  }

  for(int j = 0; j<size;j++)
  {      for(int i =0 ; i< size-1;
i++){      if(arr[i]>arr[i+1])
    {        temp =
arr[i];        arr[i]
=arr[i+1];
arr[i+1] = temp;
    }
   }
  }
  cout<<"Your sorted array(in ascending order): "<<endl;
for(int i =0;i<size;i++)
  {
    cout<<arr[i]<<endl;
  }

  for(int j = 0; j<size;j++)
  {      for(int i =0 ; i< size-1;
i++){     if(arr[i]<arr[i+1])
    {        temp =
arr[i];        arr[i]
```

```
=arr[i+1];
arr[i+1] = temp;
     }
    }
   }

   cout<<"Your sorted arrayl(in decending order): "<<endl;
for(int i =0;i<size;i++)
   {
     cout<<arr[i]<<endl;
   }
}
```



8.Second Largest Element**

*** Find the second largest number without sorting.**

```
#include <bits/stdc++.h>

#define MAX 500 using

namespace std;


int main() {
```

```
    int size;     int
arr[MAX];

    cout << "Enter the size of the array: " << endl;
cin >> size;

    if (size <= 1) {         cout << "Need at least 2 elements to find second
largest." << endl;         return 0;
    }

    cout << "Enter the elements in array: " << endl;
for (int i = 0; i < size; i++) {         cin >> arr[i];
    }

    cout << "Your array: " << endl;
for (int i = 0; i < size; i++) {
cout << arr[i] << " ";
    }
    cout << endl;

    // Find largest and second largest     int largest =
INT_MIN, secondLargest = INT_MIN;

   for (int i = 0; i < size; i++) {        if (arr[i] > largest) {
secondLargest = largest;   // update second largest
largest = arr[i];          // update largest
        }
        else if (arr[i] > secondLargest && arr[i] != largest) {
secondLargest = arr[i];
        }
    }
```

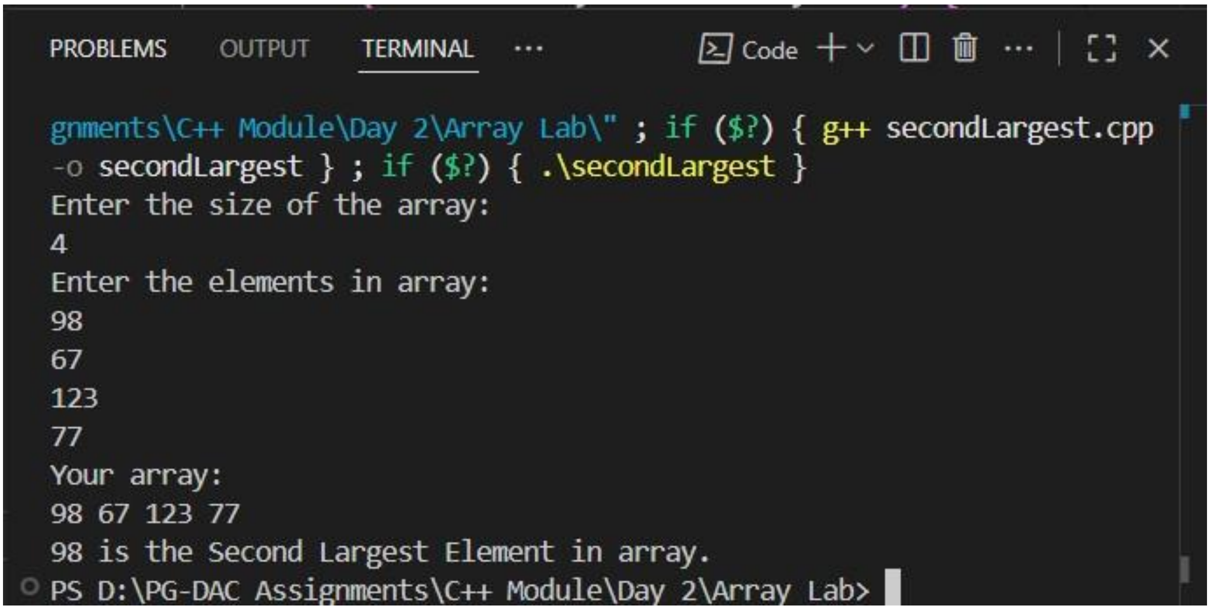
if (secondLargest == INT_MIN) { cout << "No second largest element found (all elements are equal)." << endl;

} else { cout << secondLargest << " is the Second Largest Element in array." << endl;

}

return 0;

}



9. Frequency of Each Element**

*** Count how many times each element occurs.**

// Frequency of Each Element**

// * Count how many times each element occurs.

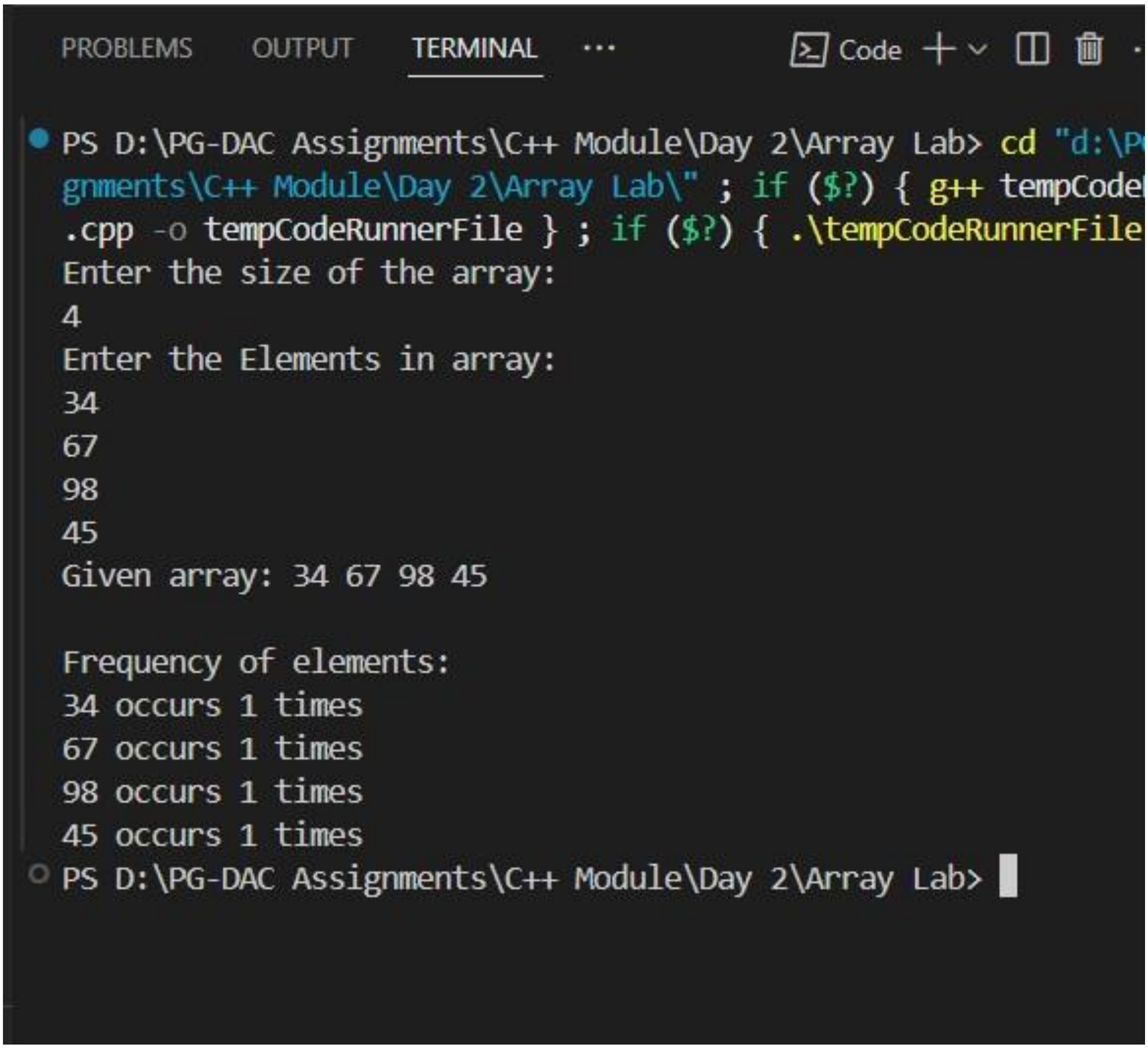
#include <bits/stdc++.h>

#define MAX 500 using

namespace std;

```
int main() {     int size, arr[MAX];     cout <<
"Enter the size of the array: " << endl;     cin >>
size;

   if (size <= 0) {        cout << "Enter +ve
size only" << endl;        return 0;
   }

   cout << "Enter the Elements in array: " << endl;
for (int i = 0; i < size; i++) {        cin >> arr[i];
   }

   cout << "Given array: ";     for (int i = 0; i <
size; i++) cout << arr[i] << " ";     cout << endl;

   cout << "\nFrequency of elements:\n";
 bool visited[MAX] = {false};

  for (int i = 0; i < size; i++) {
if (visited[i]) continue;

      int cnt = 1;        for (int j = i +
1; j < size; j++) {            if (arr[i] ==
arr[j]) {                cnt++;
visited[j] = true;
         }
      }
      cout << arr[i] << " occurs " << cnt << " times\n";
   }

   return 0;
```

}



10. Merge Two Arrays**

*** Take two arrays and merge them into a third array.**

```
#include<bits/stdc++.h>

#define MAX 500 using

namespace std;


int main(int argc, char const *argv[])

{    int size1, size2;    int

arr1[MAX],arr2[MAX];
```

```
  // first array    cout<<"Enter the size of the first
array: "<<endl;    cin>> size1;    if(size1 <= 0)
  {
    cout<<"Enter +ve size only"<<endl;
    return 0;
  }
  cout<<"Enter the Elements in first array: "<<endl;
for(int i = 0; i< size1; i++)
  {
cin>>arr1[i];
  }

 cout<<"Your first array: "<<endl;
for(int i = 0; i< size1; i++)
 {
cout<<arr1[i]<<endl;
 }

 // second array    cout<<"Enter the size of the second
array: "<<endl;    cin>> size2;    if(size2 <= 0)
  {
    cout<<"Enter +ve size only"<<endl;
    return 0;
  }
  cout<<"Enter the Elements in second array: "<<endl;
for(int i = 0; i< size2; i++)
  {
cin>>arr2[i];
  }
```

```
  cout<<"Your second array: "<<endl;
for(int i = 0; i< size2; i++)
  {
    cout<<arr2[i]<<endl;
  }

  int size3 = size1+ size2, arr3[size3];
//    while(arr1[size1-1] != 0 )
//    {
//       int temp = size1-1;
//       arr3[i++] = arr1[temp--];
//    }

//    while(arr2[size2-1] != 0 )
//    {
//       int temp = size2-1;
//       arr3[size1++] = arr2[temp--];
//    }
//    cout<<"Merged Final array: "<<endl;
//    for(int i = 0; i< size1+size2; i++)
//    {
//      cout<<arr3[i]<<endl;
//    }

 for(int i = 0; i<size1;i++)
 {     arr3[i] =arr1[i];  }
for(int i = 0; i < size2; i++)
 {    arr3[size1+i] =
arr2[i];
 }
```

```
  cout<<"Merged Final array: "<<endl;
for(int i = 0; i< size3; i++)
  {
   cout<<arr3[i]<<endl;
  }
return 0; }
```

PROBLEMS OUTPUT TERMINAL ··· Code

```
 -o mergeTwoArrays } ; if ($?) { .\mergeTwoArrays }
Enter the size of the first array:
3
Enter the Elements in first array:
1
2
3
Your first array:
1
2
3
Enter the size of the second array:
4
Enter the Elements in second array:
4
5
6
7
Your second array:
4
5
6
7
Merged Final array:
1
2
3
4
5
6
7
PS D:\PG-DAC Assignments\C++ Module\Day 2\Array Lab>
```

Day 3

1:Write a program to create student class with data members rollno, marks1,mark2,mark3.

Accept data (acceptInfo()) and display using display member function.

Also display total,percentage and grade.

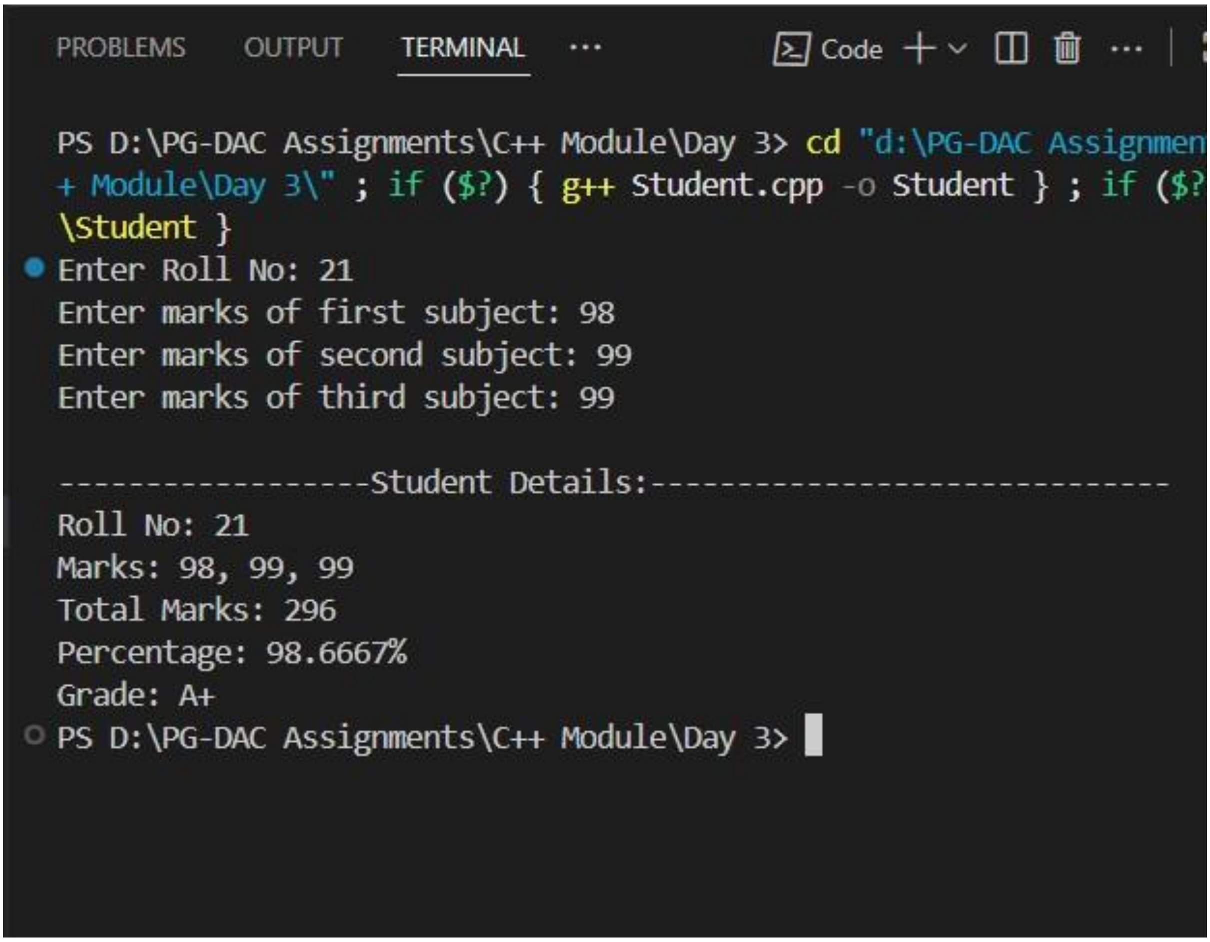
```
#include <bits/stdc++.h> using
namespace std;

class Student { private:
   int rollno;
int mark1;     int
mark2;     int
mark3;

public:
   int total;     double
percentage;     char
grade;

   void acceptInfo() {        cout
<< "Enter Roll No: ";        cin
>> rollno;

      cout << "Enter marks of first subject: ";
cin >> mark1;

      cout << "Enter marks of second subject: ";
cin >> mark2;

      cout << "Enter marks of third subject: ";
cin >> mark3;
```

```
        total = mark1 + mark2 + mark3;         percentage =
(double)total / 3.0;   // correct calculation
    }

    void Display() {         cout << "\n------------------Student Details:------------------
------------"<< endl;         cout << "Roll No: " << rollno << endl;         cout <<
"Marks: " << mark1 << ", " << mark2 << ", " << mark3 << endl;         cout <<
"Total Marks: " << total << endl;         cout << "Percentage: " << percentage <<
"%" << endl;

        // Assign grade based on percentage         if
(percentage >= 91 && percentage <= 100) {
cout << "Grade: A+" << endl;         } else if
(percentage >= 81) {             cout << "Grade: A" <<
endl;         } else if (percentage >= 71) {
cout << "Grade: B" << endl;         } else if
(percentage >= 61) {             cout << "Grade: C" <<
endl;         } else if (percentage >= 51) {
cout << "Grade: D" << endl;         } else if
(percentage >= 41) {             cout << "Grade: E" <<
endl;
        } else {             cout << "Grade: F
(Fail)" << endl;
        }
    }
};

int main() {
Student s;
    s.acceptInfo();   // take input from user
```

s.Display(); // display info

return 0;

}



1. Create a class Person with data members as name, age, city. Write getters and setters for all the data

members. Also add the display function. Create Default and Parameterized constructors. Create the object of this class in main method and invoke all the methods in that class. #include<bits/stdc++.h> using namespace std;

class Person

{

// Data Members or Properties

private:

```
            string name;
int age;              string
city;

    //Methods or Behaviour
public:
          // Default Constructor
          Person()
          {
             cout<<"------------Default Constructor Called---------------------------------"<<endl;
name = "Vishwajit";               age = 23;               city = "Pune";
          }

          //Parameterized Constructor
          Person(string name,int age, string city)
          {
              cout<<"--------------------------Parameterized Constructor Called-----------------------
--------"<<endl;               this-
>name = name;
this->age = age;
this->city = city;
          }
         //setters
void setName()
          {
              cout<<"Enter your name: "<<endl;
cin>>name;
           }

           void setCityName()
           {
```

```
        cout<<"Enter the city Name: "<<endl;
cin>>city;
      }

      void setAge()
      {
        cout<<"Enter your Age: "<<endl;
cin>>age;


      }
    // Getters
string getName()
      {
return name;
      }

       int getAge()
       {
return age;
       }
       string getCity()
       {
return city;
       }

   //Display Function
void Display()
      {
        cout<<"-------------------------Details of the Person------------------------------
"<<endl;          cout<<"Name of the Person:
"<<name<<endl;          cout<<"Your age is
```

```
"<<age<<endl;                cout<<"your city is
"<<city<<endl;
            }
};

int main(int argc, char const *argv[])
{
   Person p1;
  //Default Constructor call
  // p1.Display();



   //Parameterized constructor call
Person p2("Sumant",22,"Mumbai");
p2.Display();

    Person p3;
   //call using setters
p3.setName();    p3.setAge();
p3.setCityName();


  //getName
p3.getName();
p3.getAge();
p3.getCity();

   p1.Display();
return 0;
}
```

```
PROBLEMS    OUTPUT    TERMINAL    ...                    Code

PS D:\PG-DAC Assignments\C++ Module\Day 3> cd "d:\PG-DAC Assignments\C+
+ Module\Day 3\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunn
erFile } ; if ($?) { .\tempCodeRunnerFile }
-----------Default Constructor Called--------------------------------
-
--------------------------Parameterized Constructor Called-----------
-------------------
-------------------------Details of the Person-----------------------
-------
Name of the Person: Sumant
Your age is 22
your city is Mumbai
-----------Default Constructor Called--------------------------------
-
Enter your name:
Vishwajit
Enter your Age:
23
Enter the city Name:
Pune
-------------------------Details of the Person-----------------------
-------
Name of the Person: Vishwajit
Your age is 23
your city is Pune
PS D:\PG-DAC Assignments\C++ Module\Day 3>
```

2. Create a class Date with data members as dd, mm, yy. Write getters and setters for all the data members. Also add the display function. Create Default and Parameterized constructors. Create the object of this class in main method and invoke all the methods in that class.

#include<bits/stdc++.h> using

namespace std;

class Date

{

// Data Members or Properties private:

```
        int dd;
int mm;
int yy; public:
        // Default Constructor
    Date()
    {
        cout<<"--------------------Default Constructor Called-------------------------"<<endl;
dd = 26;        mm = 8;        yy = 2025;
    }

    //Parameterized Constructors
    Date(int dd, int mm, int yy)
    {
        cout<<"--------------Parameterized Constructor Called------------------------"<<endl;
this->dd = dd;        this->mm = mm;

        this->yy = yy;
   }

   //setters    void
setDay()
    {
        cout<<"Enter the day: "<<endl;
cin>>dd;
    }

    void setMonth()
    {
        cout<<"Enter the Month: "<<endl;
cin>>mm;
    }
```

```
    void setYear()
    {
        cout<<"Enter the Year: "<<endl;
cin>>yy;
    }

    //Getters

    int getDay()
    {       return
dd;
    }

    int getMonth()
    {       return
mm;
    }

    int getYear()
    {       return
yy;
    }

    //Display Function
void Display()
    {
        cout<<"Date you Entered is: "<<endl;
cout<<dd<<"/"<<mm<<"/"<<yy<<endl;
    }
```

```
// ~Date() {}
}; int main(int argc, char const
*argv[])
{
  Date d1;
  //Display Function called with default constructor
d1.Display();

  //Setters    cout<<"----------------Setters-----------------------------
--"<<endl;    Date d2;     d2.setDay();     d2.setMonth();
   d2.setYear();

    //Getters      cout<<"-----------------------------------Getters-----------------------
----"<<endl;      cout<<"Day "<<d2.getDay()<<endl;      cout<<"Month:
"<<d2.getMonth()<<endl;      cout<<"Year: "<<d2.getYear()<<endl;

  //Display Function
d2.Display();

   //Parameterized  Constructor
Date                d3(7,11,2002);
d3.Display();

   return 0;
}
```

```
PS D:\PG-DAC Assignments\C++ Module\Day 3> cd "d:\PG-DAC Assignments\C+
+ Module\Day 3\" ; if ($?) { g++ Date.cpp -o Date } ; if ($?) { .\Date
}
--------------------Default Constructor Called-------------------------
Date you Entered is:
26/8/2025
----------------Setters--------------------------------
--------------------Default Constructor Called-------------------------
Enter the day:
1
Enter the Month:
9
Enter the Year:
2025
------------------------------------Getters----------------------------
Day 1
Month: 9
Year: 2025
Date you Entered is:
1/9/2025
--------------Parameterized Constructor Called------------------------
Date you Entered is:
7/11/2002
PS D:\PG-DAC Assignments\C++ Module\Day 3>
```

3. Create a class Book with data members as bname,id,author,price. Write getters and setters for all the

data members. Also add the display function. Create Default and Parameterized constructors. Create the object of this class in main method and invoke all the methods in that class.

```
#include<bits/stdc++.h> using
namespace std;


class Book
{
   //Data Members or Properties private:
        string bookName;
string author;
```

```
        int bookId;
double price; public:
    //Default Constructor
        Book()
        {
            cout<<"------------------Default Constructor Called--------------"<<endl;
bookName = "Autobiography of Yogi";             author = "Paramhansa
Yoganand";             bookId= 1927;             price = 220;
        }
    //Parameterized Constructor
        Book(string bookName,string author,int bookId,double price)
        {
            cout<<"------------------Parameterized Constructor Called-------------"<<endl;
this->bookName = bookName;             this->author = author;             this->bookId =
bookId;             this->price = price;
        }

    //Setters
    void setBookName()
    {
        cout<<"Enter the Book Name: "<<endl;
getline(cin,bookName);
    }

    void setAuthor()
    {    cout<<"Enter the name of author:
"<<endl;         getline(cin,author);
    }

    void setBookId()
```

```
    {
      cout<<"Enter the book Id"<<endl;
cin>>bookId;
    }

    void setPrice()
    {
      cout<<"Enter the Price of the Book:"<<endl;
cin>>price;
    }

    //Getters      string
getBookName()
    {
      return bookName;
    }

    string getAuthor()
    {        return
author;
    }

    int getBookId()
    {
     return bookId;
    }

    double getPrice()
    {        return
price;
```

```
    }

    //Display Function
void display()
    {
      cout<<"-------------------------------Book Details-----------------------------"<<endl;
cout<<"Name of the Book: "<<bookName<<endl;        cout<<"Author of the Book:
"<<author<<endl;        cout<<"Book Id: "<<bookId<<endl;        cout<<"Price of the
Book: "<<price<<endl;
    }

}; int main(int argc, char const
*argv[])
{
   Book b1;
b1.display();

   //Direct Value set using parameterized constructor
   Book b2("Billaniare's Mindset","Vishwajit Nikam",2002,750);
b2.display();

   //Value set using setters cout<<"----------------Using
   Setters---------------"<<endl; Book b3;
   b3.setBookName();
   b3.setAuthor();
b3.setBookId();    b3.setPrice();

   //Getters     cout<<"-------------------Display Using Getters------------------
---"<<endl;     cout<<"Book Name: "<<b3.getBookName()<<endl;
cout<<"Author of The Book"<<b3.getAuthor()<<endl;     cout<<"Book Id:
"<<b3.getBookId()<<endl;     cout<<"Book Price: "<<b3.getPrice()<<endl;
```

```
    return 0;
}
```

```
PROBLEMS    OUTPUT    TERMINAL    ...                    Code  +  ...

PS D:\PG-DAC Assignments\C++ Module\Day 3> cd "d:\PG-DAC Assignments\C+
+ Module\Day 3\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o tempCodeRunn
erFile } ; if ($?) { .\tempCodeRunnerFile }
------------------Default Constructor Called--------------
-----------------------------Book Details-----------------------------
--
Name of the Book: Autobiography of Yogi
Author of the Book: Paramhansa Yoganand
Book Id: 1927
Price of the Book: 220
------------------Parameterized Constructor Called-------------
-----------------------------Book Details-----------------------------
--
Name of the Book: Billaniare's Mindset
Author of the Book: Vishwajit Nikam
Book Id: 2002
Price of the Book: 750
----------------Using Setters---------------
------------------Default Constructor Called--------------
Enter the Book Name:
Build don't Talk
Enter the name of author:
Raj shamani
Enter the book Id
34
Enter the Price of the Book:
1550
-------------------Display Using Getters---------------------
Book Name: Build don't Talk
Author of The BookRaj shamani
Book Id: 34
Book Price: 1550
PS D:\PG-DAC Assignments\C++ Module\Day 3> 
```

4. Create a class Point with data members as x,y. Create Default and Parameterized constructors. Write

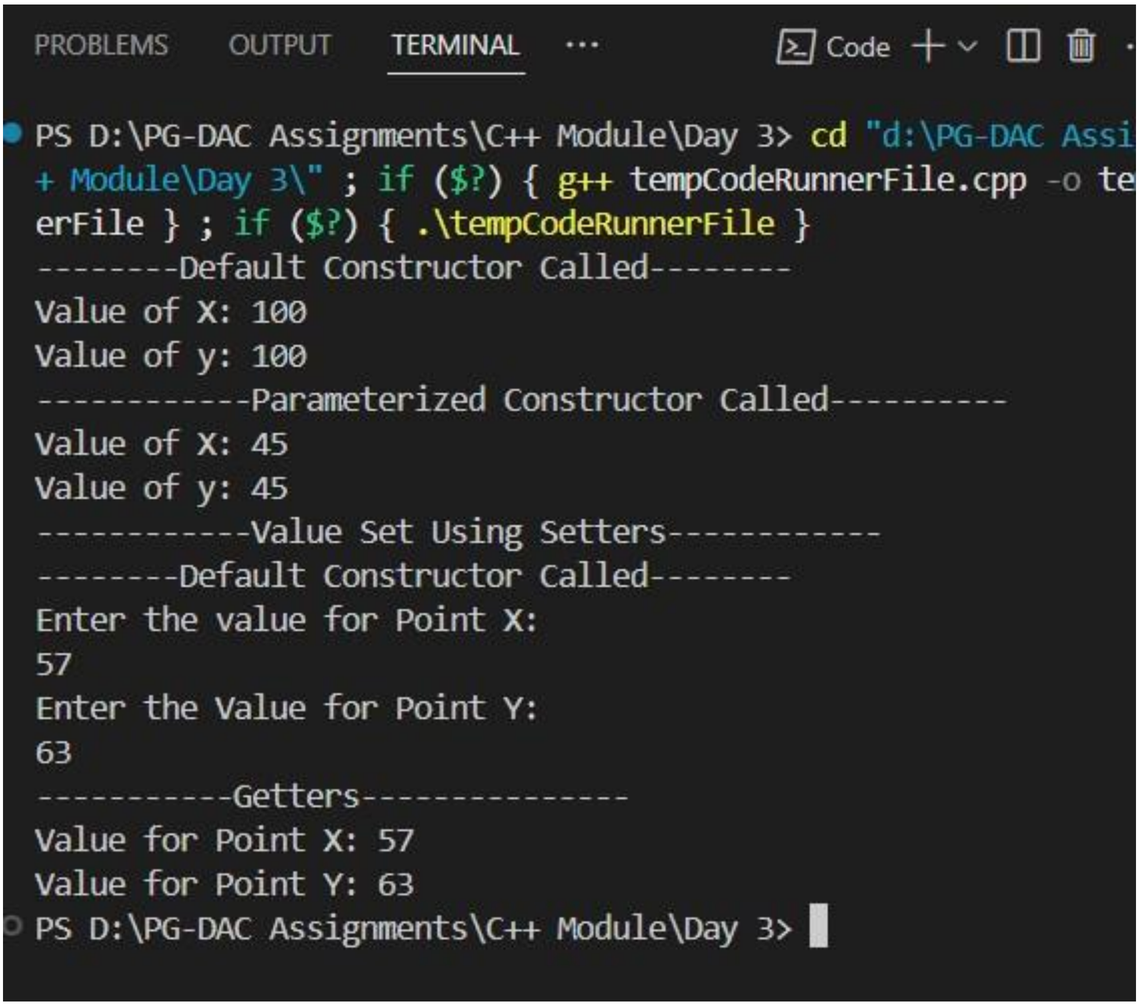
getters and setters for all the data members. Also add the display function. Create the object of this class in main method and invoke all the methods in that class.

```
#include<bits/stdc++.h> using

namespace std;
```

```
class Point
{ private:
        int x;
int y;

public:
      //Default Constructor
        Point()
        {
            cout<<"--------Default  Constructor  Called--------"<<endl;
x = 100;             y  = 100;
        }

        //Parameterized Constructor
        Point(int x, int y)
        {
            cout<<"------------Parameterized Constructor Called----------"<<endl;
this->x=x;             this->y=y;
        }

    //setters     void
setPointX()
    {
       cout<<"Enter the value for Point X: "<<endl;
        cin>>x;
    }

    void setPointY()
    {
```

```
        cout<<"Enter the Value for Point Y: "<<endl;
cin>>y;
    }

    //Getters     int
getPointX()
    {
return x;
    }

    int getPointY()
    {
return y;
    }

    //Display Function
void Display()
    {
        cout<<"Value of X: "<<x<<endl;
cout<<"Value of y: "<<y<<endl;
    } }; int main(int argc, char const
*argv[])
{
   Point p1;
    p1.Display();

    Point p2(45,45);
p2.Display();

  cout<<"------------Value Set Using Setters------------"<<endl;
Point p3;    p3.setPointX();    p3.setPointY();
```

```
cout<<"-----------Getters---------------"<<endl;
cout<<"Value for Point X: "<<p3.getPointX()<<endl;
cout<<"Value for Point Y: "<<p3.getPointY()<<endl;

return 0;
}
```



```
PROBLEMS   OUTPUT   TERMINAL   ···   Code
PS D:\PG-DAC Assignments\C++ Module\Day 3> cd "d:\PG-DAC Assi
+ Module\Day 3\" ; if ($?) { g++ tempCodeRunnerFile.cpp -o te
erFile } ; if ($?) { .\tempCodeRunnerFile }
--------Default Constructor Called--------
Value of X: 100
Value of y: 100
------------Parameterized Constructor Called----------
Value of X: 45
Value of y: 45
------------Value Set Using Setters------------
--------Default Constructor Called--------
Enter the value for Point X:
57
Enter the Value for Point Y:
63
-----------Getters---------------
Value for Point X: 57
Value for Point Y: 63
PS D:\PG-DAC Assignments\C++ Module\Day 3>
```

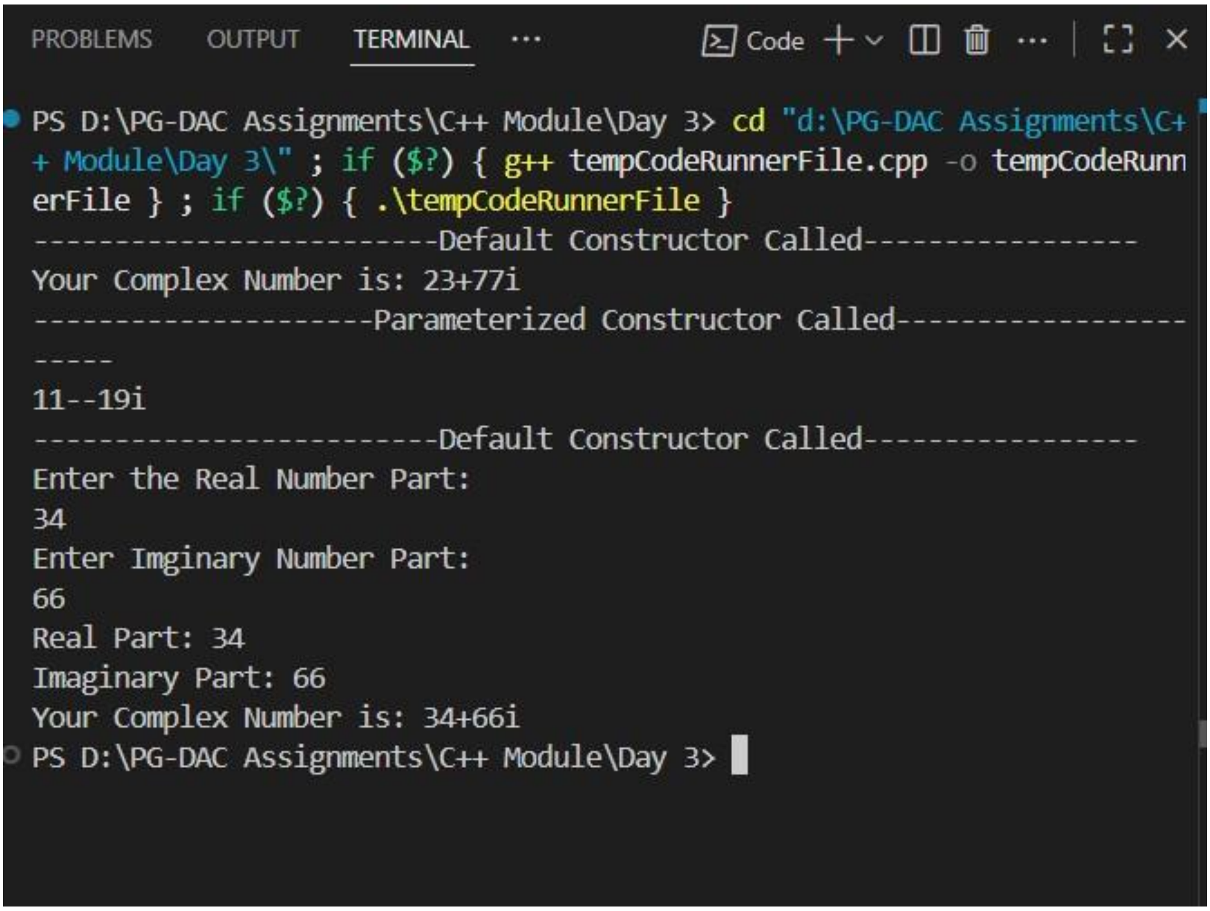
5. Create a class ComplexNumber with data members real, imaginary. Create Default and Parameterized constructors. Write getters and setters for all the data members. Also add the display function. Create the object of this class in main method and invoke all the methods in that class. #include<bits/stdc++.h> using namespace std;

```
class complexNumber
{
```

```
   //Data Members or Properties private:
       int real;
int img; public:
       //Default Constructor
        complexNumber()
        {
          cout<<"-------------------------Default Constructor Called-----------------"<<endl;
real = 23;             img = 77;
        }

        //Parameterized Constructor
complexNumber(int real,int img)
        {
          cout<<"---------------------Parameterized Constructor Called-----------------------
"<<endl;
          this->real = real;
this->img = img;
        }

        //Setters
void setRealPart()
        {
          cout<<"Enter the Real Number Part: "<<endl;
cin>>real;
        }

        void setImgPart()
        {
          cout<<"Enter Imginary Number Part: "<<endl;
cin>>img;
        }
```

```
        //Getters
int getRealPart()      {
return real;
        }

         int getImgPart()
         {
return img;
         }

        //Display Function

        void displayComplexNumber()
        {
if(img>0)
           {
           cout<<"Your Complex Number is: "<<real<<"+"<<img<<"i"<<endl;
           }
else
           {
              cout<<real<<"-"<<img<<"i"<<endl;
           }
        }


}; int main(int argc, char const
*argv[])
{    complexNumber c1;
c1.displayComplexNumber();
```

```
complexNumber c2(11,-19);
c2.displayComplexNumber();

  complexNumber c3;
   c3.setRealPart();
c3.setImgPart();

   cout<<"Real Part: "<<c3.getRealPart()<<endl;
cout<<"Imaginary Part: "<<c3.getImgPart()<<endl;
c3.displayComplexNumber();
    return 0;
}
```



Day 4

Lab 5

1:Create Date class with members day,month ,year.

Write no argument and parameterised constructor .Create two object s and initialize them using no argument and parameterised constructor respectively.Print date using display function.

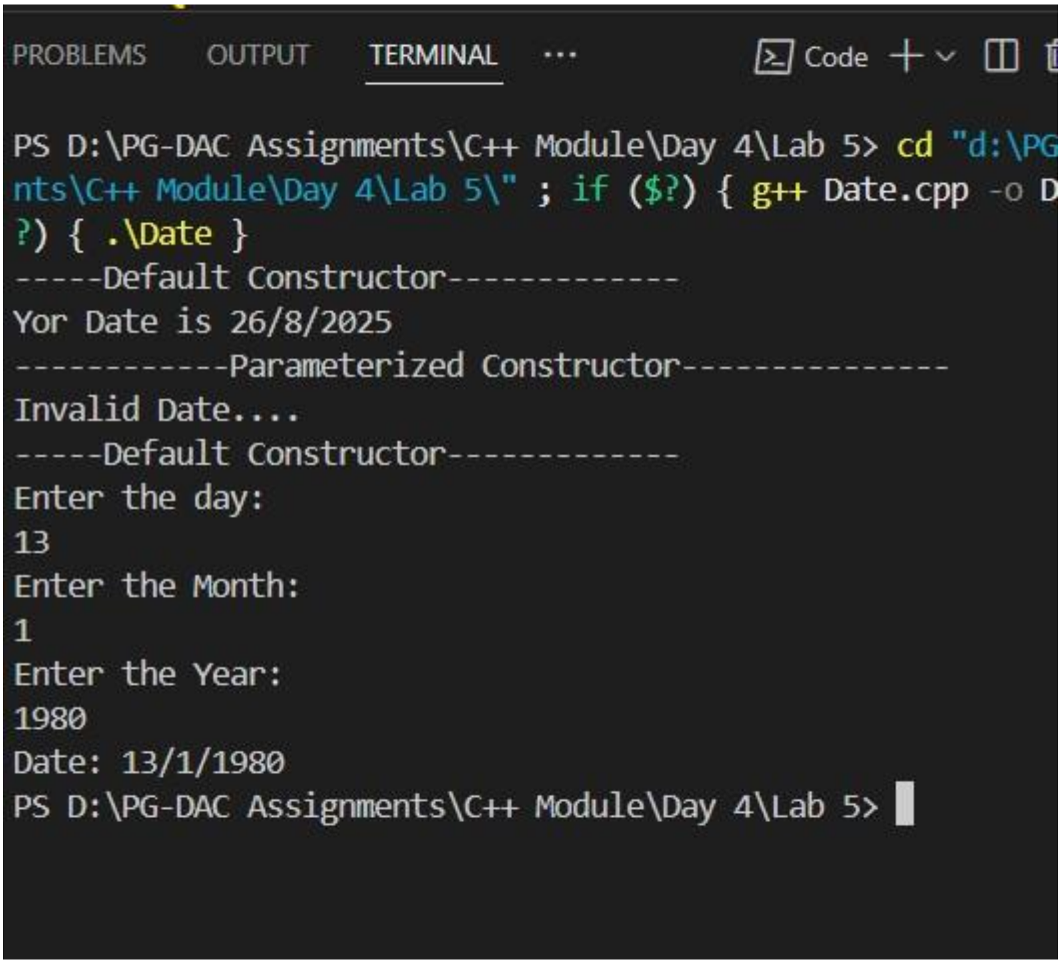
```
#include<bits/stdc++.h> using

namespace std;


class Date

{

  //Data Members or Properties private:
```

```
      int day;
int month;
int year;

      //Methods or Behaviour public:
   //Default Constructor
    Date()
    {
       cout<<"-----Default Constructor-------------"<<endl;
      day = 26;
month = 8;        year
= 2025;
    }

    //Parameterized Constructor
    Date(int day,int month,int year)
    {
       cout<<"------------Parameterized Constructor---------------
"<<endl;     this->day =day;         this->month = month;
       this->year = year;
    }

    //Setters
void setDay()
    {
      cout<<"Enter the day: "<<endl;
      cin>>day;        if(day<=0 || day
> 31)         cout<<"Enter Valid
Day"<<endl;
    }
```

```
    }

    void setMonth()
    {
        cout<<"Enter the Month: "<<endl;
cin>>month;        if(month<1 || month
> 12)        cout<<"Enter Valid
Month"<<endl;
    }

    void setYear()
    {
        cout<<"Enter the Year: "<<endl;
        cin>>year;     if(year <= 00
    and year > 2026)
        {
            cout<<"Enter the Valid Year..."<<endl;
        }
    }

    //Getters
int getDay()
    {
        return day;
    }

    int getMonth()
    {
        return month;
    }

    int getYear()
```

```
    {
        return year;
    }

    void displayDate()
    {
        if(month == 2 && day > 28 && year%4 != 0)
        {
            cout<<"Invalid Date...."<<endl;
        }
        else if(month == 2 && day > 29 && year%4 == 0)
        {
            cout<<"InValid Date...!"<<endl;
        }
        else if ((month == 1 || month == 3 || month == 5 || month == 7 || month == 8 ||
month == 10 || month == 12) && day<=31)
        {
            cout<<"Yor Date is "<<day<<"/"<<month<<"/"<<year<<endl;
        }
        else if((month == 4 || month == 6 || month == 9 || month == 11) && (day<=30))
        {
            cout<<"Yor Date is "<<day<<"/"<<month<<"/"<<year<<endl;
        }

    }

};
int main(int argc, char const *argv[])
{
```

```
  // Default Constructors
Date d1;    d1.displayDate();

   //Parameterized Constructors
Date d2(29,2,2015);
d2.displayDate();

  // Setters
   Date d3;
   d3.setDay(); d3.setMonth();
   d3.setYear();

   // Getters    cout<<"Date:
"<<d3.getDay()<<"/"<<d3.getMonth()<<"/"<<d3.getYear()<<endl;

    return 0;
}
```



```
PROBLEMS    OUTPUT    TERMINAL    ···                Code  +

PS D:\PG-DAC Assignments\C++ Module\Day 4\Lab 5> cd "d:\PG
nts\C++ Module\Day 4\Lab 5\" ; if ($?) { g++ Date.cpp -o D
?) { .\Date }
-----Default Constructor------------
Yor Date is 26/8/2025
------------Parameterized Constructor---------------
Invalid Date....
-----Default Constructor------------
Enter the day:
13
Enter the Month:
1
Enter the Year:
1980
Date: 13/1/1980
PS D:\PG-DAC Assignments\C++ Module\Day 4\Lab 5>
```

2:Create Employee class with members id(int),name(string),dob(Date). Use above created Date class.

Write default and parameterised constructor in Employee Class.Write accept() function to accept information and display() to display emp information.

```
#include<bits/stdc++.h>

using namespace std;


// Date Class class

Date

{

   //Data Members or Properties private:
```

```
    int day;
int month;
int year;

    //Methods or Behaviour public:
  //Default Constructor
   Date()
   {
     cout<<"-----Default Constructor-------------"<<endl;
    day = 26;
month = 8;       year
= 2025;
   }

   //Parameterized Constructor
   Date(int day,int month,int year)
   {
     cout<<"------------Parameterized Constructor---------------
"<<endl;        this->day =day;        this->month = month;
     this->year = year;
   }

   //Setters
void setDay()
   {
    cout<<"Enter the day: "<<endl;
    cin>>day;       if(day<=0 || day
> 31)        cout<<"Enter Valid
Day"<<endl;
   }
```

```
   void setMonth()
   {
      cout<<"Enter the Month: "<<endl;
cin>>month;        if(month<1 || month
> 12)        cout<<"Enter Valid
Month"<<endl;
   }

   void setYear()
   {
      cout<<"Enter the Year: "<<endl;
      cin>>year;        if(year <= 1900
and year > 2026)
      {
         cout<<"Enter the Valid Year..."<<endl;
      }
   }

   //Getters int
   getDay()
   {
     return day;
  }

   int getMonth()
   {
      return month;
   }

   int getYear()
```

```
    {
        return year;
    }

    void displayDate()
    {
        if(month == 2 && day > 28 && year%4 != 0)
        {
            cout<<"Invalid Date...."<<endl;
        }
        else if(month == 2 && day > 29 && year%4 == 0)
        {
            cout<<"InValid Date...!"<<endl;
        }
        else if ((month == 1 || month == 3 || month == 5 || month == 7 || month == 8 || month ==
10 || month == 12) && day<=31 && (year>=1900))
        {
            cout<<"Your Birth Date is "<<day<<"/"<<month<<"/"<<year<<endl;
        }
          else if((month == 4 || month == 6 || month == 9 || month == 11) && (day<=30) &&
year>=1900)
        {
            cout<<"Your Birth Date is "<<day<<"/"<<month<<"/"<<year<<endl;
        }
        else
        {
            cout<<"Enter Valid DOB Please..!"<<endl;
        }

    }
```

```
};

//Employee Class

class Employee
{
private:
        int empId;
        string empName;
         Date *dob;

public:
       int dd,mm,yy;
   //Default Constructor
    Employee()
    {
       cout<<"-----------------Default Constructor Called-----------------------"<<endl;
     empId = 1; empName =
   "Vishwajeet";    dob = new
   Date(7,11,2002);
   }

    //------------------Parameterized Constructor-------------
    Employee(int empId,string empName, Date *dob)
    {
       cout<<"---------------Parameterized Constructor Called-----------------"<<endl;         this-
>empId = empId;
       this->empName = empName;
       this->dob = dob;
       //Rethink Again
```

```
        // this->dob = new Date(dd,mm,yy);
    }

    //Accept Function
void acceptDetails()
    {
        cout<<"---------------Accepting Employee Details-----------------------
"<<endl;         cout<<"Enter the Employee Id: "<<endl;         cin>>empId;
        //cin.ignore() :-->This prevents skipping new line to
getline         cout<<"Enter the Name: "<<endl;
getline(cin>>ws ,empName);         cout<<"\nEnter the Date of
Birth(dd-mm-yyyy): "<<endl;         cin>>dd>>mm>>yy;
        //New Object made because Pointer don't points wrong object
dob = new Date(dd,mm,yy);

        //  dob->setDay();
        //  dob->setMonth();
        //  dob->setYear();

   }

    //Disply Fuction

    void displayDetails()
    {
        cout<<"-----------------------Employee Details----------------"<<endl;
        cout<<"Employee Id of the Employee: "<<empId<<endl;
cout<<"Name of the Employee: "<<empName<<endl;
        dob->displayDate();

    }
```

```
 };
int main(int argc, char const *argv[])
{
  //Default Values Call
Employee e1;
e1.displayDetails();

   //Using accept Function
Employee e2;    e2.acceptDetails();
e2.displayDetails(); //Direct Value
passing
   Employee e3(3,"Namdev Nikam",new Date(13,1,1980));
   e3.displayDetails();

   return 0;
}
```

```
PROBLEMS    OUTPUT    TERMINAL    ...                    Code
tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
-----------------Default Constructor Called------------------------
------------Parameterized Constructor---------------
----------------------Employee Details----------------
Employee Id of the Employee: 1
Name of the Employee: Vishwajeet
Your Birth Date is 7/11/2002
-----------------Default Constructor Called------------------------
------------Parameterized Constructor---------------
---------------Accepting Employee Details-----------------------
Enter the Employee Id:
1
Enter the Name:
Vishwajit Nikam

Enter the Date of Birth(dd-mm-yyyy):
7
11
2002
------------Parameterized Constructor---------------
----------------------Employee Details----------------
Employee Id of the Employee: 1
Name of the Employee: Vishwajit Nikam
Your Birth Date is 7/11/2002
------------Parameterized Constructor---------------
---------------Parameterized Constructor Called-----------------
----------------------Employee Details----------------
Employee Id of the Employee: 3
Name of the Employee: Namdev Nikam
Your Birth Date is 13/1/1980
PS D:\PG-DAC Assignments\C++ Module\Day 4\Lab 5>
```

3:Consider that payroll software needs to be developed for computerization of operations of an ABC organization. The organization has employees.

3.1. Construct a class Employee with following members using private access

specifies:

Employee Id integer

Employee Name string

Basic Salary double

HRA double

Medical double=1000

PF double

PT double

Net Salary double

Gross Salary double

Please use following expressions for calculations://Note:Don't accept HRA,PF PT from user

*** HRA = 50% of Basic Salary**

*** PF = 12% of Basic Salary**

*** PT = Rs. 200**

3.2. Write methods to display the details of an employee and calculate the gross and net salary.

*** Goss Salary = Basic Salary + HRA + Medical**

*** Net Salary = Gross Salary – (PT + PF)**

Create Object of employee class and assign values and display Details.

```
#include <iostream>
using namespace std;

class Employee {
private:
   int empId;
string empName;
double
basicSalary;
```

```
double HRA;
double medical =
1000;     double
PF;     double PT
= 200;     double
grossSalary;
double netSalary;

public:
   void setDetails(int id, string name, double
salary) {
       empId = id;
empName = name;
       basicSalary = salary;
       HRA = 0.5 *
basicSalary;         PF = 0.12 *
basicSalary;
calculateSalary();
   }

   void calculateSalary() {         grossSalary
= basicSalary + HRA + medical;
netSalary = grossSalary - (PF + PT);
   }

   void displayDetails() {         cout <<
"\nEmployee ID: " << empId << endl;         cout
<< "Employee Name: " << empName << endl;
       cout << "Basic Salary: Rs. " << basicSalary
<< endl;         cout << "HRA: Rs. " << HRA <<
```

```
endl;        cout << "Medical: Rs. " << medical <<
endl;        cout << "PF: Rs. " << PF << endl;
cout << "PT: Rs. " << PT << endl;
        cout << "Gross Salary: Rs. " << grossSalary
<< endl;        cout << "Net Salary: Rs. " <<
netSalary << endl;
    }
};

int main() {
Employee emp;
    int id;
    string name;
double salary;

    cout << "Enter Employee ID: ";
    cin >> id;
    cin.ignore(); // To ignore leftover
newline     cout << "Enter Employee
Name: ";     getline(cin, name);     cout
<< "Enter Basic Salary: ";     cin >>
salary;

    emp.setDetails(id, name, salary);
emp.displayDetails();

    return 0;
}
```

```
PS D:\PG-DAC Assignments\C++ Module\Day 4\Lab 5> cd "d:\PG-DAC Assignments\C++ Module\Day 4\Lab 5\" ; if ($?) { g++ payRoll.cpp -o payRoll } ; if ($?) { .\payRoll }
Enter Employee ID: 1
Enter Employee Name: Vikramaditya
Enter Basic Salary: 23000000

Employee ID: 1
Employee Name: Vikramaditya
Basic Salary: Rs. 2.3e+007
HRA: Rs. 1.15e+007
Medical: Rs. 1000
PF: Rs. 2.76e+006
PT: Rs. 200
Gross Salary: Rs. 3.4501e+007
Net Salary: Rs. 3.17408e+007
PS D:\PG-DAC Assignments\C++ Module\Day 4\Lab 5>
```

Lab 6

1 Solve this.

Fresh business scenario to apply inheritance , polymorphism to emp based organization scenario.

Create Emp based organization structure --- Emp , Mgr , Worker

1.1 Emp state--- id(int), name, deptId , basicSalary(double)

Accept all of above in constructor arguments.

Methods ---

1.2 compute net salary ---ret 0

(eg : public double computeNetSalary(){return 0;})

1.2 Mgr state ---id,name,basic,deptId , perfBonus

Add suitable constructor

Methods ----

1. compute net salary (formula: basic+perfBonus) -- override computeNetSalary

1.3 Worker state --id,name,basic,deptId,hoursWorked,hourlyRate

Methods :

1. compute net salary (formula: = basic+(hoursWorked*hourlyRate) --override computeNetSalary

2. get hrlyRate of the worker -- add a new method to return hourly rate of a worker.(getter)

Create suitable array to store organization details.

Provide following options

1. Hire Manager

I/P : all manager details

2. Hire Worker

I/P : all worker details

3. Display information of all employees net salary (by invoking computeNetSal),

4. Exit

--

```
#include <bits/stdc++.h> using
namespace std;

// ================= Base Employee Class =================
class Employee { private:
   int empId;
string empName;
   int deptId;     double
basicSalary;

public:
   // Constructor
   Employee(int empId, string empName, int deptId, double basicSalary)
      : empId(empId), empName(empName), deptId(deptId), basicSalary(basicSalary) {}

   // Getters     int getEmpId() { return empId;
}     string getEmpName() { return empName;
}     int getDeptId() { return deptId; }
   double getBasicSalary() { return basicSalary; }
```

```
    // Virtual Function      virtual double
computeNetSalary() = 0;

    // Virtual Destructor (important for cleanup of derived objects)
virtual ~Employee() {}
};

// ================= Manager Class =================
class Manager : public Employee { private:
    double perfBonus;

public:
    Manager(int empId, string empName, int deptId, double basicSalary, double perfBonus)
        : Employee(empId, empName, deptId, basicSalary), perfBonus(perfBonus) {}

    double computeNetSalary() override {
return getBasicSalary() + perfBonus;
    }
};

// ================= Worker Class =================
class Worker : public Employee { private:
    int hoursWorked;
double hourlyRate;

public:
    Worker(int empId, string empName, int deptId, double basicSalary,
int hoursWorked, double hourlyRate)         : Employee(empId,
empName, deptId, basicSalary),           hoursWorked(hoursWorked),
hourlyRate(hourlyRate) {}
```

```
    double computeNetSalary() override {         return
getBasicSalary() + (hoursWorked * hourlyRate);
    }

    double getHourlyRate() { return hourlyRate; }
};

// ================= Main Function =================
int main() {
    Employee* employees[100];
    int index = 0;
int choice;

    do {         cout << "\n-----------------------Menu-----------------
----\n";         cout << "1) Hire Manager\n";         cout << "2)
Hire Worker\n";         cout << "3) View Employee Details\n";
cout << "4) Exit\n";         cout << "Enter Choice: ";         cin
>> choice;

        if (choice == 1) {
int id, dept;             string
name;             double
basic, bonus;

            cout << "Enter Manager ID: ";
cin >> id;             cout << "Enter Name: ";
cin.ignore();             getline(cin, name);
cout << "Enter Department ID: ";             cin
>> dept;             cout << "Enter Basic Salary:
```

```
";            cin >> basic;            cout << "Enter
Performance Bonus: ";            cin >> bonus;

            employees[index++] = new Manager(id, name, dept, basic, bonus);

        } else if (choice == 2) {
int id, dept, hrs;
string name;            double
basic, rate;

            cout << "Enter Worker ID: ";
cin >> id;            cout << "Enter Name:
";            cin.ignore();
getline(cin, name);            cout <<
"Enter Department ID: ";            cin >>
dept;            cout << "Enter Basic
Salary: ";            cin >> basic;
            cout << "Enter Hours Worked: ";
cin >> hrs;            cout << "Enter
Hourly Rate: ";            cin >> rate;

            employees[index++] = new Worker(id, name, dept, basic, hrs, rate);

        } else if (choice == 3) {            cout << "\n------
Employee Details ------\n";            for (int i = 0; i <
index; i++) {                cout << "ID: " <<
employees[i]->getEmpId()
                 << ", Name: " << employees[i]->getEmpName()
                 << ", Dept: " << employees[i]->getDeptId()
                 << ", Net Salary: " << employees[i]->computeNetSalary()
```

```
                    << endl;
            }
        } else if (choice == 4) {
cout << "Exiting...\n";
        } else {            cout <<
"Invalid choice!\n";
        }
    } while (choice != 4);

    // cleanup     for (int i = 0; i <
index; i++) {         delete
employees[i];
    }

    return 0;
}
```

```
PROBLEMS    OUTPUT    TERMINAL    ...                Code  +  ...
----------------------Menu--------------------
1) Hire Manager
2) Hire Worker
3) View Employee Details
4) Exit
Enter Choice: 1
Enter Manager ID: 1
Enter Name: Vishwajit
Enter Department ID: 1
Enter Basic Salary: 250000
Enter Performance Bonus: 25000

----------------------Menu--------------------
1) Hire Manager
2) Hire Worker
3) View Employee Details
4) Exit
Enter Choice: 2
Enter Worker ID: 2
Enter Name: Kashi
Enter Department ID: 2
Enter Basic Salary: 25000
Enter Hours Worked: 2400
Enter Hourly Rate: 200

----------------------Menu--------------------
1) Hire Manager
2) Hire Worker
3) View Employee Details
4) Exit
Enter Choice: 3

------ Employee Details ------
ID: 1, Name: Vishwajit, Dept: 1, Net Salary: 275000
ID: 2, Name: Kashi, Dept: 2, Net Salary: 505000

----------------------Menu--------------------
```

2:Create cpp application for bank account handling.

2.1. Create a class BankAccount -- acct no(int),customer name(string),balance(double)

Add constr. (2 constrs : first to accept all details)

2.2 Add Business logic methods

Methods public void

withdraw(double amt) public

void deposit(double amt)

2.3: Create object of account class and test withdraw and deposit methods.

```
// Create cpp application for bank account handling.
// 2.1. Create a class BankAccount -- acct no(int),customer name(string),balance(double) //
Add  constr. (2 constrs : first to accept all details )

// 2.2 Add Business logic methods
// Methods
// public void withdraw(double amt)
// public void deposit(double amt)

// 2.3: Create object of account class and test withdraw and deposit methods.

#include<bits/stdc++.h>
#define MAX 100000 using
namespace std;

class bankAccount
{ private:
        long long accountNumber;
        string accountHolder; double
        balance;

        double amount;
```

```
public:
bankAccount()
    {
      cout<<"---------------Default Constructor Called------------------"<<endl;
this->accountNumber = 7219436696;           this->accountHolder =
"Mr.Vishwajit Namdev Nikam";           this->balance = 250000;
    }

    // Parameterized Constructor        bankAccount(int
accountNumber,string accountHolder,double balance)
    {
      cout<<"------------------Parameterized Constructor Called-----------------"<<endl;
this->accountNumber = accountNumber;           this->accountHolder = accountHolder;
this->balance =balance;
    }

    //setter         void
setBalance()
    {
      cout<<"Enter the Account Balance: "<<endl;
cin>>balance;
    }

    void setAccountHolderName()
    {    cout<<"Enter the name of Account Holder:
    "<<endl;    cin.ignore();
    getline(cin,accountHolder);
    }
```

```
        double getBalance()
        {             return
balance;
        }

        string getAccountHolderName()
        {
           return accountHolder;
        }

        long long getAccountNumber()
        {
           return accountNumber;
        }

        //Withdraw Function
void withdraw()
        {
           cout<<"Enter the Amount that you want to Withdraw"<<endl;
cin>>amount;             if(getBalance() - amount <= 0)
           {
              cout<<"Insufficient Balance."<<endl;
           }
         else    {
         cout<<
         amount
         <<"
         withdra
         wed
         and
```

```
        "<<get
        Balanc
        e() -
        amount
        <<"
        Balanc
        e is
remaining."<<endl;
          }
        }

        //Deposit Function
void Deposit()
        {
          cout<<"Enter the amount that you want to deposit: "<<endl;
cin>>amount;             balance += amount;
        }

        //Display Function
void getBankDetails()
        {
         cout<<"Account Holder Name: "<<getAccountHolderName()<<endl;
cout<<"Account Number: "<<getAccountNumber()<<endl;           cout<<"Account
Balance: "<<getBalance()<<endl;
        } }; int main(int argc, char const
*argv[])
{    bankAccount
accounts;    int
choice,index=0;
```

```
    do{       cout<<"----------Menu---------------
"<<endl;       cout<<"1) Withdraw "<<endl;
cout<<"2) Deposit "<<endl;       cout<<"3)
Check Bank Details"<<endl; cout<<"4)
Exit"<<endl;


      cout<<"Enter the choice: "<<endl;
cin>>choice;       switch(choice)
      {
case 1:
               cout<<"--------Withdrawl------"<<endl;
accounts.withdraw();                break;
         case 2:
               accounts.Deposit();
cout<<"Deposited."<<endl;
               break;
         case 3:
accounts.getBankDetails();
break;

case 4:
                cout<<"Exited..."<<endl;
return 0;

default:
                cout<<"Enter  the Valid Choice"<<endl;
      }
    }while(choice != 4);

  return 0;
```

}

```
----------Menu---------------
1) Withdraw
2) Deposit
3) Check Bank Details
4) Exit
Enter the choice:
3
Account Holder Name: Mr.Vishwajit Namdev Nikam
Account Number: 7219436696
Account Balance: 250000
----------Menu---------------
1) Withdraw
2) Deposit
3) Check Bank Details
4) Exit
Enter the choice:
1
--------Withdrawl------
2) Deposit
3) Check Bank Details
4) Exit
Enter the choice:
1
--------Withdrawl------
4) Exit
Enter the choice:
1
--------Withdrawl------
Enter the Amount that you want to Withdraw
Enter the choice:
1
--------Withdrawl------
Enter the Amount that you want to Withdraw
1
--------Withdrawl------
Enter the Amount that you want to Withdraw
--------Withdrawl------
Enter the Amount that you want to Withdraw
2300
Enter the Amount that you want to Withdraw
2300
```

Day 6

Basic Template Assignments

Function Template – Swap

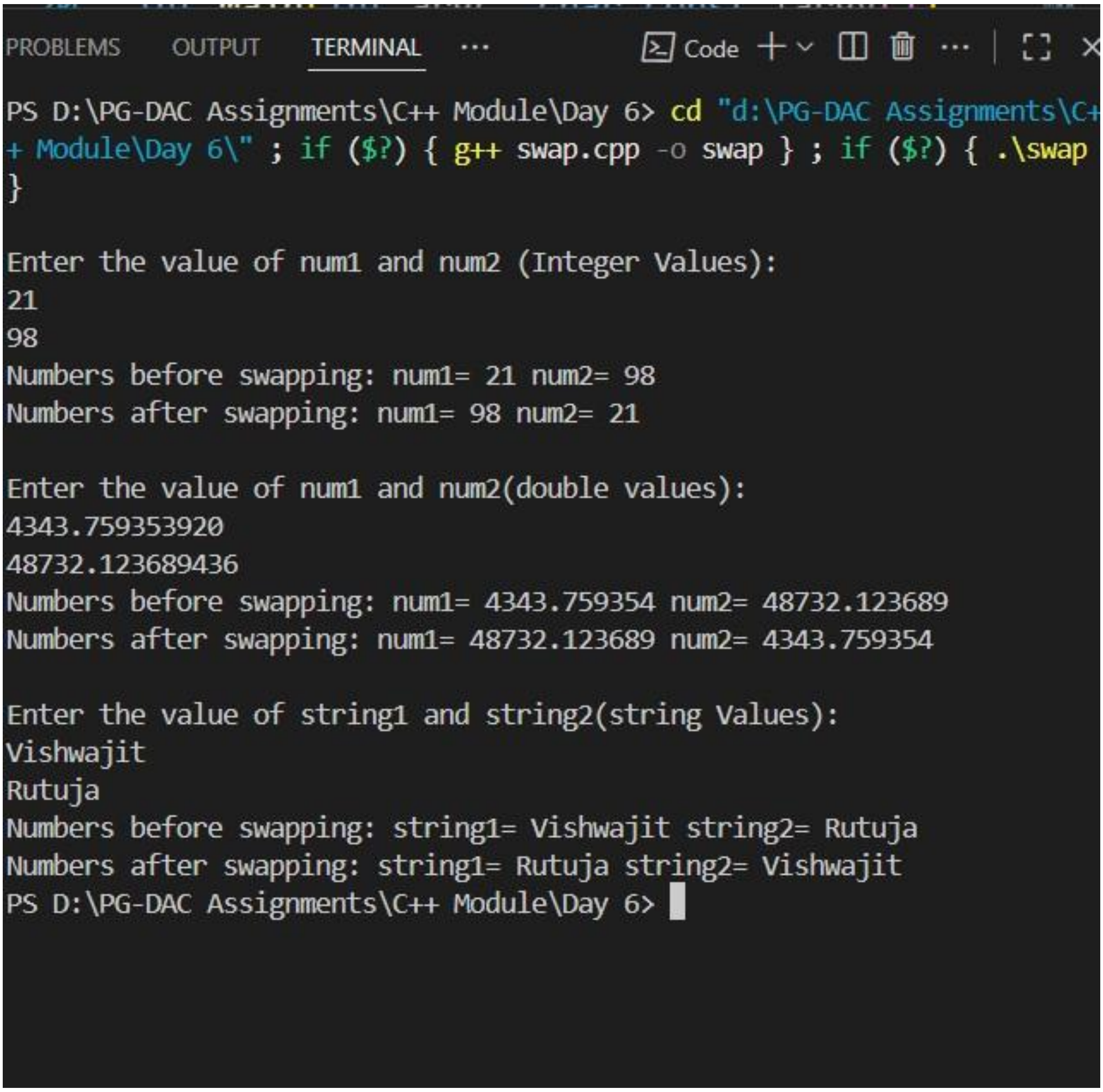
Write a function template swapValues() that swaps two variables of any type.

Test with int, double, and

string. #include<bits/stdc++.h>

```
using namespace std;

//Template Function template<class
Type>

Type swapValues(Type &m, Type &n)
{
        Type temp = m;
m = n;          n =
temp;
}

int main(int argc, char const *argv[])
{     int num1,num2;
double num3, num4;
string string1,string2;

    cout<<fixed<<setprecision<<endl;

    cout<<"Enter the value of num1 and num2 (Integer Values): "<<endl;
cin>>num1>>num2;

    //Integers swapping     cout<<"Numbers before swapping: num1= "<<num1<<"
num2= "<<num2<<endl;     swapValues(num1,num2);     cout<<"Numbers after
```

```
swapping: num1= "<<num1<<" num2= "<<num2<<endl;
cout<<fixed<<setprecision<<endl;
   //Double swapping      cout<<"Enter the value of num1 and num2(double values):
"<<endl;     cin>>num3>>num4;     cout<<"Numbers before swapping: num1=
"<<num3<<" num2= "<<num4<<endl;     swapValues(num3,num4);
cout<<"Numbers after swapping: num1= "<<num3<<" num2= "<<num4<<endl;

   //string swapping      cout<<fixed<<setprecision<<endl;      cout<<"Enter the value of
string1 and string2(string Values): "<<endl;     cin>>string1>>string2;     cout<<"Numbers
before swapping: string1= "<<string1<<" string2= "<<string2<<endl;
swapValues(string1,string2);     cout<<"Numbers after swapping: string1= "<<string1<<"
string2= "<<string2<<endl;
   return 0;
}
```



```
PROBLEMS    OUTPUT    TERMINAL    ...                Code  +  ...

PS D:\PG-DAC Assignments\C++ Module\Day 6> cd "d:\PG-DAC Assignments\C+
+ Module\Day 6\" ; if ($?) { g++ swap.cpp -o swap } ; if ($?) { .\swap
}

Enter the value of num1 and num2 (Integer Values):
21
98
Numbers before swapping: num1= 21 num2= 98
Numbers after swapping: num1= 98 num2= 21

Enter the value of num1 and num2(double values):
4343.759353920
48732.123689436
Numbers before swapping: num1= 4343.759354 num2= 48732.123689
Numbers after swapping: num1= 48732.123689 num2= 4343.759354

Enter the value of string1 and string2(string Values):
Vishwajit
Rutuja
Numbers before swapping: string1= Vishwajit string2= Rutuja
Numbers after swapping: string1= Rutuja string2= Vishwajit
PS D:\PG-DAC Assignments\C++ Module\Day 6>
```

Function Template – Maximum

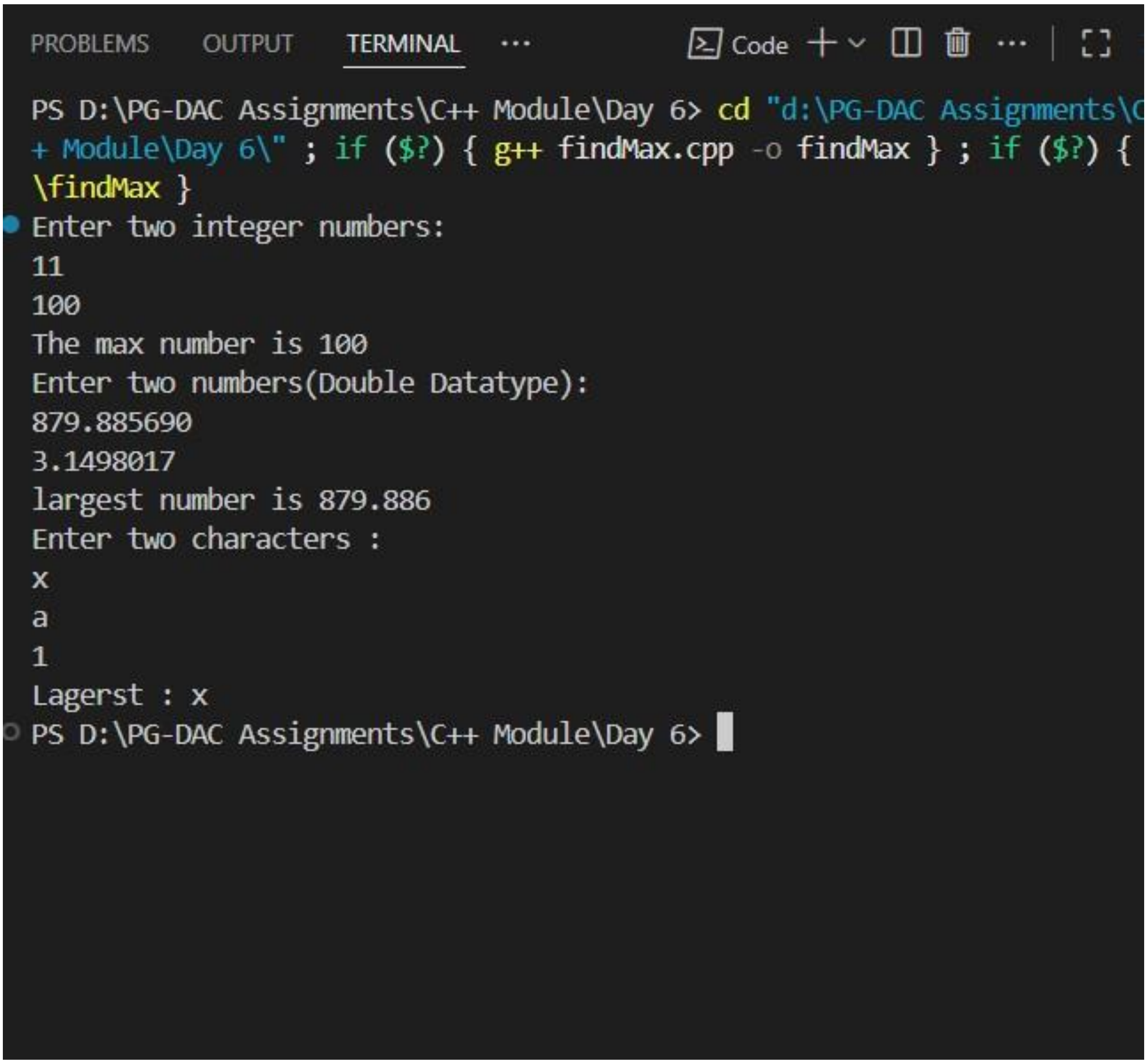
Write a function template findMax() that returns the maximum of two values.

Test with int, float, and char.

```
// Function Template – Maximum

//  Write a function template findMax() that returns the maximum of two values.


#include<bits/stdc++.h> using

namespace std;


template <class Type>
```

```
Type findMax(Type &x,Type &y)
{
if(x>=y)
return x;
else
return y;
}

int main(int argc, char const *argv[])
{    int
num1,num2;
   //Integer Number
   //cout<<fixed<<setprecision<<endl;     cout<<"Enter two
integer numbers: "<<endl;     cin>>num1>>num2;
cout<<"The max number is "<<findMax(num1,num2)<<endl;
   //cout<<fixed<<setprecision<<endl;
   //Double     double num3,num4;     cout<<"Enter two
numbers(Double Datatype): "<<endl;
cin>>num3>>num4;
   //cout<<fixed<<setprecision<<endl;     cout<<"largest number is
"<<findMax<double>(num3,num4)<<endl;

   //Characters
char ch1,ch2;
  //cout<<fixed<<setprecision<<endl; cout<<"Enter
   two characters : "<<endl; cin>>ch1>>ch2;
   cout<<fixed<<setprecision<<endl;
   cout<<"Lagerst :
   "<<findMax<char>(ch1,ch2)<<endl;
```

```
   // cout<<fixed<<setprecision<<endl;
   return 0;
}
```



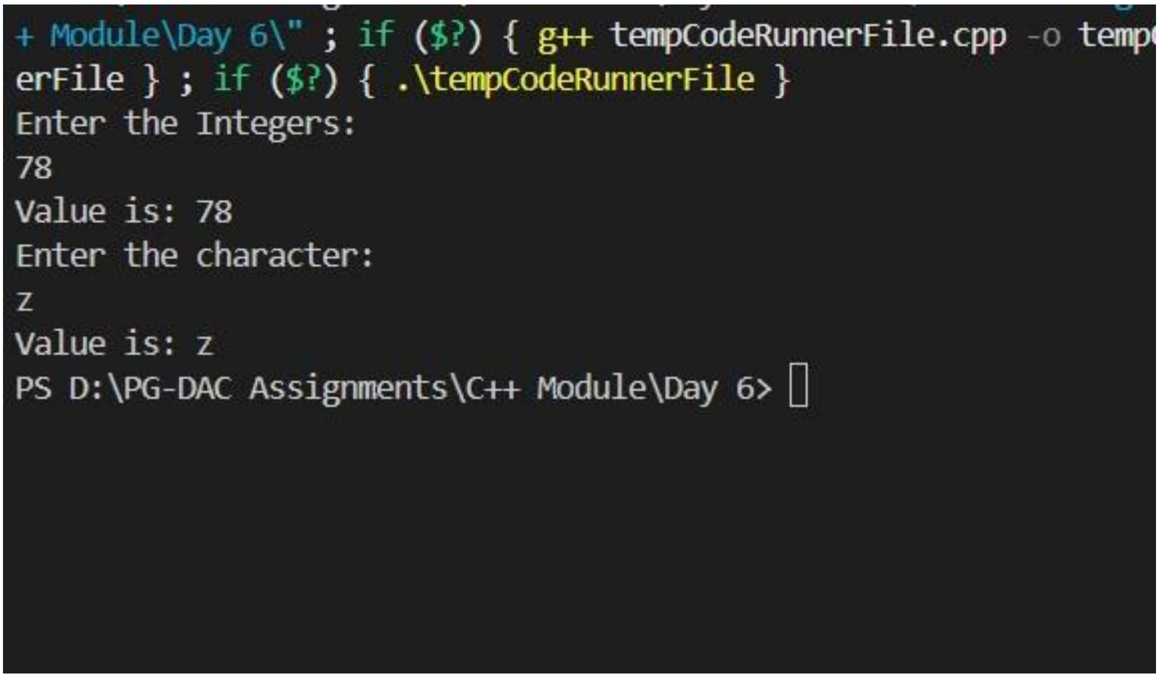
3: Class Template – Box

Implement a class template Box<T> that stores one value of any type and provides getValue() and setValue() methods.

```
#include <iostream> using
namespace std;


template <typename T>
```

```
class Box
{ private:
   T value;

public:
   // Default Constructor
   Box() {}

   // Parameterized Constructor
   Box(T val)
   {       value =
val;    }
   // Setters     void
setValue(T val)
   {       value =
val;
   }

   T getValue()
   {       return
value;
   } }; int
main()
{ int num;
   char ch;
   // integer Testing
   cout << "Enter the Integers: " << endl;
   cin >> num; Box<int> b;
   b.setValue(num); cout << "Value is: " <<
   b.getValue() << endl;
```

```
    // character Testing      cout << "Enter
the character: " << endl;      cin >> ch;
Box<char> c;
    c.setValue(ch);      cout << "Value is: " <<
c.getValue() << endl;      return 0;
}
```



Class Template – Calculator

Create a class template Calculator<T> with functions: add(), subtract(), multiply(), divide().

Test with int and double.

```
// Class Template – Calculator
//  Create a class template Calculator<T> with functions:
// add(), subtract(), multiply(), divide().
//  Test with int and double.

#include <iostream> using
namespace std;
```

```
// Template class template <typename
Type>class Calculator
{ private:
    Type num1;
    Type num2;

public:
    // Default Constructor
    Calculator() {};

    // Parameterized Constructor
    Calculator(T val1, T val2)
    {         num1 =
val1;         num1 =
val2;
    }

   // add
    Type add(Type val1, Type val2)
    {     return val1 +
    val2; }
    // substract
    Type sub(Type val1, Type val2)
    {     return val1 -
    val2;
    }
    // Multiply
    Type mult(Type val1, Type val2)
```

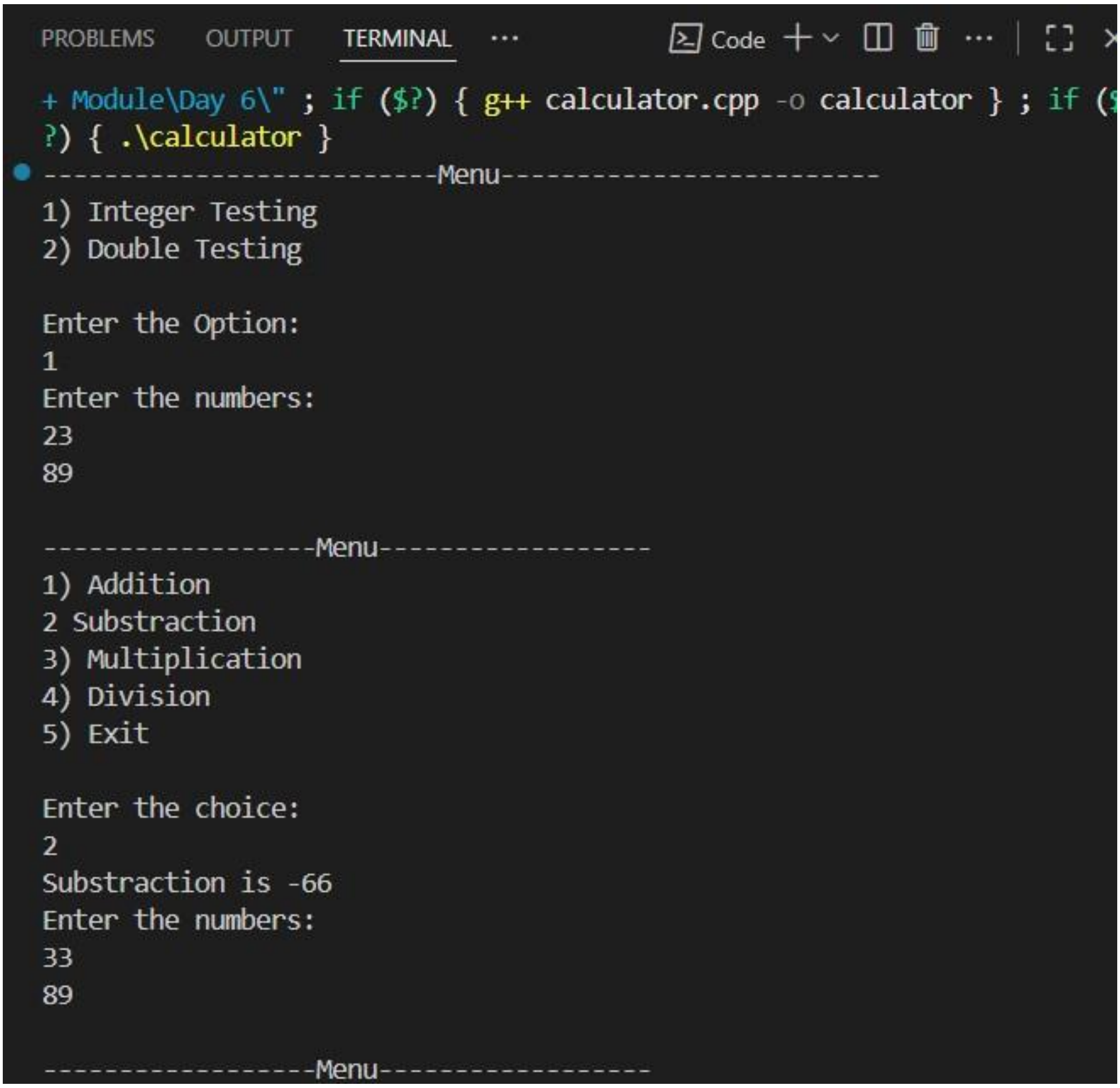
```
{       return val1 *
val2;
}
// Divide
Type div(Type val1, Type val2)
{       return val1 /
val2;
}
};

int main() {    int num1,
num2, choice;    double
num3, num4;
int option;

do   {       cout << "--------------------------Menu-------------------------"
<< endl;    cout << "1) Integer Testing" << endl;       cout << "2)
Double Testing " << endl;

cout << endl;

cout << "Enter the Option: " << endl;
cin >> option;   switch (option)
{
case 1:
// Integer Testing
do
{
cout << "Enter the numbers: " << endl;
cin >> num1 >> num2;
```

```
cout << endl;

cout << "------------------Menu------------------" << endl;
cout << "1) Addition " << endl;                cout << "2
Substraction " << endl;                cout << "3) Multiplication " <<
endl;                cout << "4) Division " << endl;                cout <<
"5) Exit" << endl;

cout << endl;

cout << "Enter the choice: " << endl;

cin >> choice;             Calculator <int> A;

switch (choice)
{
case 1:
// Calculator<int>A;              cout << "Sum is " <<
A.add(num1, num2) << endl;
break;
case 2:
// Calculator<int>A;                  cout << "Substraction is "
<< A.sub(num1, num2) << endl;
break;
case 3:
// Calculator<int>A;                  cout << "Multiplication is "
<< A.mult(num1, num2) << endl;
break;
case 4:
// Calculator<int>A;                  cout << "Division is "
<< A.div(num1, num2) << endl;
```

```
                    break;
case 5:
                    cout << "Exiting the Program..." << endl;
                    return 0;
break;
                default:
                  cout << "Enter the Valid Choice..!" << endl;
                }
            } while (choice != 5);
break;

        case 2:
            // Double Testing
    do
            {                   cout << "Enter the numbers: "
<< endl;                  cin >> num3 >> num4;


                cout << endl;


                cout << "------------------Menu------------------" << endl;
cout << "1) Addition " << endl;                  cout << "2
Substraction " << endl;                  cout << "3) Multiplication " <<
endl;                  cout << "4) Division " << endl;


                cout << endl;


                cout << "Enter the choice: " << endl;
cin >> choice;


                Calculator<double> B;
```

```
                switch (choice)
                {
 case 1:
                  // Calculator<int>A;                cout << "Sum is " <<
  B.add(num3, num4) << endl;
                  break;
               case 2:
                  // Calculator<int>A;                cout << "Substraction is "
   << B.sub(num3, num4) << endl;
                   break;


                case 3:
                   // Calculator<int>A;                    cout << "Multiplication is "
<< B.mult(num3, num4) << endl;
                   break;
                 case 4:
                   // Calculator<int>A;                    cout << "Division is "
<< B.div(num3, num4) << endl;
                   break;
case 5:
                   cout << "Exiting the Program..." << endl;
                   return 0;
break;
                 default:
                   cout << "Enter the Valid Choice..!" << endl;
                }
             } while (choice != 5);
break;


      case 3:
```

```
            cout << "Exit Operation..!" << endl;
return 0;           break;

        default:
            cout << "Enter valid Option..." << endl;
        }
    } while (option != 3);
return 0;
}
```



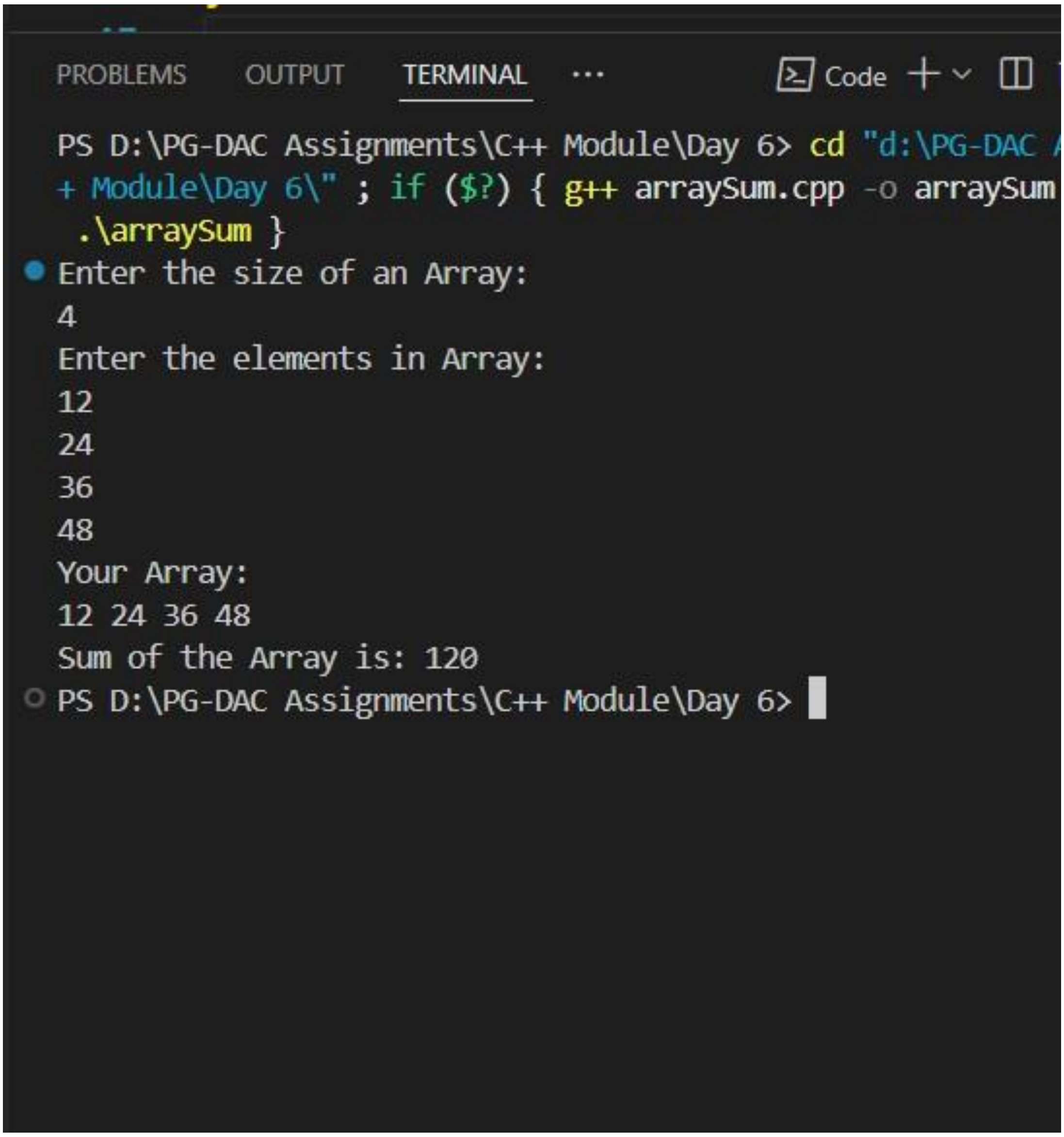
```
PROBLEMS    OUTPUT    TERMINAL    ...                    Code  +  ...

+ Module\Day 6\" ; if ($?) { g++ calculator.cpp -o calculator } ; if ($
?) { .\calculator }
-------------------------Menu------------------------
1) Integer Testing
2) Double Testing

Enter the Option:
1
Enter the numbers:
23
89

-----------------Menu-----------------
1) Addition
2 Substraction
3) Multiplication
4) Division
5) Exit

Enter the choice:
2
Substraction is -66
Enter the numbers:
33
89

-----------------Menu-----------------
```

Function Template – Array Sum

Write a function template sumArray() that accepts an array of any type and returns the sum of its elements.

```
#include <bits/stdc++.h>

#define MAX 1000 using

namespace std;


// Function Template template<typename

T>

T arraySum(T arr[], int size)

{     T sum = 0;     for (int i =

0; i < size; i++)

    {         sum +=

arr[i];

    }

    return sum;

}


int main() {

int size;     int

arr[MAX];


    cout << "Enter the size of an Array: " << endl;

cin >> size;


    cout << "Enter the elements in Array: " << endl;

for (int i = 0; i < size; i++)

    {         cin >>

arr[i];

    }
```

```
    cout << "Your Array: " << endl; for
    (int i = 0; i < size; i++)
    {       cout << arr[i] <<
" ";
    }
    cout << endl;

    cout << "Sum of the Array is: " << arraySum(arr, size) << endl;

    return 0;
}
```



6: Class Template – Stack

Implement a class template Stack<T> with functions: push(), pop(), peek(), isEmpty().Test with int and string.

```
#include <bits/stdc++.h>
#define MAX 1000 using
namespace std;

template <typename T>
class myStack { private:
    T arr[MAX];
```

```
    int top;

public:
myStack() {
        top = -1;
    }

    void push(T value) {          if (top >= MAX
- 1) {              cout << "Stack Overflow..." <<
endl;
        } else {
arr[++top] = value;
        }
    }

    T pop() {         if (top < 0) {             cout <<
"Stack Underflow..." << endl;             return
T(); // return default value
        } else {
return arr[top--];
        }
    }

    T peek() {         if (top < 0) {             cout
<< "Stack is Empty..." << endl;            return
T();
       } else {
           return arr[top];
       }
    }
```

```
    bool isEmpty() {
return (top < 0);
    }

    void Display() {         cout <<
"Your Stack: " << endl;         for (int i
= top; i >= 0; i--) {             cout <<
arr[i] << endl;
        }
    }
};

int main() {     myStack<int> s1;     s1.push(10);
s1.push(20);     s1.push(30);     s1.Display();
cout << "Popped: " << s1.pop() << endl;     cout
<< "Top Element: " << s1.peek() << endl;

    myStack<string> s2;
s2.push("Hello");     s2.push("World");
s2.Display();    cout << "Popped: " <<
s2.pop() << endl; return 0;

}
```

```
Your Stack:
30
20
10
Popped: 30
Top Element: 20
Your Stack:
World
Hello
Popped: World
PS D:\PG-DAC Assignments\C++ Module\Day 6>
```

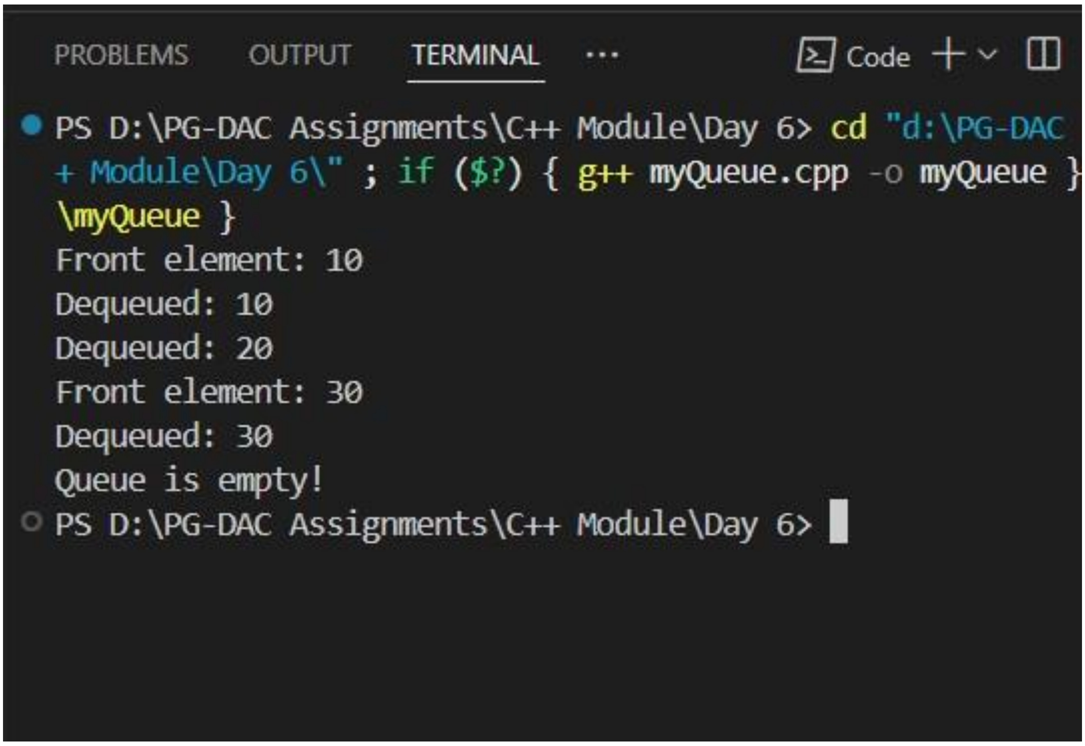
7Class Template – Queue

Implement a class template Queue<T> with functions: enqueue(), dequeue(), front(), isEmpty().

```
#include <iostream> using
namespace std;

template <class T> class Queue {
private:    int frontIndex, rearIndex,
capacity;
    T* arr;
public:
    Queue(int size = 10) {
capacity = size;        arr =
new T[capacity];
frontIndex = -1;
rearIndex = -1;
    }
```

```
    ~Queue() {
        delete[] arr;
    }

    bool isEmpty() {          return (frontIndex == -1 ||
frontIndex > rearIndex);
    }

    bool isFull() {          return (rearIndex
== capacity - 1);
    }

    void enqueue(T data) {
        if (isFull()) {              cout <<
"Queue is full!" << endl;
            return;
        }
        if (frontIndex == -1) frontIndex = 0;
arr[++rearIndex] = data;
    }

    void dequeue() {
        if (isEmpty()) {              cout <<
"Queue is empty!" << endl;
            return;
        }
        cout << "Dequeued: " << arr[frontIndex++] << endl;
    }
```

```
    T front() {          if (isEmpty()) {              throw
runtime_error("Queue is empty!");
        }
        return arr[frontIndex];
    }
};

int main() {
    Queue<int> q(5);

    q.enqueue(10);
    q.enqueue(20);
    q.enqueue(30);

    cout << "Front element: " << q.front() << endl;

    q.dequeue();
    q.dequeue();

    cout << "Front element: " << q.front() << endl;

    q.dequeue();
    q.dequeue(); // extra dequeue to test empty case

    return 0;
}
```



Day 7 Lab

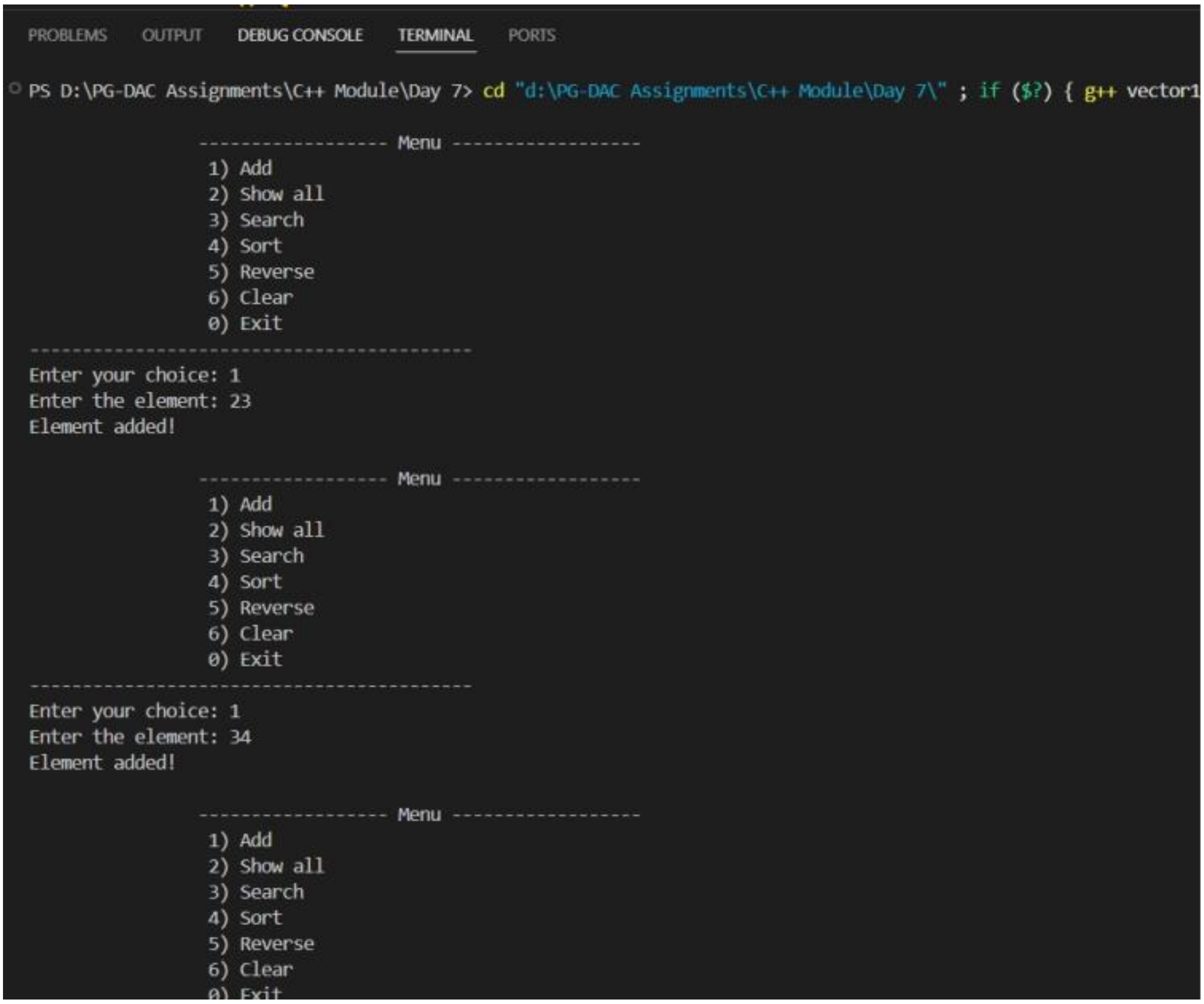
Assignments on STL and File IO

1:Create an application for storing int values in vector.
Create menu drivin app for following menu;
1:add 2:show all 3:search 4:sort 5:reverse 6:clear

```
#include <bits/stdc++.h>
using namespace std;
```

```
int main() {
    vector<int> v;
    int choice, val;

    do {
        cout << "\n\t\t------------------ Menu ------------------\n";
        cout << "\t\t 1) Add\n";
        cout << "\t\t 2) Show all\n";
        cout << "\t\t 3) Search\n";
        cout << "\t\t 4) Sort\n";
        cout << "\t\t 5) Reverse\n";
        cout << "\t\t 6) Clear\n";
        cout << "\t\t 0) Exit\n";
        cout << "------------------------------------------\n";

        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {
            case 1: {
                cout << "Enter the element: ";
                cin >> val;
                v.push_back(val);
                cout << "Element added!\n";
                break;
            }
            case 2: {
                if (v.empty()) {
                    cout << "Vector is empty.\n";
                } else {
                    cout << "Your Vector: ";
                    for (int num : v) cout << num << " ";
                    cout << "\n";
                }
                break;
            }
            case 3: {
                if (v.empty()) {
                    cout << "Vector is empty.\n";
                } else {
                    int s;
                    cout << "Enter the element to search: ";
                    cin >> s;
                    auto it = find(v.begin(), v.end(), s);
                    if (it != v.end())
```

```
                cout << "Element found at position " << (it - v.begin()) << "\n";
            else
                cout << "Element not found!\n";
            }
            break;
        }
        case 4: {
            sort(v.begin(), v.end());
            cout << "Vector sorted!\n";
            break;
        }
        case 5: {
            reverse(v.begin(), v.end());
            cout << "Vector reversed!\n";
            break;
        }
        case 6: {
            v.clear();
            cout << "Vector cleared!\n";
            break;
        }
        case 0:
            cout << "Exiting program...\n";
            break;
        default:
            cout << "Invalid choice! Please try again.\n";
    }

    } while (choice != 0);

    return 0;
}
```



```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\PG-DAC Assignments\C++ Module\Day 7> cd "d:\PG-DAC Assignments\C++ Module\Day 7\" ; if ($?) { g++ vector1

                ----------------- Menu -----------------
                 1) Add
                 2) Show all
                 3) Search
                 4) Sort
                 5) Reverse
                 6) Clear
                 0) Exit
-----------------------------------------
Enter your choice: 1
Enter the element: 23
Element added!

                ----------------- Menu -----------------
                 1) Add
                 2) Show all
                 3) Search
                 4) Sort
                 5) Reverse
                 6) Clear
                 0) Exit
-----------------------------------------
Enter your choice: 1
Enter the element: 34
Element added!

                ----------------- Menu -----------------
                 1) Add
                 2) Show all
                 3) Search
                 4) Sort
                 5) Reverse
                 6) Clear
                 0) Exit
```

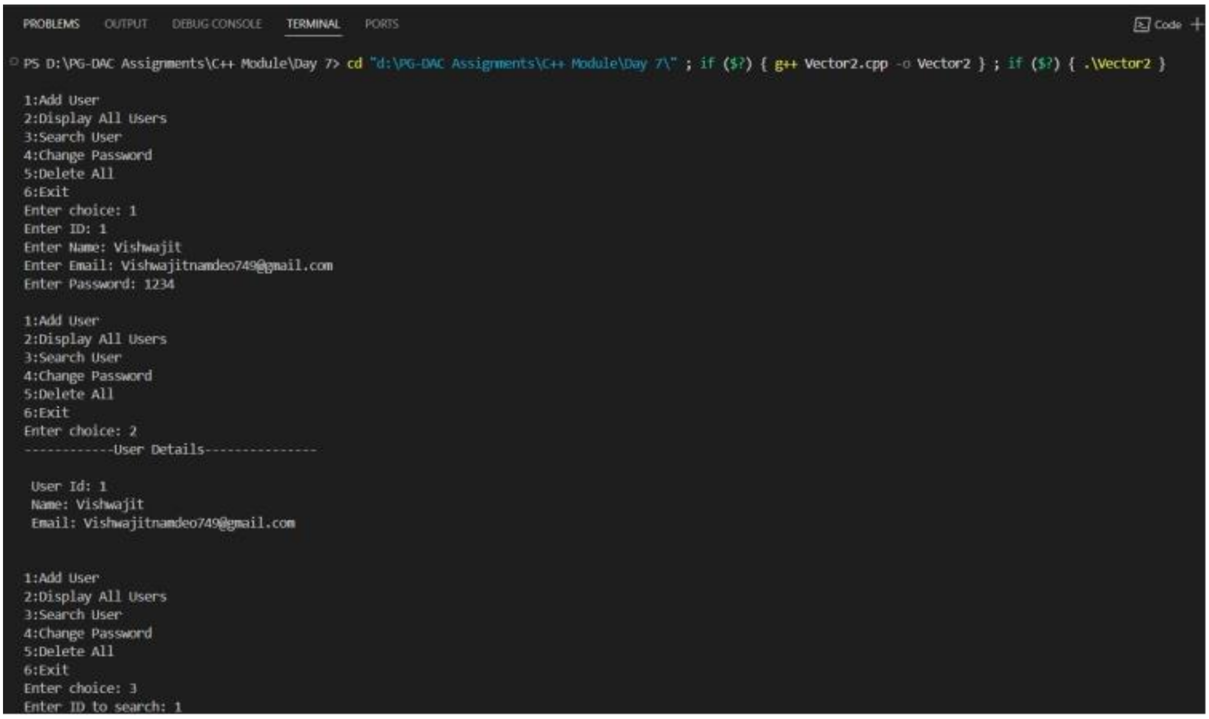
2:Create an application for storing user information in vector.
(Hint:User class with data member userid,name,email,pwd)
Create Menu Driven app
1:add user
2:display all users
3:search user
4:change pwd
5:delete all

```
#include <bits/stdc++.h>
using namespace std;

class User {
    int userid;
    string name, email, pwd;

public:
    User() {} // default constructor

    User(int id, string n, string e, string p) {
        userid = id;
        name = n;
```

```
        email = e;
        pwd = p;
    }

    void setData() {
        cout << "Enter ID: ";
        cin >> userid;
        cout << "Enter Name: ";
        cin >> name;
        cout << "Enter Email: ";
        cin >> email;
        cout << "Enter Password: ";
        cin >> pwd;
    }

    int getId() const { return userid; }
    string getName() const { return name; }
    string getEmail() const { return email; }
    string getPwd() const { return pwd; }
    void setPwd(string p) { pwd = p; }

    void display() const {
        cout <<" User Id: "<< userid <<endl;
        cout<<" Name: " << name<<endl;
        cout<<" Email: " << email << endl;
        cout<<fixed<<endl;
    }
};

int main() {
    vector<User> users;
    int choice;

    do {
        cout << "\n1:Add User\n2:Display All Users\n3:Search User\n4:Change
Password\n5:Delete All\n6:Exit\nEnter choice: ";
        cin >> choice;

        switch (choice) {
        case 1: {
            User u;
            u.setData();
            users.push_back(u);
            break;
        }
        case 2: {
```

```
      cout<<"------------User Details---------------"<<endl;
      cout<<fixed<<endl;
    if (users.empty()) cout << "No users found\n";
    else {
        for (int i = 0; i < users.size(); i++) {
            users[i].display();
        }
    }
    break;
}
case 3: {
    int id, found = 0;
    cout << "Enter ID to search: ";
    cin >> id;
    for (int i = 0; i < users.size(); i++) {
        if (users[i].getId() == id) {
            users[i].display();
            found = 1;
            break;
        }
    }
    if (!found) cout << "User not found\n";
    break;
}
case 4: {
    int id, found = 0;
    cout << "Enter ID to change password: ";
    cin >> id;
    for (int i = 0; i < users.size(); i++) {
        if (users[i].getId() == id) {
            string newpwd;
            cout << "Enter new password: ";
            cin >> newpwd;
            users[i].setPwd(newpwd);
            cout << "Password updated\n";
            found = 1;
            break;
        }
    }
    if (!found) cout << "User not found\n";
    break;
}
case 5: {
    users.clear();
    cout << "All users deleted\n";
    break;
```

```
        }
        case 6:
            cout << "Exiting...\n";
            break;
        default:
            cout << "Invalid choice\n";
        }
    } while (choice != 6);

    return 0;
}
```



3:Create an application using set .
Accept name of city from user and store in set
Create Menu drivien app
1:add city
2:display all city
3: serach city

```
#include <bits/stdc++.h>
using namespace std;

int main() {
    set<string> cities;
    int choice;

    do {
        cout<<"-----------------Menu------------------"<<endl;
        cout<<fixed<<endl;
```

```
    cout << "\n1:Add City\n2:Display All Cities\n3:Search City\n4:Exit"<<endl;
    cout<<fixed<<endl;
     cout<<" Enter choice: "<<endl;
    cin >> choice;

    switch (choice) {
    case 1: {
        string city;
        cout << "Enter city name: ";
        cin >> city;
        auto result = cities.insert(city);
        if (!result.second)
            cout << "City already exists\n";
        else
            cout << "City added\n";
        break;
    }
    case 2: {
        if (cities.empty())
            cout << "No cities stored\n";
        else {
            cout <<"----Cities--------"<<endl;
            cout<<fixed<<endl;
            for (auto &c : cities)
                cout << c << endl;
        }
        break;
    }
    case 3: {
        string city;
        cout<<fixed<<endl;
        cout << "Enter city name to search: ";
        cin >> city;
        if (cities.find(city) != cities.end())
            cout << "City found: " << city << endl;
        else
            cout << "City not found\n";
        break;
    }
    case 4:
        cout << "Exiting...\n";
        break;
    default:
        cout << "Invalid choice\n";
    }
} while (choice != 4);
```

```
    return 0;
}
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

City added
-----------------Menu------------------


1:Add City
2:Display All Cities
3:Search City
4:Exit

 Enter choice:
1
Enter city name: Bengaluru
City added
-----------------Menu------------------


1:Add City
2:Display All Cities
3:Search City
4:Exit

 Enter choice:
2
----Cities--------

Bengaluru
Karad
Pune
-----------------Menu------------------


1:Add City
2:Display All Cities
3:Search City
4:Exit
```

4:Create an application using map for storing key and value
key:int
value:Account type
Create Account class with actid ,name,balance
Create Menu driven app
1:Add Account
2:Display all
3:Search account by actid;

4:Remove all

```
#include <iostream>
#include <map>
#include <string>
using namespace std;

class Account {
    int actid;
    string name;
    double balance;

public:
    Account() {}
    Account(int id, string n, double b) {
        actid = id;
        name = n;
        balance = b;
    }

    int getId() const { return actid; }
    string getName() const { return name; }
    double getBalance() const { return balance; }

    void display() const {
        cout << "Account ID: " << actid << " Name: " << name << " Balance: " <<
balance << endl;
    }
};

int main() {
    map<int, Account> accounts;
    int choice;

    do {
        cout << "\n1:Add Account\n2:Display All\n3:Search Account by ID\n4:Remove
All\n5:Exit\nEnter choice: ";
        cin >> choice;

        switch (choice) {
        case 1: {
            int id;
            string name;
            double bal;
            cout << "Enter Account ID: ";
            cin >> id;
            cout << "Enter Name: ";
```

```
    cin >> name;
    cout << "Enter Balance: ";
    cin >> bal;

    if (accounts.find(id) != accounts.end()) {
        cout << "Account ID already exists!\n";
    } else {
        accounts[id] = Account(id, name, bal);
        cout << "Account added successfully!\n";
    }
    break;
}
case 2: {
    if (accounts.empty())
        cout << "No accounts available\n";
    else {
        cout << "Accounts List:\n";
        for (auto &entry : accounts) {
            cout << "Key: " << entry.first << " -> ";
            entry.second.display();
        }
    }
    break;
}
case 3: {
    int id;
    cout << "Enter Account ID to search: ";
    cin >> id;
    auto it = accounts.find(id);
    if (it != accounts.end()) {
        cout << "Account found!\n";
        it->second.display();
    } else {
        cout << "Account not found\n";
    }
    break;
}
case 4: {
    accounts.clear();
    cout << "All accounts removed\n";
    break;
}
case 5:
    cout << "Exiting...\n";
    break;
default:
```

```
            cout << "Invalid choice\n";
        }
    } while (choice != 5);

    return 0;
}
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

1:Add Account
2:Display All
3:Search Account by ID
4:Remove All
5:Exit
Enter choice: 1
Enter Account ID: 1
Enter Name: Vishwajit
Enter Balance: 25000
Account added successfully!

1:Add Account
2:Display All
3:Search Account by ID
4:Remove All
5:Exit
Enter choice: 2
Accounts List:
Key: 1 -> Account ID: 1 Name: Vishwajit Balance: 25000

1:Add Account
2:Display All
3:Search Account by ID
4:Remove All
5:Exit
Enter choice: 3
Enter Account ID to search: 1
Account found!
Account ID: 1 Name: Vishwajit Balance: 25000

1:Add Account
2:Display All
3:Search Account by ID
4:Remove All
5:Exit
Enter choice:
```

5: Create an File IO application for basic operation
1:Write file:accept data from user and store in file
2:Read file:display line by line
3:copy data from one file into another file

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {
    int choice;
    string filename1 = "file1.txt";
    string filename2 = "file2.txt";

    do {
        cout << "\n1:Write File\n2:Read File\n3:Copy Data (file1 →
file2)\n4:Exit\nEnter choice: ";
        cin >> choice;
        cin.ignore(); // clear buffer

        switch (choice) {
        case 1: {
            ofstream fout(filename1, ios::app);
            if (!fout) {
                cout << "Error opening file for writing\n";
                break;
            }
            string data;
            cout << "Enter text (type END to stop):\n";
            while (true) {
                getline(cin, data);
                if (data == "END") break;
                fout << data << endl;
            }
            fout.close();
            cout << "Data written successfully!\n";
            break;
        }
        case 2: {
            ifstream fin(filename1);
            if (!fin) {
                cout << "Error opening file for reading\n";
                break;
            }
            string line;
            cout << "File Contents:\n";
            while (getline(fin, line)) {
                cout << line << endl;
            }
            fin.close();
```

```
            break;
        }
        case 3: {
            ifstream fin(filename1);
            ofstream fout(filename2);
            if (!fin || !fout) {
                cout << "Error opening file(s)\n";
                break;
            }
            string line;
            while (getline(fin, line)) {
                fout << line << endl;
            }
            fin.close();
            fout.close();
            cout << "Data copied from " << filename1 << " to " << filename2 << "
successfully!\n";
            break;
        }
        case 4:
            cout << "Exiting...\n";
            break;
        default:
            cout << "Invalid choice\n";
        }
    } while (choice != 4);

    return 0;
}
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\PG-DAC Assignments\C++ Module\Day 7> cd "d:\PG-DAC Assignments\C++ Modul

1:Write File
2:Read File
3:Copy Data (file1 ΓåÆ file2)
4:Exit
Enter choice: 1
Enter text (type END to stop):
Vishwajit Nikam(Founder & CEO Nikam Technologies).
END
Data written successfully!

1:Write File
2:Read File
3:Copy Data (file1 ΓåÆ file2)
4:Exit
Enter choice: 2
File Contents:
Vishwajit Nikam(Founder & CEO Nikam Technologies).

1:Write File
2:Read File
3:Copy Data (file1 ΓåÆ file2)
4:Exit
Enter choice: 3
Data copied from file1.txt to file2.txt successfully!

1:Write File
2:Read File
3:Copy Data (file1 ΓåÆ file2)
4:Exit
Enter choice: 4
Exiting...
PS D:\PG-DAC Assignments\C++ Module\Day 7>
```

6: Create CRUD Shop Application Using File
Write class Product with data member prdid,name,qty,price;
Menus:
1:Add Prd
2:Display Prds
3:Search Prd
4:Update/Modify prd
5:delete prd

```
#include <iostream>
#include <fstream>
```

```
#include <string>
#include <cstring> // for strcpy
using namespace std;

class Product {
    int prdid;
    char name[50];   // fixed-size array for safe file storage
    int qty;
    double price;

public:
    Product() {}
    Product(int id, const char* n, int q, double p) {
        prdid = id;
        strcpy(name, n);
        qty = q;
        price = p;
    }

    int getId() const { return prdid; }
    const char* getName() const { return name; }
    int getQty() const { return qty; }
    double getPrice() const { return price; }

    void setData() {
        cout << "Enter Product ID: ";
        cin >> prdid;
        cout << "Enter Name: ";
        cin.ignore(); // clear leftover newline
        cin.getline(name, 50);
        cout << "Enter Quantity: ";
        cin >> qty;
        cout << "Enter Price: ";
        cin >> price;
    }

    void display() const {
        cout << "ID: " << prdid
             << " | Name: " << name
             << " | Qty: " << qty
             << " | Price: " << price << endl;
    }
};

// File name
const string FILENAME = "products.dat";
```

```
// Add product
void addProduct() {
   Product p;
   p.setData();
   ofstream fout(FILENAME, ios::binary | ios::app);
   fout.write((char*)&p, sizeof(p));
   fout.close();
   cout << "✅ Product added successfully!\n";
}

// Display all products
void displayProducts() {
   ifstream fin(FILENAME, ios::binary);
   if (!fin) {
      cout << "⚠ No products found!\n";
      return;
   }
   Product p;
   cout << "\n--- Product List ---\n";
   while (fin.read((char*)&p, sizeof(p))) {
      p.display();
   }
   fin.close();
}

// Search product by ID
void searchProduct() {
   ifstream fin(FILENAME, ios::binary);
   if (!fin) {
      cout << "⚠ File not found!\n";
      return;
   }
   int id, found = 0;
   cout << "Enter product ID to search: ";
   cin >> id;
   Product p;
   while (fin.read((char*)&p, sizeof(p))) {
      if (p.getId() == id) {
         cout << " Product found!\n";
         p.display();
         found = 1;
         break;
      }
   }
   if (!found) cout << " Product not found!\n";
```

```
    fin.close();
}

// Update product
void updateProduct() {
    fstream file(FILENAME, ios::binary | ios::in | ios::out);
    if (!file) {
        cout << " File not found!\n";
        return;
    }
    int id, found = 0;
    cout << "Enter product ID to update: ";
    cin >> id;
    Product p;
    while (file.read((char*)&p, sizeof(p))) {
        if (p.getId() == id) {
            cout << "Current details:\n";
            p.display();
            cout << "Enter new details:\n";
            Product newP;
            newP.setData();
            file.seekp(-sizeof(p), ios::cur);
            file.write((char*)&newP, sizeof(newP));
            cout << " Product updated!\n";
            found = 1;
            break;
        }
    }
    if (!found) cout << " Product not found!\n";
    file.close();
}

// Delete product
void deleteProduct() {
    ifstream fin(FILENAME, ios::binary);
    if (!fin) {
        cout << " File not found!\n";
        return;
    }
    ofstream fout("temp.dat", ios::binary);
    int id, found = 0;
    cout << "Enter product ID to delete: ";
    cin >> id;
    Product p;
    while (fin.read((char*)&p, sizeof(p))) {
        if (p.getId() == id) {
```

```
            cout << "🗑 Deleting product: ";
            p.display();
            found = 1;
        } else {
            fout.write((char*)&p, sizeof(p));
        }
    }
    fin.close();
    fout.close();
    remove(FILENAME.c_str());
    rename("temp.dat", FILENAME.c_str());
    if (found) cout << " Product deleted!\n";
    else cout << " Product not found!\n";
}

int main() {
    int choice;
    do {
        cout << "\n===== Shop Menu =====\n";
        cout << "1: Add Product\n";
        cout << "2: Display Products\n";
        cout << "3: Search Product\n";
        cout << "4: Update/Modify Product\n";
        cout << "5: Delete Product\n";
        cout << "6: Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        switch (choice) {
        case 1: addProduct(); break;
        case 2: displayProducts(); break;
        case 3: searchProduct(); break;
        case 4: updateProduct(); break;
        case 5: deleteProduct(); break;
        case 6: cout << " Exiting...\n"; break;
        default: cout << " Invalid choice!\n";
        }
    } while (choice != 6);

    return 0;
}
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS D:\PG-DAC Assignments\C++ Module\Day 7> cd "d:\PG-DAC Assignmen

===== Shop Menu =====
1: Add Product
2: Display Products
3: Search Product
4: Update/Modify Product
5: Delete Product
6: Exit
Enter choice: 1
Enter Product ID: 1
Enter Name: Laptop
Enter Quantity: 1
Enter Price: 500000
Γ£à Product added successfully!

===== Shop Menu =====
1: Add Product
2: Display Products
3: Search Product
4: Update/Modify Product
5: Delete Product
6: Exit
Enter choice: 3
Enter product ID to search: 1
 Product found!
ID: 1 | Name: -■a | Qty: 6422248 | Price: 1.82281e-307

===== Shop Menu =====
1: Add Product
2: Display Products
3: Search Product
4: Update/Modify Product
```

7:Create Student class in namespace CDAC namespace .
Create another Student class in IACSD namespace.
Try to access both student classes using namespace

```
#include <iostream>
using namespace std;
```

```cpp
namespace CDAC {
    class Student {
        int roll;
        string name;
    public:
        Student(int r, string n) {
            roll = r;
            name = n;
        }
        void display() {
            cout << "[CDAC] Roll: " << roll << " Name: " << name << endl;
        }
    };
}

namespace IACSD {
    class Student {
        int roll;
        string name;
    public:
        Student(int r, string n) {
            roll = r;
            name = n;
        }
        void display() {
            cout << "[IACSD] Roll: " << roll << " Name: " << name << endl;
        }
    };
}

int main() {
    // Fully qualified names
    CDAC::Student s1(101, "Rahul");
    IACSD::Student s2(201, "Sneha");

    s1.display();
    s2.display();

    // Using namespace alias
    namespace C = CDAC;
    namespace I = IACSD;

    C::Student s3(102, "Amit");
    I::Student s4(202, "Priya");

    s3.display();
```

```
    s4.display();

    return 0;
}
```